

MAT 157 Analysis I

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2025 Fall

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0.1 Introduction

The course is mainly based on the book *Spivak Calculus* covers chapter 10 of *A readable introduction to real mathematics* by *Daniel Rosenthal et al.*

Chapter 1

Week 1

Note:-

Additional practice problems on cardinality, foundations, and functions can be found in Chapter 23, Sections 23.1 and 23.2.

1.1 Cardinality

Definition 1.1.1: Cardinality

Cardinality is the "number of things" in a set. Example: $A = \{1, 2, 3\}$ has cardinality 3, $B = \{1, 2, 3, \dots\}$ has infinite cardinality. To be specific, when counting the numbers in the set, the mapping of numbers to the elements need to be **bijective**.

Definition 1.1.2

Given two sets S and T , if there is a $f : S \rightarrow T$, that's bijective, S and T have the same cardinality, we write $|S| = |T|$.

Definition 1.1.3

Given a bijective function $f : S \rightarrow T$ then S and T have the same cardinality

Note:-

Note that the concepts of injective, surjective and bijective functions are noted down in course note MAT 240, thus disregard

Example 1.1.1 (Bijection from $[1, 3] \rightarrow (30, 40]$)

(This is a thinking question at the end of the week 1 lecture) Construct a bijection between $[1, 3]$ to $(30, 40]$

A solution could be that

$$f(x) = \begin{cases} 30 + \frac{1}{2}, & x = 1, \\ 30 + \frac{1}{2^{n+1}}, & x = 1 + \frac{2^{-n}}{5}, n \in \mathbb{N} \\ 5x + 25, & \text{otherwise.} \end{cases}$$

The basic idea is that the sequence $\frac{1}{2^n}, n \in \mathbb{N}$ is strictly decreasing from $\frac{1}{2}$ approaching 0. This allows us to construct a sequence $30 + \frac{1}{2^n}$ which is decreasing from $30 + \frac{1}{2}$ to 30. By shifting it to the left, we have $30 + \frac{1}{2^{n+1}}$, which gives a vacant spot for assigning the case where $x = 1$ (the idea of Hilbert's hotel).

The second case's condition may look strange. The since we have pushed te $30 + \frac{1}{2^n}$ sequence back by one, we need to remap the preimage of it to the new one. The original function is $f(x) = 5x + 25$, so the condition derives from solving the equation $30 + \frac{1}{2^n} = 5x + 25$, which gives $x = 1 + \frac{2^{-n}}{5}$

In short, the key idea is to find a decreasing sequence in the codomain that's approaching the open end, and shift it to allow a space for $x = 1$.

Definition 1.1.4

A set A is finite if there is a $n \in \mathbb{N}$ and a $f : \{1, 2, \dots, n\} \rightarrow A$ which is bijective. We say $|A| = n$

Definition 1.1.5

If A is the empty set then $|A| = |\emptyset| = |\{\}| = 0$

Definition 1.1.6

A set A is **countable** if A is infinite or $|A| = |\mathbb{N}|$. There is $f : \mathbb{N} \rightarrow A$ is bijective

Example 1.1.2 (Even natural numbers have same cardinality as all natural numbers)

Question 1

Let $\mathbb{E}$ = even natural numbers = $\{2, 4, 6, 8, \dots\}$. $|\mathbb{E}| = |\mathbb{N}|$ and E is countable. Find a function $f : \mathbb{N} \rightarrow \mathbb{E}$, where f is bijective.

Solution:

$$f(n) = 2n$$

Claim 1.1.1 $f(n) = 2n$ is injective

Proof: Assume $f(n_1) = f(n_2)$ where $n_1, n_2 \in \mathbb{N}$. Then $2n_1 = 2n_2$ and so $2(n_1 - n_2) = 0$. Therefore $n_1 - n_2 = 0$ and so $n_1 = n_2$. Thus f is injective. ☺

Claim 1.1.2 $f(n) = 2n$ is surjective

Proof: Let $t \in \mathbb{E}$. Then $t = 2n$ for some $n \in \mathbb{N}$. Therefore $f(n) = t$. And f is onto $\mathbb{E}$. ☺

Example 1.1.3 (The set of even natural numbers and the set of odd natural numbers have the same cardinality.)

Proof: Let the set of even natural numbers $E = \{2, 4, \dots, 2n, \dots\}$ and the set of odd natural numbers $O = \{1, 3, \dots, 2n - 1, \dots\}$. To satisfy the definition 1.1.3, we need to show that there is an injective function taking E onto O . Define a function $f : E \rightarrow O$, simply note that $k_1 - 1 = k_2 - 1$ implies $k_1 = k_2$. Also, f is surjective, because for any $m \in O$, $m = 2n - 1$ for some $n \in \mathbb{N}$. Therefore $f(2n) = 2n - 1 = m$. Thus, f is bijective and $|E| = |O|$. ☺

Example 1.1.4 (The set of natural numbers and the set of nonnegative integers have the same cardinality.)

Proof: Let S denote the set of nonnegative integers. We want to construct a bijective function $f : S \rightarrow \mathbb{N}$. Let f be $f(n) = n + 1, n \in S$. This makes f surjective. Also, $f(n_1) = f(n_2)$ implies $n_1 + 1 = n_2 + 1$, giving $n_1 = n_2$. Since f is bijective, $\mathbb{N}$ and S have the same cardinality. ☺

Theorem 1.1.1 The set of natural numbers and the set of positive rational numbers have the same cardinality

Proof: To prove the theorem, we first construct the array:

$$\begin{array}{cccccccc} \frac{1}{1} & \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \frac{1}{6} & \frac{1}{7} & \dots \\ \frac{2}{1} & \frac{2}{2} & \frac{2}{3} & \frac{2}{4} & \frac{2}{5} & \frac{2}{6} & \frac{2}{7} & \dots \\ \frac{3}{1} & \frac{3}{2} & \frac{3}{3} & \frac{3}{4} & \frac{3}{5} & \frac{3}{6} & \frac{3}{7} & \dots \\ \frac{4}{1} & \frac{4}{2} & \frac{4}{3} & \frac{4}{4} & \frac{4}{5} & \frac{4}{6} & \frac{4}{7} & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \end{array}$$

Let the following:

- $f(1) = \frac{1}{1}$
- $f(2) = \frac{1}{2}$
- $f(3) = \frac{2}{1}$
- $f(4) = \frac{3}{1}$
- $f(5) = \frac{1}{3}$
- $f(6) = \frac{3}{4}$

- $f(7) = \frac{2}{3}$
- $f(8) = \frac{3}{2}$
- $f(9) = \frac{4}{1}$
- $f(10) = \frac{5}{1}$
- ...

We can observe that for the first diagonal of equivalent elements $1/1, 2/2, \dots$ it moves 1 down 1 right, $1d1r$ in short. Next for $1/2$ it moves $1d2r$ and $1d3r$ for $1/3$, and so on. Moreover, for $2/1, 4/2, \dots$ it move $2d1r$, and $3/1, 6/2, \dots$ moves $3d1r$, and so on. According to this sequence, we can pair natural numbers to the positive rational numbers indicated in the "zigzagging" manner. Therefore $|\mathbb{Q}^+| = |\mathbb{N}|$

To be precise, consider the set

$$S = \{(m, n) \in \mathbb{N} \times \mathbb{N} : \gcd(m, n) = 1\}$$

Next define the function f :

$$f : \mathbb{N} \rightarrow \mathbb{Q}_{>0}, f(x) = \frac{m_x}{n_x}$$

Claim 1.1.3 f is injective

Proof: If $f(x) = f(y)$,

$$\frac{m_x}{n_x} = \frac{m_y}{n_y}$$

since both fraction have $\gcd = 1, x = y$

☺

Claim 1.1.4 f is surjective

Proof: Take any $q \in \mathbb{Q}_+$, it can be expressed as $q = \frac{m}{n}$ in lowest terms. Then $(m, n) \in S$, which equals to some a_x, b_x , thus $f(x) = q$

☺

Since the function f is bijective, $|\mathbb{N}| = |\mathbb{Q}_{>0}|$

☺

Another proof (during lecture)

Proof: Define

$$R = \left\{ \frac{q}{p} : p, q \in \mathbb{N} \right\}$$

Clearly $\mathbb{Q}^+ \subsetneq R$, since, for example $\frac{2}{4} \in R$ but when considered as a set of rationals we have $2/4 = 1/2 \in \mathbb{Q}^+$.

Lemma 1.1.1

Any subset of a countable set is either finite or countably infinite.

By proving R is countable, it follows that $\mathbb{Q}^+$ is countable as well. Consider the map

$$\begin{aligned}\psi : \mathbb{N} \times \mathbb{N} &\rightarrow R \\ \psi(p, q) &= \frac{q}{p}\end{aligned}$$

This is a **surjection**. Now we need to prove that $\mathbb{N} \times \mathbb{N}$ is countable. We define the **Cantor pairing function** $g : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$,

$$g(i, j) = \frac{(i + j - 2)(i + j - 1)}{2} + j$$

This enumerates all lattice points (i, j) by diagonals of constant sum $i + j$. By checking it's bijective (skip), $\mathbb{N} \times \mathbb{N}$ is countable. Taking the inverse $f = g^{-1} : \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$. Since ψ is surjective mapping $\mathbb{N} \times \mathbb{N}$ to R , $\varphi = \psi \circ f : \mathbb{N} \rightarrow R$ is a surjection. Therefore R is countable. Applying the lemma, it implies that $\mathbb{Q}^+$ is countable. ☺

Definition 1.1.7: Pigeonhole principle

If S is a **finite set**, then a function $g : S \rightarrow S$ is injective, if and only if it is onto. This principle fails for **infinite sets**.

Proof:

☺

1.2 Countable Sets and Uncountable Sets

Definition 1.2.1: Countable

A set is **countable** (or denumerable or enumerable) if it is either

- **finite**
- or has the **same cardinality** as the set of natural numbers.

Definition 1.2.2: Closed interval

For a and b real numbers with $a \leq b$, the *closed interval from a to b* is the set of all real numbers between a and b , **including them**. It's denoted $[a, b] = \{x : a \leq x \leq b\}$

Theorem 1.2.1

The closed interval $[0, 1]$ is uncountable.

Proof: We will show that there is no function f maps $\mathbb{N}$ onto $[0, 1]$. The following is *Cantor diagonal argument*. Suppose that f is any function taking $\mathbb{N} \rightarrow [0, 1]$. To prove that f cannot be onto, we can imagine the following construction where we write out all

the values of f in a list, as follows:

$$\begin{aligned} f(1) &= .a_{11}a_{12}a_{13}a_{14}a_{15} \dots \\ f(2) &= .a_{21}a_{22}a_{23}a_{24}a_{25} \dots \\ f(3) &= .a_{31}a_{32}a_{33}a_{34}a_{35} \dots \\ f(4) &= .a_{41}a_{42}a_{43}a_{44}a_{45} \dots \\ f(5) &= .a_{51}a_{52}a_{53}a_{54}a_{55} \dots \\ &\vdots \end{aligned}$$

In other words, for $f(x) = 0.a_1a_2\dots$, where

$$a_k = \lfloor 10^k x \rfloor - 10 \lfloor 10^{k-1} x \rfloor$$

Now we need to construct a number in $[0, 1]$ but not in the range of function f . We consider the following sequence $x = .b_1b_2\dots$ where if $a_{ii} = 3, b_i = 4$, and if $a_{ii} \neq 3, b_i = 3$. In this way, we can construct a number x that differs $f(j)$ in its j^{th} digit. Thus $f(j) \neq x \forall j$, so x is not in the range of f . Moreover, if we try to adding the number x into the range of f by having a new function g where $g(1) = x$ and $g(n) = f(n-1)$ for $n \geq 2$ (basically shifting every element in f back by 1). However, the construction above could be used again on $g(x)$ and construct a new x that is not in the range of g . Thus this proves that $[0, 1]$ is not countable. ☺

To be more precise,

Proof: Every $y \in [0, 1]$ has binary expansion

$$y = \sum_{k \geq 1} a_k 2^{-k}, a_k \in 0, 1$$

Some y has two expansions, e.g. $0.10000\dots_2$ and $0.01111\dots_2$. For each $n \in \mathbb{N}$, a **convention** is that to write $f(n)$ in its binary expansion that's not eventually all 1's.

We can write

$$f(n) = 0.a_{n1}a_{n2}a_{n3}\dots, a_{nk} \in \{0, 1\}$$

Define a binary sequence where $b_k \geq 1$ by $b_k := 1 - a_{kk}$, let

$$x := 0.b_1b_2b_3\dots = \sum_{k \geq 1} b_k 2^{-k}$$

Then $x \in [0, 1]$. This makes x and $f(n)$ differ in the n -th digit, so $x \neq f(n)$. Since this holds for every n , we conclude $x \notin f(\mathbb{N})$. Therefore, no function $f : \mathbb{N} \rightarrow [0, 1]$ is surjective. Hence $[0, 1]$ is uncountable. ☺

Theorem 1.2.2

If a and b are real numbers and $a < b$, the $[a, b]$ and $[0, 1]$ have the **same cardinality**.

Proof: The theorem will be established if there is a bijective function $f : [0, 1] \rightarrow [a, b]$.

Let $f(x) = a + (b - a)x$. Then $f(0) = a$ and $f(1) = b$.

To prove it's **injective**, we have

$$\begin{aligned} f(x_1) &= f(x_2) \\ a + (b - a)x_1 &= a + (b - a)x_2 \\ (b - a)x_1 &= (b - a)x_2 \\ x_1 &= x_2 \end{aligned}$$

To show that f is **surjective**, let y be any element of $[a, b]$. Let $x = \frac{y-a}{b-a}$. Then $x \in [0, 1]$ and $f(x) = y$. Thus, the f is bijective, and $|[0, 1]| = |[a, b]|$ ☺

Theorem 1.2.3 The intervals $[0, 1]$ and $(0, 1]$ have the same cardinality

Similar proof to the thinking question 1.1.1, thus disregard.

Theorem 1.2.4 If $|S| = |T|$ and $|T| = |U|$, then $|S| = |U|$

Proof: Let $f : S \rightarrow T$ and $g : T \rightarrow U$ be **bijective**, and $h = g \circ f$. Given $u \in U$, since g is surjective, there exists a $t \in T$ such that $g(t) = u$.

Step 1: proving surjectivity. Since f is surjective, there is an $s \in S$ such that $f(s) = t$. Then $h(s) = g(f(s)) = g(t) = u$. Thus, h is surjective.

Step 2: proving injectivity. Suppose that $h(s_1) = h(s_2)$, $g(f(s_1)) = g(f(s_2))$, so $f(s_1) = f(s_2)$ since g is injective. Since f is also injective, so $s_1 = s_2$. This proves that h is bijective, and $|S| = |U|$. ☺

Theorem 1.2.5 If a, b, c, d are real numbers with $a < b$ and $c < d$, then $(a, b]$ and $(c, d]$ have the same cardinality

Proof: f is defined by $f(x) = a + (b - a)x$ which is an injective function mapping $(0, 1]$ onto $(a, b]$, as mentioned in the theorem 1.2.2. Hence, $|(0, 1]| = |(a, b]|$. Similarly g is defined by $g(x) = c + (d - c)x$ which is an injective function mapping $(0, 1]$ onto $(c, d]$. Following theorem 1.2.2, $|(a, b]| = |(c, d]|$ ☺

Theorem 1.2.6 The cardinality of the set of nonnegative real numbers is the same as the cardinality of the unit interval $[0, 1]$

Proof: We aim to show the set $S = \{x : x \geq 1\}$ has same cardinality as $(0, 1]$.

Step 1.1: show surjectivity. Note that the function f is defined by $f(x) = \frac{1}{x}$ maps S into $(0, 1]$; for if $x \geq 1$, then $\frac{1}{x} \leq 1$. Also, f maps S onto $(0, 1]$; for if $y \in (0, 1]$, then $\frac{1}{y} \geq 1$ and $f(\frac{1}{y}) = y$.

Step 1.2: show injectivity. To see that f is injective, suppose that $f(x_1) = f(x_2)$, then $\frac{1}{x_1} = \frac{1}{x_2}$, so $x_1 = x_2$. Hence, f is bijective. Thus, $|S| = |(0, 1]|$.

Step 2 Now let $T = \{x : x \geq 0\}$. Define the function g by $g(x) = x - 1$. Then g is a bijective function mapping $S \rightarrow T$. Hence, $|T| = |S|$. Thus, by theorem 1.2.4, $|T| = |(0, 1]|$.

By theorem 1.2.3,

$$|[0, 1]| = |(0, 1]| \Rightarrow |T| = |[0, 1]|$$



Theorem 1.2.7 The union of a countably many countable sets is countable

That is, if $\{S_i : i \in \mathbb{N}\}$ is a countable collection of sets and each S_i is countable, then

$$S = \bigcup_{i=1}^{\infty} S_i$$

is countable.

The proof varies from the book

Proof: For each $i \in \mathbb{N}$, since S_i is countable, there exists a **surjection**

$$f_i : \mathbb{N} \rightarrow S_i$$

Now defining a function

$$\begin{aligned} F : \mathbb{N} \times \mathbb{N} &\rightarrow S \\ F(i, j) &= f_i(j) \end{aligned}$$

This is a **surjection** because for any $s \in S$, exists $s \in S_k$ for some k , so $s = f_k(j)$ for some j . Hence $s = F(k, j)$.

Step 1: Countability of $\mathbb{N} \times \mathbb{N}$. Previously proven with **Cantor's pairing function** that it's countable. We can define

$$\begin{aligned} g : \mathbb{N} \times \mathbb{N} &\rightarrow \mathbb{N} \\ g(i, j) &= \frac{(i + j - 2)(i + j - 1)}{2} + j \end{aligned}$$

this function is a bijection, hence $\mathbb{N} \times \mathbb{N}$ is countable.

Step 2: Composition. Since g is bijective, it has an inverse $f = g^{-1} : \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$, by compositing it with F gives:

$$h = F \circ f : \mathbb{N} \rightarrow S$$

Then h is a surjection from $\mathbb{N} \rightarrow S$. By definition, a set is countable is there is a surjection from $\mathbb{N}$ onto it. Thus, the function h constructed above means that S is countable. ☺

Theorem 1.2.8 Cantor-Bernstein Theorem

If there are injections $f : A \rightarrow B$ and $g : B \rightarrow A$, then there exists a bijection $h : A \rightarrow B$.

Chapter 2

Week 2

Note:-

Additional practice on field properties, order axioms, and induction can be found in Chapter 23, Section 23.1.

Note:-

This is the end of contents in *Rosenthal's* book. The following contents are mainly based on *Spivak's* Calculus

Definition 2.0.1: Basic properties of numbers

The properties are listed in notes for Rudin's analysis, rewritten in the format of

1. **Associativity** For any $a, b, c \in A$, $a + (b + c) = (a + b) + c$
 2. **Additive identity** There is $0 \in A$ so that for any $a \in A$, $0 + a = a + 0 = a$
 3. **Additive inverse** For any $a \in A$, there is $-a \in A$ so that $a + (-a) = (-a) + a = 0$
 4. **Commutativity** For any $a, b \in A$, $a + b = b + a$
 5. **Associativity of multiplication** For any $a, b, c \in A$, $a(bc) = (ab)c$
 6. **Multiplicative identity** There is $1 \in A$, $1 \neq 0$ so that for any $a \in A$, $1a = a1 = a$
 7. **Multiplicative inverse** For any $a \in A$, $a \neq 0$, there is $a^{-1} \in A$ so that $aa^{-1} = a^{-1}a = 1$
 8. **Multiplicative commutativity** For any $a, b \in A$, $ab = ba$
 9. **Distributive law** For any $a, b, c \in A$, $a(b + c) = (ab) + (ac)$
 10. **Trichotomy law** Consider the collection of all positive numbers, P . For every number a , one and only one of the following holds:
 - (a) $a = 0$
 - (b) a is in the collection P
 - (c) $-a$ is in the collection P
 11. **Closure under addition** If a and b are in P , then so is $a + b$
 12. **Closure under multiplication** If a and b are in P , then so is $a \cdot b$
- P is the collection of all positive numbers.

Claim 2.0.1 If $(A, +)$ satisfies (1) – (3), then 0 is unique

Proof: Suppose $\tilde{0} \in A$ also satisfies the identity property, i.e.

$$a + \tilde{0} = a = \tilde{0} + a \quad \text{for all } a \in A.$$

Let $a \in A$ be arbitrary. Then

$$a + \tilde{0} = a.$$

By property (3), the inverse $-a$ exists. Adding $-a$ to both sides gives

$$(-a) + (a + \tilde{0}) = (-a) + a.$$

By associativity (1),

$$((-a) + a) + \tilde{0} = (-a) + a.$$

By property (3), $(-a) + a = 0$. Hence

$$0 + \tilde{0} = 0.$$

Finally, by the identity property (2),

$$\tilde{0} = 0.$$

Thus the additive identity is unique. ☺

Question 1

- uniqueness of $-a \Rightarrow -(-a) = a$
- prove $0 \cdot a = 0$ for all $a \in A$
- prove 1 is unique
- $(-a) \cdot b = -(a \cdot b)$
- $(-a) \cdot (-b) = a \cdot b$

Spivak's numbers means a set that contains $\mathbb{Z}$ and satisfies (1)-(9).

Definition 2.0.2: Ordering on A

If $a, b \in A$ and A, P satisfies (10)-(12) then,

- $a > b$ means $a + (-b) \in P$
- $a \geq b$ means $a + (-b) \in P$ or $a + (-b) = 0$
- $a < b$ means $b + (-a) \in P$
- $a \leq b$ means $b + (-a) \in P$ or $b + (-a) = 0$

In particular, $a > 0$ if and only if $a \in P$.

Claim 2.0.2

Lemma 2.0.1 If $a \in P \Rightarrow a + 0 \in P$

- If $a < b$ and $c > 0$ then $a \cdot c < b \cdot c$
- If $a < b$ and $c \in A$ then $a + c < b + c$

- If $a < b$ and $c < 0 \Rightarrow a \cdot c > b \cdot c$

Proof: Proof of (2), from definition of $<$, $a < b \Rightarrow b + (-a) \in P$.

$$\begin{aligned} &\Rightarrow b + (-a) + 0 \in P \\ &b + (-a) + ((-c) + c) \in P \\ &(b + (-c)) + ((-a) + c) \in P \\ &(b + (-c)) + (-(a + (-c))) \in P \end{aligned}$$



Claim 2.0.3 If $0 < a < b \Rightarrow a^2 < b^2$

Proof: Proof 1:

$$b^2 + (-a^2) = (b \cdot b) + (-(a \cdot a)) = (b + (-a))(b + a)$$

Since $b > a$, $b - a > 0$ and $b + a > 0$. This proves $(b - a)(b + a) > 0 \Rightarrow (b^2 - a^2) > 0 \Rightarrow b^2 > a^2$
Proof 2:

$$\begin{aligned} a^2 &< a \cdot b \\ a \cdot b &< b^2 \\ a^2 &< b^2 \end{aligned}$$



Claim 2.0.4 If $a > 0, b > 0, a^2 < b^2 \Rightarrow a < b$

Claim 2.0.5 If $a < b$ then $a^3 < b^3$

Claim 2.0.6 If $0 < a < b$, $a, b \in \mathbb{R}$, then $a < \sqrt{a \cdot b} < \frac{a+b}{2} < b$

Assume that for any $c > 0, c \in \mathbb{R}$ there is a unique $d \in \mathbb{R}, d > 0$ so that $d^2 = c$, call d " $\sqrt{c}$ ".

Chapter 3

Week 3

Note:-

Additional practice on induction, well-ordering, and combinatorial identities can be found in Chapter 23, Section 23.1.

3.1 Peano Axiom

The Peano's axioms are proposed by **Giuseppe Peano (1889)** to rigorously define the natural numbers $\mathbb{N} = \{0, 1, 2, 3, \dots\}$ and their arithmetic.

Definition 3.1.1: Natural numbers

The natural numbers $\mathbb{N}$ are a set equipped with a successor operation $S : \mathbb{N} \rightarrow \mathbb{N}$ such that the following Peano axioms hold:

1. **Existence of the first element:** 1 is a natural number.
2. **Successor property:** If $n \in \mathbb{N}$, then $S(n) \in \mathbb{N}$.
3. **Injectivity of successor:** If $m, n \in \mathbb{N}$ and $S(m) = S(n)$, then $m = n$.
4. **Non-predecessorship of 1:** There is no $n \in \mathbb{N}$ such that $S(n) = 1$.
5. **Induction axiom:** If $A \subseteq \mathbb{N}$ such that $1 \in A$, and whenever $n \in A$ implies $S(n) \in A$, then $A = \mathbb{N}$.

Using these axioms, we can define addition and multiplication recursively via S .

Note:-

Peano originally had $\mathbb{N}$ start with 1, later $\mathbb{N}$ started with 0. We take $\mathbb{N}$ start with 1 for the course.

3.2 Induction

Here are some examples for mathematical induction during lecture:

Example 3.2.1 (For all $n \in \mathbb{N}$: $2 + 4 + 6 + \dots + 2n = n(n + 1)$)

The ideas:

- Let $K = \{n \in \mathbb{N}, 2 + 4 + \dots + 2n = n(n+1)\}$
- show $1 \in K$
- assume $n \in K$ show $n+1 \in K$
- Conclude $K = \mathbb{N}$ by peano axiom that $S(n) = n+1$

Proof: Let $K = \{n \in \mathbb{N}, 2 + 4 + \dots + 2n = \sum_{i=1}^n 2i = n(n+1)\}$. Take $n = 1$, then $2 = 1(1+1)$ is true. So $1 \in K$.

Assume $n \in K$,

$$2 + 4 + 6 + \dots + 2n + 2(n+1) = n(n+1) + 2(n+1)$$

because $n \in K$

So

$$2 + 4 + \dots + 2(n+1) = (n+1)((n+1)+1)$$

is true, and so $n+1 \in K$. Therefore K satisfies Peano axiom 5, and so $K = \mathbb{N}$.

☺

Example 3.2.2

Let $a \in \mathbb{R}$ with $-k < a < 0$ and $a < -1$ For all $n \in \mathbb{N}$

$$(1+a)^n < 1 + na + \frac{n^2 a^2}{2}$$

Proof: Let

$$K = \{n \in \mathbb{N} : (1+a)^n < 1 + na + \frac{n^2 a^2}{2}\}$$

We first show $1 \in K$,

$$(1+a) < 1 + 1 \cdot a + \frac{1^2 a^2}{2}$$

because $\frac{1^2 a^2}{2} > 0$ and $a \neq 0$. Therefore $1 \in K$.

Assume $n \in K$, so

$$(1+a)^n < 1 + na + \frac{n^2 a^2}{2}$$

We want to show that $n+1 \in K$.

$$(1+a)^{n+1} = (1+a)^n(1+a) < (1 + na + \frac{n^2 a^2}{2})(1+a)$$

because $n \in K$ and $1+a > 0$. By expanding product,

$$\begin{aligned} (1+a)^{n+1} &< 1 + (n+1)a + na^2 + \frac{n^2 a^2}{2} + \frac{n^2 a^3}{2} \\ &= 1 + (n+1)a + \frac{(n+1)^2 a^2}{2} - \frac{(n+1)^2 a^2}{2} + na^2 + \frac{n^2 a^2}{2} + \frac{n^2 a^3}{3} \end{aligned}$$

by several steps of deduction, $= 1 + (n+1)a + \frac{(n+1)^2 a^2}{2} + \frac{a^2(an^2-1)}{2}$

$$\leq 1 + (n+1)a + \frac{(n+1)^2 a^2}{2}$$

because $a < 0 \Rightarrow \frac{a^2(an^2-1)}{2} < 0$. Therefore $n+1 \in K$ and $K = \mathbb{N}$

☺

Theorem 3.2.1 (AM–GM Inequality)

Let $x_1, x_2, \dots, x_n \geq 0$. Then

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n},$$

with equality if and only if $x_1 = x_2 = \dots = x_n$.

Proof: We proceed by induction on n .

Base case: $n = 1$. Trivially $\frac{x_1}{1} = x_1 = \sqrt[1]{x_1}$.

Base case: $n = 2$. We need to show

$$\frac{x_1 + x_2}{2} \geq \sqrt{x_1 x_2}.$$

Consider the nonnegative square

$$0 \leq (x_1 - x_2)^2 = x_1^2 - 2x_1 x_2 + x_2^2.$$

Rearrange:

$$x_1^2 + 2x_1 x_2 + x_2^2 \geq 4x_1 x_2 \implies (x_1 + x_2)^2 \geq 4x_1 x_2.$$

Taking the positive square root (since both sides are nonnegative) and then dividing by 2 gives the desired inequality. Equality holds only when $(x_1 - x_2)^2 = 0$, i.e. $x_1 = x_2$.

Induction step. Assume the result holds for all lists of length n . Let $x_1, x_2, \dots, x_{n+1} \geq 0$. Write

$$\alpha = \frac{x_1 + \dots + x_{n+1}}{n+1}.$$

If all $x_i = \alpha$, then both sides (AM and GM) equal α , and the result holds with equality. Otherwise, not all x_i are equal.

We can reorder so that x_{n+1} is one of the “extreme” elements, and consider the first n elements $x_1, \dots, x_n$. By the induction hypothesis,

$$\frac{x_1 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n}.$$

Now compare the arithmetic mean of all $n+1$ with the geometric mean:

$$\alpha = \frac{1}{n+1}(x_1 + \dots + x_n + x_{n+1}) = \frac{n}{n+1} \cdot \frac{x_1 + \dots + x_n}{n} + \frac{1}{n+1} x_{n+1}.$$

Applying the (already established) two-term AM–GM to the two numbers

$$\frac{x_1 + \dots + x_n}{n} \quad \text{and} \quad x_{n+1},$$

we get

$$\frac{1}{2} \left(\frac{x_1 + \dots + x_n}{n} + x_{n+1} \right) \geq \sqrt{\frac{x_1 + \dots + x_n}{n} \cdot x_{n+1}}.$$

One arranges the coefficients so that this gives

$$\alpha \geq \sqrt[n+1]{x_1 x_2 \dots x_{n+1}}.$$

A more detailed algebraic manipulation is usually given in textbooks (or see the Wikipedia “induction proof” section). Equality can hold only if each inequality used in the induction is an equality, which forces $x_1 = \dots = x_{n+1}$.



Theorem 3.2.2 (AM–GM an alternative proof)

For any $n \in \mathbb{N}$ and any positive real numbers $b_1, \dots, b_n$,

$$\frac{b_1 + \dots + b_n}{n} \geq (b_1 b_2 \dots b_n)^{1/n},$$

with equality iff $b_1 = \dots = b_n$.

Proof: We split the proof in two steps.

Step 1 (Inductive lemma for product = 1). *Claim.* If $a_1, \dots, a_n > 0$ and $a_1 \dots a_n = 1$, then

$$a_1 + \dots + a_n \geq n,$$

with equality iff $a_1 = \dots = a_n = 1$.

Proof by induction on n .

- **Base $n = 1$.** From $a_1 \cdot 1 = 1$ we get $a_1 = 1$, hence $a_1 \geq 1$ with equality.
- **Inductive step.** Assume the statement holds for some $n \geq 1$. Let $a_1, \dots, a_{n+1} > 0$ satisfy $a_1 \dots a_{n+1} = 1$. Reindex so that

$$a_n = \min\{a_1, \dots, a_{n+1}\} \quad \text{and} \quad a_{n+1} = \max\{a_1, \dots, a_{n+1}\}.$$

Because the product is 1, we must have $a_n \leq 1 \leq a_{n+1}$. Hence

$$(a_n - 1)(a_{n+1} - 1) \leq 0 \implies a_n + a_{n+1} \geq 1 + a_n a_{n+1}. \quad (*)$$

Now define the n -tuple

$$b_1, \dots, b_{n-1}, b_n := a_1, \dots, a_{n-1}, a_n a_{n+1}.$$

Then $b_1 \dots b_n = (a_1 \dots a_{n-1})(a_n a_{n+1}) = a_1 \dots a_{n+1} = 1$. By the inductive hypothesis,

$$a_1 + \dots + a_{n-1} + a_n a_{n+1} \geq n. \quad (\dagger)$$

Adding $(*)$ and $(\dagger)$ gives

$$a_1 + \dots + a_{n+1} = (a_1 + \dots + a_{n-1} + a_n a_{n+1}) + (a_n + a_{n+1} - a_n a_{n+1}) \geq n + 1.$$

Equality case. In $(*)$ equality forces $a_n = 1$ and $a_{n+1} = 1$; then $(\dagger)$ forces $a_1 = \dots = a_{n-1} = 1$ by the inductive hypothesis. Thus equality holds iff all $a_i = 1$.

This completes the induction and proves the claim. $\triangle$

Step 2 (Normalization and conclusion). Let $b_1, \dots, b_n > 0$ and set

$$G := (b_1 b_2 \dots b_n)^{1/n}, \quad a_i := \frac{b_i}{G} \quad (1 \leq i \leq n).$$

Then $a_1 \dots a_n = \frac{b_1 \dots b_n}{G^n} = 1$, so the lemma yields

$$\sum_{i=1}^n a_i = \sum_{i=1}^n \frac{b_i}{G} \geq n.$$

Multiplying by G gives

$$b_1 + \dots + b_n \geq n G = n (b_1 b_2 \dots b_n)^{1/n},$$

and dividing by n proves AM–GM.

Equality case. Equality in Step 1 requires $a_1 = \dots = a_n = 1$, i.e. $b_1 = \dots = b_n = G$. ☺

Definition 3.2.1: Comparison: Standard vs. Strong Induction

Standard (Weak) Induction. To prove a statement $P(n)$ for all $n \geq n_0$:

- **Base case:** Show $P(n_0)$ is true.
- **Inductive step:** Assume $P(k)$ is true for some $k \geq n_0$ (the induction hypothesis). Prove $P(k+1)$ is true.

Here the hypothesis only assumes $P(k)$ to prove $P(k+1)$.

Strong Induction. To prove a statement $P(n)$ for all $n \geq n_0$:

- **Base case:** Show $P(n_0)$ is true (and sometimes $P(n_0+1), \dots, P(n_0+r)$ depending on the problem).
- **Inductive step:** Assume $P(n_0), P(n_0+1), \dots, P(k)$ are all true (the strong induction hypothesis). Prove $P(k+1)$ is true.

Here the hypothesis allows you to assume *all* earlier cases up to k , not just the immediate predecessor.

Summary.

- Standard induction: assume $P(k)$ to prove $P(k+1)$.
- Strong induction: assume all $P(j)$ for $n_0 \leq j \leq k$ to prove $P(k+1)$.
- Both principles are logically equivalent, but strong induction is often easier when $P(k+1)$ depends on several earlier cases.

Definition 3.2.2: Fundamental Theorem of Arithmetic

If $n \in \mathbb{N}, n \geq 2$ can be written as a product of primes.

Proof: We proceed by *strong induction* on n .

Base case ($n = 2$). 2 is prime, so it is already a product of primes (just itself).

Induction hypothesis. Assume that every integer m with $2 \leq m \leq k$ can be written as a product of primes.

Inductive step. We prove the statement for $k+1$.

- If $k+1$ is prime, then it is its own prime factorization.
- If $k+1$ is composite, then there exist integers a, b with $2 \leq a, b \leq k$ such that $k+1 = ab$. By the induction hypothesis, both a and b can be written as products of primes. Multiplying these factorizations gives a prime factorization of $k+1$.

Conclusion. By the principle of strong induction, every $n \geq 2$ has a prime factorization. ☺

3.2.1 Recursion Relations

Example 3.2.3 (Recursion Relations)

Generate a sequence $\{c_1, c_2, \dots\} \subseteq \mathbb{N}$ as follows:

$$c_1 = 0, \quad c_2 = 1, \quad c_{n+1} = 5c_n - 6c_{n-1} \quad (n \geq 2).$$

Claim 3.2.1

For all $n \geq 1$, the closed form is

$$c_n = 3^{n-1} - 2^{n-1}.$$

Proof: We proceed by induction on n .

Base cases. For $n = 1$: $c_1 = 0$, and the formula gives $3^0 - 2^0 = 1 - 1 = 0$. For $n = 2$: $c_2 = 1$, and the formula gives $3^1 - 2^1 = 3 - 2 = 1$. So the formula holds for $n = 1, 2$.

Induction hypothesis. Assume for some $n \geq 2$ that

$$c_n = 3^{n-1} - 2^{n-1}, \quad c_{n-1} = 3^{n-2} - 2^{n-2}.$$

Inductive step. Using the recurrence:

$$c_{n+1} = 5c_n - 6c_{n-1}.$$

Substitute the hypothesis:

$$c_{n+1} = 5(3^{n-1} - 2^{n-1}) - 6(3^{n-2} - 2^{n-2}).$$

Simplify:

$$= 5 \cdot 3^{n-1} - 5 \cdot 2^{n-1} - 6 \cdot 3^{n-2} + 6 \cdot 2^{n-2}.$$

Factor powers:

$$\begin{aligned} &= 3^{n-2}(15 - 6) - 2^{n-2}(10 - 6) = 9 \cdot 3^{n-2} - 4 \cdot 2^{n-2}. \\ &= 3^n - 2^n. \end{aligned}$$

Thus the formula holds for $n + 1$.

Conclusion. By induction, the closed form $c_n = 3^{n-1} - 2^{n-1}$ is valid for all $n \geq 1$. ☺

Example 3.2.4 (Binary Strings of length m)

How many binary strings of length n has k 1s?

$$B_{k,m} = \text{number of binary strings with } k \text{ 1s}$$

Example 3.2.5

Consider path that go north by length 1 or go northeast with length $\sqrt{2}$. How many ways can you go from $(0, 0)$ to (n, k) , $n, k \in \mathbb{N}$?

$$W_{k,n} = \text{number of such path from } (0, 0) \text{ to } (k, n)$$

$$W_{k,n} = W_{k,n-1} + W_{k-1,n-1} \quad W_{0,n} = 1, W_{k,0} = 0 (k > 0)$$

Example 3.2.6

Let $A_{k,n}$ be the number (equivalently, the coefficient) of the term $x^k y^{n-k}$ in the expansion of $(x+y)^n$.

Claim. $A_{k,n} = \binom{n}{k}$ for $0 \leq k \leq n$ (and $A_{k,n} = 0$ otherwise).

Reason 1 (Combinatorial). In $(x+y)^n$, to get $x^k y^{n-k}$ you must choose exactly k of the n factors to contribute an x (the others contribute y). There are $\binom{n}{k}$ such choices.

Reason 2 (Pascal recurrence). Define $A_{k,n}$ from $(x+y)^1 = x+y$: $A_{0,1} = 1$, $A_{1,1} = 1$. Multiply by $(x+y)$ to go from n to $n+1$:

$$(x+y)^{n+1} = (x+y)(x+y)^n.$$

The coefficient of $x^k y^{(n+1)-k}$ in $(x+y)^{n+1}$ comes from: - taking x and the term $x^{k-1} y^{n-(k-1)}$ from $(x+y)^n$ (count $A_{k-1,n}$ ways), - taking y and the term $x^k y^{n-k}$ from $(x+y)^n$ (count $A_{k,n}$ ways).

Thus

$$A_{k,n+1} = A_{k-1,n} + A_{k,n} \quad \text{for } 1 \leq k \leq n,$$

with edge conditions

$$A_{0,n} = 1, \quad A_{n,n} = 1.$$

This is Pascal's triangle, so $A_{k,n} = \binom{n}{k}$.

Conclusion. The number of terms of the form $x^k y^{n-k}$ in $(x+y)^n$ (i.e., its coefficient) is

$$A_{k,n} = \binom{n}{k}.$$

Example 3.2.7

How many subsets of size k does $\{1, 2, \dots, n\}$ have?

Solution. A subset of size k is obtained by choosing exactly k distinct elements from the n -element set. The number of ways to do this is the binomial coefficient

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Reasoning. Each subset of size k corresponds to a k -element combination out of n . Order does not matter, so we divide out the $k!$ permutations of the chosen elements.

Conclusion. The set $\{1, 2, \dots, n\}$ has

$$\binom{n}{k}$$

distinct subsets of size k .

Theorem 3.2.3

If $S_{k,n}$ is the k th entry in the $(n+1)$ st rows of pascal's triangle then $S_{k,n} = \frac{n!}{k!(n-k)!}$. We also write $S_{k,n} = \binom{n}{k}$, and we say "n choose k".

Proof: By induction,

$P(n)$ = "For all $k \in \{0, \dots, n\}$ the k th entry in the $(n+1)$ st row of Pascals triangle equals:

$$\frac{n!}{k!(n-k)!}$$

⊕

Note:-

(P1)-(P12) are satisfied by both $\mathbb{Q}$ and $\mathbb{R}$

Lemma 3.2.1

1. $n \in \mathbb{N} \Rightarrow n$ is even or odd
2. n even $\Rightarrow n^2$ even
3. n odd $\Rightarrow n^2$ odd
4. n^2 even $\Rightarrow n$ even
5. n^2 odd $\Rightarrow n$ odd

Claim 3.2.2

$\sqrt{2}$ is not rational.

The proof is in notes for baby rudin.

Chapter 4

Week 4

Note:-

Additional practice on the supremum axiom, the Archimedean property, and density of rationals can be found in Chapter 23, Section 23.1.

4.1 Upper and lower bounds

Note:-

This section is covered in baby Rudin notes. The following are course notes

Idea: Given a set that has an ordering, we should be able to talk about upper bounds, lower bounds, least upper bounds, greatest lower bounds.

Example 4.1.1

$$A = \{2n : n \in \mathbb{N}\} \subseteq \mathbb{Z} \quad \mathbb{Z} \text{ has ordering}$$

- upperbounds:
- lowerbounds: 2,-5,

Example 4.1.2 (special case)

$$A = \emptyset \subset \mathbb{Q}$$

- GLB LUB doesn't exist
- upperbound & lowerbound: $\mathbb{Q}$

Definition 4.1.1: LUB/GLB/bounded above/bounded below

- **Bounded above:** A set $A \subseteq \mathbb{R}$ is *bounded above* if there exists $M \in \mathbb{R}$ such that $a \leq M$ for all $a \in A$. Such an M is called an *upper bound* of A .
- **Bounded below:** A set $A \subseteq \mathbb{R}$ is *bounded below* if there exists $m \in \mathbb{R}$ such that $m \leq a$ for all $a \in A$. Such an m is called a *lower bound* of A .
- **Least upper bound (LUB) / Supremum:** A number $s \in \mathbb{R}$ is called the *least upper bound* (or *supremum*) of A if
 - s is an upper bound of A , and
 - if M is any upper bound of A , then $s \leq M$.

We write $s = \sup A$.

- **Greatest lower bound (GLB) / Infimum:** A number $t \in \mathbb{R}$ is called the *greatest lower bound* (or *infimum*) of A if
 - t is a lower bound of A , and
 - if m is any lower bound of A , then $t \geq m$.

We write $t = \inf A$.

- **Maximum:** A number $M \in A$ is called the *maximum* of A if M is an upper bound of A . Equivalently, $M = \sup A$ and $M \in A$. We write $M = \max A$.
- **Minimum:** A number $m \in A$ is called the *minimum* of A if m is a lower bound of A . Equivalently, $m = \inf A$ and $m \in A$. We write $m = \min A$.

Theorem 4.1.1

If A has a least upper bound, then it's unique.

Proof: Assume x and y are both LUB.

- $x \leq y$ because y is an upper bound and x is a least upper bound.
- $y \leq x$ because x is an upper bound and y is a least upper bound.

Therefore $x \leq y$ and $y \leq x$, so $x = y$.

Note:-

For quiz, do this with properties (P10-12)



Definition 4.1.2: (P13) Least upper bound property

For $A \subseteq \mathbb{R}$ and A is **not empty**. If A is bounded above then A has least upper bound. $\mathbb{R}$ with usual ordering satisfies (P13).

Definition 4.1.3

For a ordered field, a is positive if $a \in P$. a is negative if $-a \in P$. Else, a is 0.

Theorem 4.1.2 Archimedean Property (natural numbers are unbounded)

The natural numbers $\mathbb{N}$ are not bounded above in $\mathbb{R}$.

Proof: Suppose, for contradiction, that $\mathbb{N}$ is bounded above in $\mathbb{R}$. By completeness of $\mathbb{R}$,

the set $\mathbb{N}$ has a least upper bound. Let

$$s = \sup \mathbb{N}.$$

- Since s is the least upper bound, the number $s - 1$ cannot be an upper bound of $\mathbb{N}$.
- Therefore, there must exist some $m \in \mathbb{N}$ with $m > s - 1$.
- But then $m + 1 \in \mathbb{N}$ and we have

$$m + 1 > s.$$

This contradicts the fact that s is an upper bound of $\mathbb{N}$, since we found a natural number larger than s .

Hence our assumption was false, and $\mathbb{N}$ has no upper bound in $\mathbb{R}$. 

Theorem 4.1.3 Archimedean property (in ϵ form)

For every real number $\epsilon > 0$ exists natural $N \in \mathbb{N}$ such that

$$\frac{1}{N} < \epsilon$$

Proof: Suppose, for contradiction, that there exists $\epsilon > 0$ such that $\forall N \in \mathbb{N}$

$$\frac{1}{N} \geq \epsilon$$

This implies

$$N \leq \frac{1}{\epsilon}, \quad \forall N \in \mathbb{N}$$



so the natural numbers are bounded above by $1/\epsilon$. Since $\mathbb{N}$ is not bounded above $\mathbb{R}$, proven above, this is a contradiction. Hence, for every $\epsilon > 0$, such an N exists.

Corollary 4.1.1 Archimedean corollary

If $x, y \in \mathbb{R}$ and $y - x > 1$, then there exists some integer $n \in \mathbb{Z}$ such that

$$x < n < y$$

Proof: Let

$$A = \{n \in \mathbb{Z} : n \leq x\} \subseteq \mathbb{R}$$

it has an upper bound x . By least upper bound theory, $\sup(A)$ exists.

$$\sup(A) = \lfloor x \rfloor \Rightarrow \sup(A) \leq x < \sup(A) + 1$$

$$\sup(A) + 1 \leq x + 1$$

Since $y - x > 1, y > x + 1, \sup(A) + 1 < y$. Thus

$$x < \sup(A) + 1 < y$$

and because $\sup(A) = \lfloor x \rfloor, \sup(A) + 1 \in \mathbb{N}$. 

Chapter 5

Week 5

Note:-

Additional practice on functions, sets, and inverses can be found in Chapter 23, Section 23.2.

5.1 Odd and even functions

Definition 5.1.1

A function is a collection of ordered pairs such that if (a, b) and (a, c) are both in the collection, then $b = c$.

Definition 5.1.2

If the domain of f is $\mathbb{R}$, then f is even if $f(-x) = f(x), \forall x \in \mathbb{R}$. f is odd if $f(-x) = -f(x), \forall x \in \mathbb{R}$.

f	g	$f + g$
even	even	even
even	odd	neither (in general)
odd	even	neither (in general)
odd	odd	odd

Claim 5.1.1

If f and g are both even, then $f + g$ is even.

Proof: From definition of even, we want to show $(f + g)(-x) = (f + g)(x)$ for all $x \in \mathbb{R}$. From the definition of the sum of functions,

$$\begin{aligned}(f + g)(-x) &= f(-x) + g(-x), \quad \forall x \in \mathbb{R} \\ &= f(x) + g(x) \\ &= (f + g)(x)\end{aligned}$$



Definition 5.1.3: Maximum/Minimum of two functions

$$\max(f, g)(x) = \frac{f(x) + g(x)}{2} + \frac{|f(x) - g(x)|}{2}$$

The domain of $\max(f, g)$ is $\text{domain } f \cap \text{domain } g$.

$$\min(f, g)(x) = \frac{f(x) + g(x)}{2} - \frac{|f(x) - g(x)|}{2}$$

Definition 5.1.4: Möbius transformations

Given $a, b, c, d \in \mathbb{R}$,

$$f(x) = \frac{ax + b}{cx + d}$$

- **domain:** $\mathbb{R} - \{-\frac{d}{c}\}$
- **range:** $\mathbb{R} - \{\frac{a}{c}\}$

Claim 5.1.2 Injectivity of Möbius transformations

Let

$$f(x) = \frac{ax + b}{cx + d}, \quad c \neq 0.$$

Then f is injective if and only if $ad - bc \neq 0$.

Proof: Let $x_1, x_2 \in \mathbb{R}$ with $x_1, x_2 \neq -\frac{d}{c}$. Suppose $f(x_1) = f(x_2)$. Then

$$\frac{ax_1 + b}{cx_1 + d} = \frac{ax_2 + b}{cx_2 + d}.$$

Cross-multiplying gives

$$(ax_1 + b)(cx_2 + d) = (ax_2 + b)(cx_1 + d).$$

Expanding both sides:

$$acx_1x_2 + adx_1 + bcx_2 + bd = acx_1x_2 + adx_2 + bcx_1 + bd.$$

Canceling terms yields

$$adx_1 + bcx_2 = adx_2 + bcx_1,$$

or equivalently,

$$(ad - bc)(x_1 - x_2) = 0.$$

- If $ad - bc \neq 0$, then $(x_1 - x_2) = 0$, so $x_1 = x_2$. Hence f is injective.
- If $ad - bc = 0$, then $ad = bc$, and we can simplify:

$$ax + b = ax + \frac{ad}{c} = \frac{a}{c}(cx + d).$$

Thus

$$f(x) = \frac{\frac{a}{c}(cx + d)}{cx + d} = \frac{a}{c},$$

for all $x \neq -\frac{d}{c}$. Hence f is a constant function, which is not injective.

Therefore, f is injective if and only if $ad - bc \neq 0$.



Theorem 5.1.1 Composition of Möbius transformations

Let

$$f(x) = \frac{a_1x + b_1}{c_1x + d_1}, \quad g(x) = \frac{a_2x + b_2}{c_2x + d_2},$$

with $a_i, b_i, c_i, d_i \in \mathbb{R}$ (or $\mathbb{C}$) and $a_id_i - b_ic_i \neq 0$. Then $f \circ g$ is also a Möbius transformation, and its matrix is given by the product of the matrices of f and g .

Proof: We compute

$$(f \circ g)(x) = f(g(x)) = \frac{a_1 \cdot g(x) + b_1}{c_1 \cdot g(x) + d_1}.$$

Substitute $g(x) = \frac{a_2x + b_2}{c_2x + d_2}$:

$$(f \circ g)(x) = \frac{a_1 \cdot \frac{a_2x + b_2}{c_2x + d_2} + b_1}{c_1 \cdot \frac{a_2x + b_2}{c_2x + d_2} + d_1}.$$

Multiply numerator and denominator by $(c_2x + d_2)$:

$$(f \circ g)(x) = \frac{a_1(a_2x + b_2) + b_1(c_2x + d_2)}{c_1(a_2x + b_2) + d_1(c_2x + d_2)}.$$

Expanding:

$$(f \circ g)(x) = \frac{(a_1a_2 + b_1c_2)x + (a_1b_2 + b_1d_2)}{(c_1a_2 + d_1c_2)x + (c_1b_2 + d_1d_2)}.$$

This is again a Möbius transformation.

Now compare with matrix multiplication:

$$M_f = \begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}, \quad M_g = \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix}.$$

Then

$$M_f M_g = \begin{pmatrix} a_1a_2 + b_1c_2 & a_1b_2 + b_1d_2 \\ c_1a_2 + d_1c_2 & c_1b_2 + d_1d_2 \end{pmatrix},$$

whose entries match exactly the coefficients of $(f \circ g)(x)$. Thus composition corresponds to matrix multiplication.

Example. Let

$$f(x) = \frac{2x + 3}{x + 1}, \quad g(x) = \frac{x + 4}{3x + 2}.$$

Their matrices are

$$M_f = \begin{pmatrix} 2 & 3 \\ 1 & 1 \end{pmatrix}, \quad M_g = \begin{pmatrix} 1 & 4 \\ 3 & 2 \end{pmatrix}.$$

Then

$$M_f M_g = \begin{pmatrix} 11 & 14 \\ 4 & 6 \end{pmatrix}.$$

Hence

$$(f \circ g)(x) = \frac{11x + 14}{4x + 6}.$$



Question 1

When can f be its own inverse? Let

$$f(x) = \frac{ax + b}{cx + d}, \quad ad - bc \neq 0.$$

Then $f(f(x)) = x$ if and only if either $a + d = 0$ (trace zero case), or $f(x) = \pm x$.

Proof: We compute

$$f(f(x)) = f\left(\frac{ax + b}{cx + d}\right) = \frac{a \cdot \frac{ax+b}{cx+d} + b}{c \cdot \frac{ax+b}{cx+d} + d}.$$

Multiplying numerator and denominator by $(cx + d)$ gives

$$f(f(x)) = \frac{a(ax + b) + b(cx + d)}{c(ax + b) + d(cx + d)}.$$

Expanding,

$$f(f(x)) = \frac{(a^2 + bc)x + (ab + bd)}{(ac + cd)x + (bc + d^2)}.$$

For $f(f(x)) = x$, we require

$$(a^2 + bc)x + (ab + bd) = (ac + cd)x^2 + (bc + d^2)x, \quad \forall x.$$

Comparing coefficients:

- Coefficient of x^2 : $ac + cd = c(a + d) = 0$.
- Coefficient of x : $a^2 + bc = bc + d^2$.
- Constant term: $ab + bd = b(a + d) = 0$.

There are two cases.

Case 1. $a + d = 0$. Then $d = -a$, and the conditions are automatically satisfied. Thus every Möbius transformation of the form

$$f(x) = \frac{ax + b}{cx - a}, \quad a^2 + bc \neq 0,$$

is an involution.

Case 2. $c = 0$ and $b = 0$. Then

$$f(x) = \frac{ax}{d}.$$

Here

$$f(f(x)) = \frac{a}{d} \cdot \frac{ax}{d} = \frac{a^2}{d^2}x.$$

For this to equal x , we need $\frac{a^2}{d^2} = 1$, i.e. $a = \pm d$. Thus

$$f(x) = x \quad (\text{identity}), \quad f(x) = -x \quad (\text{reflection}).$$

Therefore, $f(f(x)) = x$ if and only if $a + d = 0$, or $f(x) = \pm x$.



5.2 Triangle inequality

Theorem 5.2.1 Triangle inequality

For all real numbers $x, y \in \mathbb{R}$,

$$|x + y| \leq |x| + |y|.$$

Proof. If $x = 0$ or $y = 0$, the result is immediate. Assume instead that $x, y \neq 0$. Recall that for any real t ,

$$|t| = \sqrt{t^2}.$$

Then,

$$|x + y|^2 = (x + y)^2 = x^2 + 2xy + y^2.$$

Since $2xy \leq 2|x||y|$, we obtain

$$|x + y|^2 \leq x^2 + 2|x||y| + y^2 = (|x| + |y|)^2.$$

Taking square roots of both sides (noting both sides are nonnegative),

$$|x + y| \leq |x| + |y|.$$



Theorem 5.2.2 Extended Triangle Inequality

For all real numbers $a, b, x \in \mathbb{R}$,

$$|a - b| \leq |a - x| + |b - x|.$$

Proof: Observe that

$$a - b = (a - x) + (x - b).$$

Taking absolute values and applying the Triangle Inequality to the right-hand side gives

$$|a - b| = |(a - x) + (x - b)| \leq |a - x| + |x - b|.$$

Since $|x - b| = |b - x|$, we obtain

$$|a - b| \leq |a - x| + |b - x|.$$



5.3 Sequences and limits

Example 5.3.1 (Inverse of permutations)

Let

$$(j, \sigma(j)) = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 1 & 4 \end{pmatrix}$$

then, the inverse would just be swapping the two rows:

$$(k, \sigma^{-1}(k)) = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}$$

Definition 5.3.1: Infinite sequence

An **infinite sequence** of real numbers is a function whose domain is $\mathbb{N}$.

Definition 5.3.2: Bound of sequence

A sequence $(a_n)_{n \in \mathbb{N}}$ is **bounded above** if there is a $M \in \mathbb{R}$ so that $a_n \leq M$ for all $n \in \mathbb{N}$. If **bounded below**, $a_n \geq M$ for all $n \in \mathbb{N}$.

Definition 5.3.3: Increasing decreasing sequences

- **Increasing** if $a_{n+1} > a_n, \forall n \in \mathbb{N}$
- **Nonincreasing** if $a_{n+1} \leq a_n, \forall n \in \mathbb{N}$
- **Nondecreasing** if $a_{n+1} \leq a_n, \forall n \in \mathbb{N}$
- **decreasing** if $a_{n+1} < a_n, \forall n \in \mathbb{N}$

If (a_n) is increasing, it is non decreasing. Not all nondecreasing sequences are increasing.

Example 5.3.2

- Decreasing

$$(c + \frac{1}{n})_{n \in \mathbb{N}}$$

- Increasing

$$(c - \frac{1}{n})_{n \in \mathbb{N}}$$

- None

$$(c + \frac{(-1)^n}{n})_{n \in \mathbb{N}}$$

- No order, i.e. $1, 1, 1, 2, \dots$

$$(\lceil \frac{n}{\pi} \rceil)_{n \in \mathbb{N}}$$

Definition 5.3.4: Limit

A sequence $(a_n)_{n \in \mathbb{N}}$ converges to $l \in \mathbb{R}$. If for every $\epsilon > 0$ there is an $N \in \mathbb{N}$ (that can depend on ϵ) so that if $n \geq N$ then $|a_n - l| < \epsilon$. We write $\lim_{n \rightarrow \infty} a_n = l$. Informally we write $a_n \rightarrow l$.

Example 5.3.3

- $a_n = \pi + \frac{1}{n^3}$
- $a_n = \pi - \frac{1}{n^3}$

- $a_n = \pi + \frac{(-1)^3}{n^3}$

The intuition is that all of these converges to π as $n \rightarrow \infty$.

$$|a_n - l| = |(\pi + \frac{1}{n^3}) - \pi| = |\frac{1}{n^3}| = \frac{1}{n^3}$$

We want $|a_n - l| < 10^{-3}$ then $n > \frac{1}{10}$ will work. Because

$$n > 10 \Rightarrow n^3 > 10^3 \Rightarrow \frac{1}{n^3} < 10^{-3}$$

Claim 5.3.1

$a + \frac{b}{n^3}$ converges to a .

Proof:

$$|(a + \frac{b}{n^3} - a)| = |\frac{b}{n^3}| = \frac{|b|}{n^3}$$

Let $\epsilon > 0$ take $N \in \mathbb{N}, N \geq (\frac{|b|}{\epsilon})^{-3}$ if $n > N$ then $n > (\frac{|b|}{\epsilon})^{-3}$ so $n^3 > \frac{|b|}{\epsilon}$ and so $\frac{|b|}{n^3} < \epsilon$ and therefore $|a_n - l| < \epsilon$. ☺

Theorem 5.3.1

If $\lim_{n \rightarrow \infty} a_n = l$ and $\lim_{n \rightarrow \infty} b_n = \bar{l}$ then $(a_n + b_n)_{n \in \mathbb{N}}$ converges and $\lim_{n \rightarrow \infty} (a_n + b_n) = l + \bar{l}$

Proof: Given $\epsilon > 0$ I want to find N so that $n > N$ then $|(a_n + b_n) - (l + \bar{l})| < \epsilon$. ☺

Theorem 5.3.2

If $\lim_{n \rightarrow \infty} a_n = l$ then $(ca_n)_{n \in \mathbb{N}}$ converges when $c \in \mathbb{R}$ and $\lim_{n \rightarrow \infty} ca_n = cl$

Theorem 5.3.3

If $\lim_{n \rightarrow \infty} a_n = l$ then $(a_n^2)_{n \in \mathbb{N}}$ converges and $\lim_{n \rightarrow \infty} a_n^2 = l^2$

Proof: Given $\epsilon > 0$, we want to find N s.t. $n > N$ implies $|a_n^2 - l^2| < \epsilon$

Note:-

I know that $a_n \rightarrow l$. Given $\epsilon > 0$ I can find N_1 s.t. $n > N_1 \Rightarrow |a_n - l| < \epsilon$. I want info about (experience of desperation lol)

$$|a_n^2 - l^2| = |(a_n - l)(a_n + l)| = |a_n - l||a_n + l| \leq |a_n - l|(|a_n| + |l|)$$

We can have $|a_n - l|$ term as small as we want. If we know $(|a_n| + |l|) < 10^6$ for example, we can make $|a_n - l| < \frac{\epsilon}{10^6}$.

By theorem on convergent sequences, there is $B \in \mathbb{R}$ so that $|a_n| \leq B$ for all n . Because $a_n \rightarrow l$, there is a $N \in \mathbb{N}$ so that $n > N \Rightarrow |a_n - l| < \frac{\epsilon}{B+|l|} \Rightarrow |a_n^2 - l^2| = |a_n - l||a_n + l| \leq |a_n - l|(|a_n| + |l|) < |a_n - l|(B + |l|) < \epsilon$ ☺

Definition 5.3.5: Convergence of sequence

A sequence $(a_n)_{n \in \mathbb{N}}$ converges to $l \in \mathbb{R}$ if for every $\epsilon > 0$ there is a $N \in \mathbb{N}$ such that $n > N \Rightarrow |a_n - l| < \epsilon$.

Definition 5.3.6: Bounded sequence

$(a_n)_{n \in \mathbb{N}}$ is a **bounded sequence** if there is a $B \in \mathbb{R}$ such that $|a_n| \leq B, \forall n \in \mathbb{N}$.

(a_n) is **bounded above** if there is a $B \in \mathbb{R}$ such that $a_n \leq B, \forall n \in \mathbb{N}$.

(a_n) is bounded $\Rightarrow$ is bounded above and bounded below

Theorem 5.3.4

Every bounded sequences has at least one convergent subsequence.

Theorem 5.3.5

If $a(n)_{n \in \mathbb{N}}$ is convergent then it is bounded.

Proof: If $(a_n)_{n \in \mathbb{N}}$ is convergent and has a limit, call it l . Take $\epsilon = 2$. There is $N \in \mathbb{N}$ such that $n > N$ then $|a_n - l| < 2$. $|a_n| = |a_n - l + l| < |a_n - l| + |l|$ by triangle inequality, and so if $n > N$, then $|a_n| < 2 + |l|$. Let $B = \max\{|a_1|, |a_2|, \dots, |a_N|, 2 + |l|\}$. Maximum exists because it is a finite set. Is (a_n) bounded by B ? I need $|a_n| \leq B, \forall n \in \mathbb{N}$,

$|a_1| \leq B$ because $|a_1| \leq B$ because $\max\{|a_1|, |a_2|, \dots, |a_N|, 2 + |l|\}$

$|a_2| \leq B$ because $|a_2| \leq \max\{|a_1|, |a_2|, \dots, |a_N|, 2 + |l|\}$ if $i \in \{1, \dots, N\}$ then $|a_i| \leq B$ because $|a_i| \leq \max\{|a_1|, |a_2|, \dots, |a_N|, 2 + |l|\}$. If $n > N$ then $|a_n| < 2 + |l| \leq B$ because $2 + |l| \leq \max\{|a_1|, |a_2|, \dots, |a_N|, 2 + |l|\}$ ☺

Theorem 5.3.6

If $(a_n)_{n \in \mathbb{N}}$ is nondecreasing and bounded above, then (a_n) is convergent and converges to its least upper bound $\sup(a_n)$

Proof: Let $\epsilon > 0$, let $a^* = \sup(a_n)$. Since (a_n) is not empty and bounded above, $a^* \in \mathbb{R}$ exists. We know that $a^* - \epsilon$ is not an upper bound because $a^* - \epsilon < a^*$ and a^* is the least upper bound. So there is some N with $a_N \geq a^* - \epsilon$. If $n > N$, then $a_n \geq a_N$ because sequence is nondecreasing, and so $n > N \Rightarrow a_n > a^* - \epsilon$. $a_n \leq a^* \Rightarrow |a_n - a^*| < \epsilon$ ☺

Example 5.3.4 (Heron's algorithms)

This is an algorithm for computing the square root of a positive number $a > 0$. We want to find x such that

$$x^2 = a$$

Start with a rough guess x_1 and improve it iteratively using

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{a}{x_n} \right)$$

We can have the following conjectures:

- $x_2 \geq \sqrt{s}$, $x_n \geq \sqrt{s}$ if $n \geq 2$
- x_n decreases to $\sqrt{s}$ for $n \geq 2$

Claim 5.3.2

If $s > 0$ and $x_1 > 0$, and we define the sequence recursively by

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{s}{x_n} \right),$$

then $x_n > 0$ for all $n \geq 1$.

Proof: We prove this by mathematical induction on n .

Base case: For $n = 1$, we are given $x_1 > 0$ by assumption.

Inductive step: Assume $x_n > 0$ for some $n \geq 1$. Then

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{s}{x_n} \right).$$

Since both $s > 0$ and $x_n > 0$, we have $\frac{s}{x_n} > 0$. Hence the sum $x_n + \frac{s}{x_n} > 0$, and multiplying by $\frac{1}{2}$ preserves positivity. Therefore $x_{n+1} > 0$.

Conclusion: By the principle of mathematical induction, $x_n > 0$ for all $n \geq 1$. ☺

Claim 5.3.3

If $x_1 \geq 0$ then $x_n^2 \geq s, \forall n \geq 2$

Proof:

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{s}{x_n} \right)$$

Is $x_{n+1} - s \geq 0$?

$$\left[\frac{1}{2} \left(x_n + \frac{s}{x_n} \right) \right]^2 - s = \frac{1}{4} \left(x_n - \frac{s}{x_n} \right)^2$$

so $x_{n+1}^2 - s \geq 0, \forall x_n \neq 0$ ☺

Claim 5.3.4

If $x_1 > 0$ then $x_{n+1} \leq x_n, \forall n \geq 2$.

Proof:

$$x_{n+1} - x_n = \frac{1}{2} \left(\frac{s - x_n^2}{x_n} \right)$$

for $x_n \neq 0$. If $x_1 > 0$ then $x_n > 0$ for all $n \geq 1$. If $x_1 \neq 0$ then $s - x_n^2 \neq 0$ for all $n \geq 2$. $x_{n+1} - x_n \leq 0, \forall n \geq 2$. $(x_n)_{n=2}^\infty$ is nonincreasing ☺

Claim 5.3.5

If $s > 0$, $x_1 > 0$, and $x_1^2 > s$, and we define

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{s}{x_n} \right),$$

then $(x_n)_{n \geq 1}$ is a nonincreasing sequence; i.e. $x_{n+1} \leq x_n$ for all $n \geq 1$.

Proof: We first compute the difference between successive terms:

$$x_{n+1} - x_n = \frac{1}{2} \left(x_n + \frac{s}{x_n} \right) - x_n = \frac{1}{2} \left(\frac{s - x_n^2}{x_n} \right),$$

which is valid for all $x_n \neq 0$.

Since $x_1 > 0$, the previous claim shows that $x_n > 0$ for all $n \geq 1$. Hence the sign of $x_{n+1} - x_n$ is determined by the numerator $s - x_n^2$.

Now, assume $x_1^2 > s$. We will show by induction that $x_n^2 > s$ for all $n \geq 1$. Indeed, if $x_n^2 > s$, then $\frac{s}{x_n} < x_n$, so

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{s}{x_n} \right) < \frac{1}{2} (x_n + x_n) = x_n,$$

and thus $x_{n+1}^2 > s$ still holds because both sides remain positive.

Therefore $s - x_n^2 < 0$ for all $n \geq 1$, which implies

$$x_{n+1} - x_n = \frac{1}{2} \left(\frac{s - x_n^2}{x_n} \right) < 0.$$

Hence (x_n) is strictly decreasing and thus nonincreasing.

Note:-

$$x_{n+1}^2 - s = \frac{1}{4} \left(x_n - \frac{s}{x_n} \right)^2 \text{ for any } n \geq 1.$$

**Claim 5.3.6**

If $x_n \rightarrow x^*$, then $x_{n+1} \rightarrow x^*$. $x^* \neq 0$, proof use subsequence

Proof: If $x_n \rightarrow x^*$ then $\frac{1}{2} \left(x_n + \frac{s}{x_n} \right) \rightarrow \frac{1}{2} \left(x^* + \frac{s}{x^*} \right)$.



Claim 5.3.7

If $x_n \rightarrow x^*$ and $x_n \rightarrow \hat{x}$ then $x^* = \hat{x}$

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{s}{x_n}\right)$$

$LHS \rightarrow x^*$, $RHS \rightarrow \frac{1}{2}(x^* + \frac{s}{x^*})$. $LHS = RHS$ for some n . Some sequence decreased differently. $x^* = \frac{1}{2}(x^* + \frac{s}{x^*})$ by 3rd claim.

$$2x^* = x^* + \frac{s}{x^*}$$

$$x^* = \frac{s}{x^*}$$

$$(x^*)^2 = s, x^* = \pm\sqrt{s}$$

Claim 5.3.8

If $a_m \rightarrow a^*$ and $b_m \rightarrow b^*$ then $(a_m + b_m) \rightarrow a^* + b^*$

Proof: Given $\epsilon > 0$ I want N such that

$$n > N \Rightarrow |(a_n + b_n) - (a^* + b^*)| < \epsilon$$

$$|(a_n + b_n) - (a^* + b^*)| \leq |a_n - a^*| + |b_n - b^*|$$

find N_1 such that $\dots < \frac{\epsilon}{2}$. Find N_2 such that $\dots < \frac{\epsilon}{2}$. Take $N = \dots$

**Claim 5.3.9**

$a_n \rightarrow a^*, b_n \rightarrow b^*$ then $a_n b_n \rightarrow a^* b^*$

Proof: Sketch: Given $\epsilon > 0$. I want N such that $n > N \Rightarrow |a_n b_n - a^* b^*| < \epsilon$.

$$|a_n b_n - a^* b_n + a^* b_n - a^* b^*| \leq |a_n - a^*| |b_n| + |a^*| |b_n - b^*|$$

b_n convergent $\Rightarrow |b_n|$ bounded. ...



Chapter 6

Week 6

Note:-

Additional practice on sequences and series can be found in Chapter 23, Section 23.3.

Theorem 6.0.1 Squeeze theorem

Let $(a_n), (b_n), (c_n)$ be real sequences. Suppose there exists an integer N such that for all $n \geq N$, $a_n \leq b_n \leq c_n$. If

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L$$

then

$$\lim_{n \rightarrow \infty} b_n = L$$

Proof: Let $\epsilon > 0$, since $\lim_{n \rightarrow \infty} a_n = L$, there exists $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$, $|a_n - L| < \epsilon$. That is,

$$L - \epsilon < a_n < L + \epsilon$$

Similarly, since $\lim_{n \rightarrow \infty} c_n = L$, there exists $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$, $|c_n - L| < \epsilon$, so

$$L - \epsilon < c_n < L + \epsilon$$

Now let $N = \max\{N_1, N_2, N\}$. Then for all $n \geq N$,

$$L - \epsilon < a_n \leq b_n \leq c_n < L + \epsilon$$

Thus, $|b_n - L| < \epsilon$ Since $\epsilon > 0$ was arbitrary,

$$\lim_{n \rightarrow \infty} b_n = L$$



6.1 Limsup and Liminf

Theorem 6.1.1

For $p \in \mathbb{R}$,

- if $p > 0$, then $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$
- if $p > 0$, then $\lim_{n \rightarrow \infty} p^{\frac{1}{n}} = 1$
- $\lim_{n \rightarrow \infty} n^{\frac{1}{n}} = 1$

Proof: • Let $\epsilon > 0$ choose $M \in \mathbb{N}$ such that

$$M > \left(\frac{1}{\epsilon}\right)^{\frac{1}{p}}$$

Then if $n \geq M$,

$$\left|\frac{1}{n^p} - 0\right| = \frac{1}{n^p} \leq \frac{1}{M^p} < \epsilon$$



6.2 Bolzano-Weierstrass Theorem

Here's a proof from Analysis I (Terence Tao). The proof uses previous propositions and lemmas that are listed below.

Lemma 6.2.1 Comparison principle

Suppose that $(a_n)_{n=m}^{\infty}$ and $(b_n)_{n=m}^{\infty}$ are two sequences of real numbers such that $a_n \leq b_n$ for all $n \geq m$. Then we have the inequalities

- $\sup(a_n)_{n=m}^{\infty} \leq \sup(b_n)_{n=m}^{\infty}$
- $\inf(a_n)_{n=m}^{\infty} \leq \inf(b_n)_{n=m}^{\infty}$
- $\limsup_{n \rightarrow \infty} a_n \leq \limsup_{n \rightarrow \infty} b_n$
- $\liminf_{n \rightarrow \infty} a_n \leq \liminf_{n \rightarrow \infty} b_n$

Proposition 6.2.1

Let $(a_n)_{n=m}^{\infty}$ be a sequence of real numbers, let L^+ be the limit superior of this sequence, and let L^- be the limit inferior of the sequence (thus both L^+ and L^- are extended real numbers, which is $\mathbb{R} \cup \{+\infty, -\infty\}$). Let $c \in \mathbb{R}$. $(a_n)_{n=m}^{\infty}$ converges to c if and only if $L^+ = L^- = c$.

Proposition 6.2.2 Subsequences related to limit points

Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers, and let L be a real number. Then the following two statements are logically equivalent.

1. L is a limit point of $(a_n)_{n=0}^{\infty}$.
2. There exists a subsequence of $(a_n)_{n=0}^{\infty}$ which converges to L .

Theorem 6.2.1 Bolzano-Weierstrass

Let $(a_n)_{n=0}^\infty$ be a bounded sequence (i.e., there exists a real number $M > 0$ such that $|a_n| \leq M$ for all $n \in \mathbb{N}$). Then there is at least one subsequence of $(a_n)_{n=0}^\infty$ which converges.

Proof: Let L be the limit superior of the sequence $(a_n)_{n=0}^\infty$. Since we have $-M \leq a_n \leq M$ for all natural numbers n , it follows from the comparison principle 6.2.1 that $-M \leq L \leq M$. In particular, L is a real number (not $+\infty$ or $-\infty$). By proposition 6.2.1, L is thus a limit point of $(a_n)_{n=0}^\infty$. Thus by proposition, there exists a subsequence of $(a_n)_{n=0}^\infty$ which converges to L . $\odot$

Here is the whole proof from scratch (without any lemma and propositions):

Theorem 6.2.2 Bolzano-Weierstrass

Every bounded sequence $(a_n)_{n=1}^\infty$ of real numbers has a convergent subsequence.

Proof: Let (a_n) be bounded; thus there exist $m, M \in \mathbb{R}$ with $m \leq a_n \leq M$ for all n . For each $n \in \mathbb{N}$, define the tail supremum and tail infimum

$$s_n := \sup_{k \geq n} a_k, \quad t_n := \inf_{k \geq n} a_k.$$

By definition of supremum/infimum and the comparison of sets $\{a_k : k \geq n+1\} \subseteq \{a_k : k \geq n\}$, we have the monotonicity

$$s_{n+1} \leq s_n \quad \text{and} \quad t_{n+1} \geq t_n \quad (n \in \mathbb{N}).$$

Moreover, $t_n \leq s_n$ for all n , and $t_n \geq m$, $s_n \leq M$ by the boundedness of (a_n) . Hence (s_n) is a decreasing sequence bounded below and (t_n) is an increasing sequence bounded above. By completeness of $\mathbb{R}$ (e.g., the supremum axiom), both limits exist:

$$\lim_{n \rightarrow \infty} s_n =: L^+ \in \mathbb{R}, \quad \lim_{n \rightarrow \infty} t_n =: L^- \in \mathbb{R},$$

with $L^- \leq L^+$. (These are the usual $\limsup a_n$ and $\liminf a_n$.)

Claim. L^+ is a limit point of the sequence (a_n) . Let $\varepsilon > 0$. Since $s_n \rightarrow L^+$, there exists N with $s_N > L^+ - \varepsilon$. By the definition of $s_N = \sup_{k \geq N} a_k$, there is some $m \geq N$ such that $a_m > L^+ - \varepsilon$. Because the same argument works for every larger N , there are *infinitely many* indices n with $a_n > L^+ - \varepsilon$. This shows that every neighborhood $(L^+ - \varepsilon, L^+ + \varepsilon)$ contains infinitely many terms of the sequence, i.e., L^+ is a limit point.

From the claim we can *construct* a subsequence converging to L^+ : choose n_1 so that $a_{n_1} > L^+ - 1$, and inductively choose $n_{k+1} > n_k$ with $a_{n_{k+1}} > L^+ - \frac{1}{k+1}$, which is possible by the claim with $\varepsilon = \frac{1}{k+1}$. Then for every k we have

$$L^+ - 1/k < a_{n_k} \leq s_{n_k},$$

and since $s_{n_k} \rightarrow L^+$ along the subsequence (as (s_n) is convergent), the squeeze yields $a_{n_k} \rightarrow L^+$: given any $\varepsilon > 0$, take K so large that both $1/K < \varepsilon$ and $|s_{n_K} - L^+| < \varepsilon$; then for all $k \geq K$,

$$L^+ - \varepsilon < a_{n_k} \leq s_{n_k} < L^+ + \varepsilon,$$

so $|a_{n_k} - L^+| < \varepsilon$.

Thus (a_{n_k}) converges to L^+ , providing the desired convergent subsequence of (a_n) . $\odot$

Theorem 6.2.3 Bolzano–Weierstrass (sequence version)

Every bounded sequence of real numbers has a convergent subsequence.

Proof: Let $(a_n)_{n=1}^\infty$ be a bounded sequence of real numbers. That means there exist real numbers m and M such that

$$m \leq a_n \leq M \quad \text{for all } n.$$

Step 1. Constructing upper and lower envelopes. For each $n \in \mathbb{N}$, define

$$s_n := \sup_{k \geq n} a_k, \quad t_n := \inf_{k \geq n} a_k.$$

Here s_n is the largest number that bounds all the terms $a_n, a_{n+1}, a_{n+2}, \dots$ from above, while t_n is the smallest number that bounds them from below.

Step 2. Monotonicity and boundedness. As n increases, the tail $\{a_k : k \geq n\}$ becomes smaller. Hence $s_{n+1} \leq s_n$ (so (s_n) is decreasing) and $t_{n+1} \geq t_n$ (so (t_n) is increasing). Since the original sequence is bounded by m and M , we have

$$m \leq t_n \leq s_n \leq M \quad \text{for all } n.$$

Thus both (s_n) and (t_n) are bounded and monotone.

Step 3. Existence of the limits. By completeness of $\mathbb{R}$ (the supremum axiom), every bounded monotone sequence converges. Therefore, both limits exist:

$$\lim_{n \rightarrow \infty} s_n =: L^+, \quad \lim_{n \rightarrow \infty} t_n =: L^-,$$

and we have $L^- \leq L^+$. These numbers are known respectively as the *limit inferior* and *limit superior* of (a_n) .

Step 4. Showing that L^+ is a limit point of (a_n) . Take any $\varepsilon > 0$. Since $s_n \rightarrow L^+$, there exists an index N such that $s_N > L^+ - \varepsilon$. By the definition of $s_N = \sup_{k \geq N} a_k$, there must be some $m \geq N$ with

$$a_m > L^+ - \varepsilon.$$

This means that no matter how small ε is, we can always find infinitely many terms a_m within ε of L^+ . Hence, L^+ is a *limit point* of the sequence.

Step 5. Constructing a convergent subsequence. We can now build a subsequence (a_{n_k}) that converges to L^+ . Choose the indices inductively as follows:

$$\begin{aligned} n_1 &\text{ such that } a_{n_1} > L^+ - 1, \\ n_2 &> n_1 \text{ such that } a_{n_2} > L^+ - \frac{1}{2}, \\ n_3 &> n_2 \text{ such that } a_{n_3} > L^+ - \frac{1}{3}, \text{ and so on.} \end{aligned}$$

We can always find such n_k because there are infinitely many terms arbitrarily close to L^+ . For this subsequence we have

$$L^+ - \frac{1}{k} < a_{n_k} \leq s_{n_k}.$$

Step 6. Convergence of the subsequence. Since $s_{n_k} \rightarrow L^+$ as $k \rightarrow \infty$, the inequalities above imply

$$L^+ - \frac{1}{k} < a_{n_k} < L^+ + \varepsilon_k$$

for some $\varepsilon_k \rightarrow 0$. Hence, by the squeeze theorem, $a_{n_k} \rightarrow L^+$.

Step 7. Conclusion. We have constructed a subsequence (a_{n_k}) that converges to L^+ . Therefore, every bounded sequence of real numbers has a convergent subsequence. ☺

Theorem 6.2.4 Bolzano–Weierstrass Theorem (Bisection Proof)

Every bounded sequence of real numbers has a convergent subsequence.

Proof: Let $(a_n)_{n=1}^\infty$ be a bounded sequence of real numbers. This means there exist real numbers m and M such that

$$m \leq a_n \leq M \quad \text{for all } n.$$

So all the terms of the sequence lie inside the closed interval $[m, M]$.

Step 1. Start with the whole interval. We begin by dividing $[m, M]$ into two halves:

$$[m, M] = [m, \frac{m+M}{2}] \cup [\frac{m+M}{2}, M].$$

At least one of these two halves must contain infinitely many terms of the sequence (a_n) , since otherwise the sequence would have only finitely many terms overall. Let's call that half *the chosen interval*, and denote it by $I_1 = [m_1, M_1]$.

Step 2. Repeat the process. We now bisect I_1 into two equal halves again:

$$I_1 = [m_1, \frac{m_1+M_1}{2}] \cup [\frac{m_1+M_1}{2}, M_1].$$

As before, at least one of these halves must contain infinitely many terms of (a_n) . Let $I_2 = [m_2, M_2]$ be that half.

Continue this process indefinitely. At each step k , we choose a subinterval $I_k = [m_k, M_k]$ that: 1. Contains infinitely many terms of (a_n) , and 2. Has half the length of the previous one:

$$M_k - m_k = \frac{M - m}{2^{k-1}}.$$

Step 3. Define a nested sequence of intervals. We now have a chain of intervals

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots,$$

each containing infinitely many terms of the sequence, and whose lengths shrink to zero:

$$|I_k| = M_k - m_k \rightarrow 0 \quad \text{as } k \rightarrow \infty.$$

Step 4. The intersection contains a single point. Because the intervals are nested and shrinking, there exists exactly one real number L that lies in all of them:

$$\{L\} = \bigcap_{k=1}^{\infty} I_k.$$

We will show that there is a subsequence of (a_n) that converges to L .

Step 5. Construct the subsequence. Since I_1 contains infinitely many a_n , choose one of them and call it a_{n_1} .

Then, since I_2 also contains infinitely many terms, we can choose a_{n_2} inside I_2 such that $n_2 > n_1$.

Continue in this way: for each k , pick a_{n_k} inside I_k such that $n_k > n_{k-1}$. This gives us a subsequence (a_{n_k}) with

$$a_{n_k} \in I_k = [m_k, M_k].$$

Step 6. Show the subsequence converges to L . Since L is contained in every I_k , and $a_{n_k} \in I_k$, we have

$$m_k \leq a_{n_k} \leq M_k \quad \text{and} \quad m_k \leq L \leq M_k.$$

Thus,

$$|a_{n_k} - L| \leq M_k - m_k = \frac{M - m}{2^{k-1}}.$$

As $k \rightarrow \infty$, the right-hand side goes to 0. Therefore, for every $\varepsilon > 0$, if we choose K large enough so that $\frac{M-m}{2^{K-1}} < \varepsilon$, we get

$$|a_{n_k} - L| < \varepsilon \quad \text{for all } k \geq K.$$

By the ε - N definition of convergence, this shows

$$a_{n_k} \rightarrow L.$$

Step 7. Conclusion. We have found a subsequence (a_{n_k}) of the bounded sequence (a_n) that converges to the real number L . Hence, every bounded sequence of real numbers has a convergent subsequence. ☺

Chapter 7

Week 7

Note:-

Additional practice on continuity and compactness can be found in Chapter 23, Section 23.4.

Definition 7.0.1: Limit

The function f **approaches the line l near a** means: for every $\epsilon > 0$, $\exists \delta > 0$ such that $0 < |x - a| < \delta$ then $|f(x) - l| < \epsilon$

Implicit definition: (by Spivak chapter 5) f is real valued, $l \in \mathbb{R}$ (need $f(x) - l$ to make sense), $a \in \mathbb{R}$ (need $x - a$ to make sense), there is a $\delta > 0$ such that

$$(a - \delta, a) \cup (a, a + \delta) \subseteq \text{domain } f$$

Example 7.0.1

Consider

$$f(x) = \begin{cases} x^2, & x \neq 2, \\ 3, & x = 2. \end{cases}$$

We claim that

$$\lim_{x \rightarrow 2} f(x) = 4.$$

Since the definition of limit does not depend on the value of $f(2)$, we only consider the case $x \neq 2$, where $f(x) = x^2$. Then

$$|f(x) - 4| = |x^2 - 4| = |(x - 2)(x + 2)| = |x - 2| |x + 2|.$$

We will find a bound for $|x + 2|$ when x is close to 2.

Assume $|x - 2| < \frac{1}{10}$. Then

$$2 - \frac{1}{10} < x < 2 + \frac{1}{10} \quad \Rightarrow \quad |x| \leq 2 + \frac{1}{10}.$$

Hence

$$|x + 2| \leq |x| + 2 \leq (2 + \frac{1}{10}) + 2 = 4.1.$$

Claim 7.0.1

$$\lim_{x \rightarrow 2} f(x) = 4.$$

Proof: Let $\epsilon > 0$. Define

$$\delta = \min \left\{ \frac{1}{10}, \frac{\epsilon}{4.1} \right\}.$$

Assume $0 < |x - 2| < \delta$. Then $x \neq 2$, so $f(x) = x^2$, and we have

$$\begin{aligned} |f(x) - 4| &= |x^2 - 4| \\ &= |(x - 2)(x + 2)| \\ &= |x - 2| |x + 2|. \end{aligned}$$

Since $\delta \leq \frac{1}{10}$, we have $|x - 2| < \frac{1}{10}$, and therefore $|x + 2| < 4.1$. Thus

$$|f(x) - 4| \leq |x - 2| |x + 2| < \delta(4.1) \leq \frac{\epsilon}{4.1}(4.1) = \epsilon.$$

Therefore, for all $\epsilon > 0$, there exists $\delta > 0$ such that

$$0 < |x - 2| < \delta \implies |f(x) - 4| < \epsilon.$$

Hence

$$\lim_{x \rightarrow 2} f(x) = 4.$$



Example 7.0.2

$$f(x) = \begin{cases} 1, & x \notin \mathbb{Q} \\ 0, & x \in \mathbb{Q} \end{cases}$$

If $a \in \mathbb{R}$, then does $\lim_{x \rightarrow a} f(x)$ exist? As $x \rightarrow a$, $f(x) \rightarrow 1$? As $x \rightarrow a$, $f(x) \rightarrow 0$? If $l \in \mathbb{R}$, $l \neq 0, 1$ as $x \rightarrow a$, $f(x) \rightarrow l$?

The general idea of proving limit:

- $x \rightarrow a$, $f(x) = l$. If given $\epsilon > 0$, there is a $\delta > 0$ so that $0 < |x - a| < \delta$ that $|f(x) - l| < \epsilon$.
- $x \rightarrow a$ does not give $f(x) \rightarrow l$. This is an $\epsilon_0 > 0$ so that for any $\delta > 0$ there is some x_0 with $0 < |x_0 - a| < \delta$ and $|f(x_0) - l| > \epsilon_0$.

Claim 7.0.2

As $x \rightarrow a$, $f(x)$ doesn't approach to 1.

Proof: Let $\epsilon_0 = \frac{1}{2}$. Let $\delta > 0$. Choose x_0 with $0 < |x_0 - a| < \delta$ so that x_0 is rational. It can be done since rationals are dense in $\mathbb{R}$. Then $|f(x_0) - 1| = |0 - 1| = 1 \geq \epsilon_0$ ☺

7.1 Continuity

Definition 7.1.1: Continuity

f is continuous at a if

- $a \in \text{domain } f$
- $\lim_{x \rightarrow a} f(x)$ exists
- $\lim_{x \rightarrow a} f(x) = f(a)$

Definition 7.1.2: Continuity (Spivak)

f is continuous at a if $\lim_{x \rightarrow a} f(x) = f(a)$

Definition 7.1.3: Continuity (useable)

f is continuous at a if given $\epsilon > 0$ then exist $\delta > 0$ so that $|x - a| < \delta \Rightarrow |f(x) - f(a)| < \epsilon$

Claim 7.1.1

If $P(x)$ is a polynomial and $a \in \mathbb{R}$ then p is continuous at a .

Sketch: Let $\epsilon > 0$. I want to control $|P(x) - P(a)|$. I have control over $|x - a|$. If I could find B so that $|x - a| < \delta \Rightarrow |P(x) - P(a)| < |x - a|B$ I would be done, take $\delta = \frac{\epsilon}{B}$. $P(x) - P(a)$ is a polynomial. It equals 0 at $x = a$.

$$P(x) - P(a) = (x - a)q(x)$$

where $q(x)$ is a polynomial of lower degree.

$$|P(x) - P(a)| = |x - a||q(x)| \leq \text{some } B \text{ using triangle inequality and } |x| \leq |a| + \frac{1}{10}$$

if $|x - a| < \frac{1}{10}$. Let $\delta = \min\{\frac{1}{10}, \frac{\epsilon}{B}\}$

Theorem 7.1.1

f and g are continuous at a , then so is

- $f + g$
- $f \cdot g$
- $\frac{f}{g}, g(a) \neq 0$

Claim 7.1.2

The function $f(x) = \sin x$ is continuous at every real number $a \in \mathbb{R}$.

Proof: We will prove that f is continuous at an arbitrary point $a \in \mathbb{R}$.

By the ϵ - δ definition of continuity, we must show that for every $\epsilon > 0$, there exists a $\delta > 0$ such that

$$0 < |x - a| < \delta \Rightarrow |\sin x - \sin a| < \epsilon.$$

Using the trigonometric identity

$$\sin x - \sin a = 2 \cos\left(\frac{x+a}{2}\right) \sin\left(\frac{x-a}{2}\right),$$

we obtain

$$|\sin x - \sin a| = 2 \left| \cos\left(\frac{x+a}{2}\right) \right| \left| \sin\left(\frac{x-a}{2}\right) \right|.$$

Since $\left| \cos\left(\frac{x+a}{2}\right) \right| \leq 1$ for all real x and a , we have

$$|\sin x - \sin a| \leq 2 \left| \sin\left(\frac{x-a}{2}\right) \right|.$$

By the inequality $|\sin u| \leq |u|$ (valid for all real u),

$$|\sin x - \sin a| \leq 2 \left| \frac{x-a}{2} \right| = |x-a|.$$

Now let $\varepsilon > 0$ be given and choose $\delta = \varepsilon$. Then whenever $0 < |x-a| < \delta$, we have

$$|\sin x - \sin a| \leq |x-a| < \varepsilon.$$

Thus, for every $\varepsilon > 0$, there exists $\delta = \varepsilon > 0$ satisfying the condition above. Therefore, $\sin x$ is continuous at $x = a$.

Hence $\sin x$ is continuous for all $a \in \mathbb{R}$.



Question 1

If I know f is continuous at 1, for example, does this give me information about f at 2?

Theorem 7.1.2

If f is continuous at a and $f(a) < 0$. Then there is a $\delta > 0$ so that $f < 0$ on $(a - \delta, a + \delta)$

Definition 7.1.4: Continuity on interval

- f is continuous on (a, b) if f is continuous at every $c \in (a, b)$.
- f is continuous on $[a, b)$ if f is continuous on (a, b) and $f(a) = \lim_{x \rightarrow a} f(x)$
- f is continuous on $[a, b]$ if f is continuous on (a, b) and $f(a) = \lim_{x \rightarrow a} f(x)$ and $f(b) = \lim_{x \rightarrow b} f(x)$

Theorem 7.1.3 Intermediate Value Theorem

If f is continuous on $[a, b]$ where $f(a) < 0$ and $f(b) > 0$, then there is a $c \in (a, b)$ with

$$f(c) = 0.$$

Sketch $f(a) < 0, f(b) > 0$. Let $a_1 = a, b_1 = b$. Let c be the midpoint $c = \frac{a_1+b_1}{2}$. If $f(c) = 0$, we're done. If $f(c) < 0$, let $a_2 = c$ and $b_2 = b_1$. If $f(c) > 0$ let $a_2 = a_1$ and $b_2 = c$.

$$\begin{aligned} a_1 &\leq a_2 & b_2 &\leq b_1 \\ b_2 - a_2 &= \frac{1}{2}(b_1 - a_1) \end{aligned}$$

Keep on going, if terminate in finitely many steps, I found my point where $f = 0$. Otherwise I found $a_1 \leq a_2 \leq \dots \leq a_n \leq a_{n+1} \leq \dots \leq b$ so $a_n \rightarrow a^*$. Similarly b_n is non increasing set, bounded below. $b_n \rightarrow b^*$ $a^* = b^*$ because of halving. $a_n \rightarrow a^*, f(a_n) < 0$ for all n . f is continuous at $a^*, f(a^*) \leq 0$. $f(b^*) \geq 0$ since $f(b_n) > 0$ for all n . So

$$0 \leq f(b^*) = f(a^*) \leq 0$$

This is very similar to binary search

Theorem 7.1.4 Intermediate value theorem (improved)

Let $a < b$ and $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function on $[a, b]$. Let y be a real number between $f(a)$ and $f(b)$, for example either

$$f(a) \leq y \leq f(b) \text{ or } f(a) \geq y \geq f(b)$$

Then there exists $c \in [a, b]$ such that $f(c) = y$.

Proof: There are two cases: $f(a) \leq y \leq f(b)$ or $f(a) \geq y \geq f(b)$. We assume the former, that $f(a) \leq y \leq f(b)$ and the latter could be proved similarly. If $y = f(a)$ or $y = f(b)$, then the claim is easy, as one could simply put $c = a$ or $c = b$. For the case where $f(a) < y < f(b)$. Let

$$E := \{x \in [a, b] : f(x) < y\}$$

Clearly E is a subset of $[a, b]$, and is hence bounded. Also, since $f(a) < y$,

☺

Theorem 7.1.5 Extreme value theorem

If f is continuous on $[a, b]$, then there exists $y \in [a, b]$ such that $f(y) \geq f(x)$ for all $x \in [a, b]$.

Proof: • **Step 1 (Bounded above).** Suppose, for contradiction, that f is not bounded above on $[a, b]$. Then for each $n \in \mathbb{N}$ there exists $x_n \in [a, b]$ with $f(x_n) > n$. The sequence (x_n) lies in the compact interval $[a, b]$, so it has a convergent subsequence $x_{n_k} \rightarrow y \in [a, b]$ (Bolzano–Weierstrass). By continuity of f at y , we have $f(x_{n_k}) \rightarrow f(y)$ (a finite limit). But $f(x_{n_k}) \geq n_k \rightarrow \infty$, a contradiction. Hence f is bounded above: there exists $B \in \mathbb{R}$ with $f(x) \leq B$ for all $x \in [a, b]$.

• **Step 2 (The least upper bound is attained).** Let $M := \sup\{f(x) : x \in [a, b]\}$, which exists by completeness of $\mathbb{R}$ and Step 1. For each $n \in \mathbb{N}$, by the definition of sup, choose $x_n \in [a, b]$ such that $M - \frac{1}{n} < f(x_n) \leq M$. Again, (x_n) has a convergent subsequence $x_{n_k} \rightarrow y \in [a, b]$. Continuity gives $f(x_{n_k}) \rightarrow f(y)$. On the other hand,

$M - \frac{1}{n_k} < f(x_{n_k}) \leq M$, so taking $k \rightarrow \infty$ yields $f(y) = M$. Thus f attains its maximum at y .

This proves the theorem.



Chapter 8

Week 8

Note:-

Additional practice on continuity, compactness, and inverse functions can be found in Chapter 23, Sections 23.4 and 23.2.

Question 1: HW4

$I, J \subseteq \mathbb{R}$ are two nonempty intervals assume both are open. $f : I \rightarrow J$ is a bijection. $f^{-1}J \rightarrow I$. The goal is to show f is continuous on I iff f^{-1} is continuous on J .

- (4a) Prove that if the bijection f is continuous, then it is strictly increasing or strictly decreasing.

4a)

Claim 8.0.1

If $f : I \rightarrow J$ is a bijection, then f strictly increasing iff f^{-1} strictly increasing; f strictly decreasing iff f^{-1} strictly decreasing.

Proof: $(\Rightarrow)$ Assume f is strictly increasing. We want to show f^{-1} is strictly increasing. That is, show that if $a, b \in J$, $a < b$ then $f^{-1}(a) < f^{-1}(b)$. Assume, for contradiction, $f^{-1}(a) \geq f^{-1}(b)$, $f^{-1}(a) = f^{-1}(b)$ is impossible because f^{-1} is one to one. $f^{-1}(a) > f^{-1}(b)$ would imply $f(f^{-1}(a)) > f(f^{-1}(b))$ because f is strictly increasing and so $a > b$, which is not true. Therefore $f^{-1}(a) < f^{-1}(b)$ as desired.

$(\Leftarrow)$ Assume f^{-1} is strictly increasing. Take $g = f^{-1}$ use $(\Rightarrow)$ argument and $(f^{-1})^{-1} = f$. (Decreasing case TBD) $\odot$

Theorem 8.0.1

If $f : I \rightarrow J$ where I and J are open, non empty intervals of $\mathbb{R}$ and f is a bijection, then f continuous on $I \Rightarrow f^{-1}$ is continuous on J .

Proof: By HW4a), f is strictly increasing or strictly decreasing. First, assume f is strictly increasing. By previous lemma, f^{-1} is strictly increasing. Let $a \in J$.

Then $f^{-1}(a) \in I$. Choose $\epsilon_1 > 0$ so that $(f^{-1}(a) - \epsilon_1, f^{-1}(a) + \epsilon_1) \subseteq I$, because J is open a is not at endpoint, similarly I is open $f^{-1}(a)$ is not at endpoint, $\epsilon_1 = \min\{1, \frac{1}{2} \text{ distance } (f^{-1}(a), \text{endpoints of } I)\}$. Let $\epsilon > 0$, we seek $\delta > 0$ so that $|y - a| < \delta \Rightarrow |f^{-1}(y) - f^{-1}(a)| < \epsilon$. Take $\tilde{\epsilon} = \min(\epsilon, \epsilon_1)$.

$$f^{-1}(a) - \tilde{\epsilon} < f^{-1}(a) < f^{-1}(a) + \tilde{\epsilon}$$

f is strictly increasing.

$$\Rightarrow f(f^{-1}(a) - \tilde{\epsilon}) < f(f^{-1}(a)) = a < f(f^{-1}(a) + \tilde{\epsilon})$$

valid because $\tilde{\epsilon} \leq \epsilon_1$. Take $\delta = \min\{f(f^{-1}(a) + \tilde{\epsilon}) - a, a - f(f^{-1}(a) - \tilde{\epsilon})\}$

☺

8.1 Derivatives

Definition 8.1.1: Derivative def 1

f is differentiable at a if

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

exists. If the limit exists, we call it $f'(a)$ and say " f is differentiable at a with derivative $f'(a)$ ". Here are something implicit in the definition:

- $a \in \text{domain } f$
- $(a-h, a+h) \subset \text{domain } f$ for all $h \leq \text{some } h_0$

Definition 8.1.2: Derivative def 2

f is differentiable at a if

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

exists.

Theorem 8.1.1

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

exists iff

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

exist. If they exist, they the same value.

Theorem 8.1.2

If f is differentiable at a then f is continuous.

Proof: It suffices to prove

$$\lim_{x \rightarrow a} (f(x) - f(a)) = 0$$

Normally $\lim_{x \rightarrow a} f(x) = f(a)$ is what you want.

$$\lim_{x \rightarrow a} [(f(x) - f(a)) + f(a)] = \lim_{x \rightarrow a} [f(x) - f(a)] + \lim_{x \rightarrow a} f(a) = 0 + f(a)$$

$$\lim_{x \rightarrow a} f(x) - f(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \lim_{x \rightarrow a} (x - a) = f'(a) \cdot 0 = 0$$

by product rule of limit, therefore,



Definition 8.1.3: Small o

In analysis, **small-o** describes how one function grows compared to another near some limit. Formally, we say $f(x) = o(g(x))$ as $x \rightarrow a$ if

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 0$$

That means $f(x)$ becomes *negligible* compared to $g(x)$ near a .

Definition 8.1.4: Derivative def 3

f is differentiable at a if there exists $c \in \mathbb{R}$ such that

$$f(x) = f(a) + c(x - a) + o(x - a)$$

as $x \rightarrow a$.

Understanding the definition: this means the function $f(x)$ can be approximated near a by a linear function

$$L(x) = f(a) + c(x - a)$$

plus an error term that's negligible compared to $(x - a)$. The term $o(x - a)$ means

$$\lim_{x \rightarrow a} \frac{o(x - a)}{x - a} = 0$$

In other words, the error becomes infinitesimally smaller than $(x - a)$.

Example 8.1.1

find $f'(a)$ for $f(x) = x^2$

Proof:

$$\begin{aligned}
 f(x) &= f(a) + c(x-a) + o(x-a) \Leftrightarrow \\
 x^2 &= a^2 + c(x-a) + o(x-a) \Leftrightarrow \\
 x^2 - a^2 - c(x-a) &= o(x-a) \Leftrightarrow \\
 ((x-a) + a)^2 - a^2 - c(x-a) &= o(x-a) \Rightarrow \\
 [(x-a)^2 + 2a(x-a) + a^2] - a^2 - c(x-a) &= o(x-a) \Leftrightarrow \\
 (x-a)^2 + (2a-c)(x-a) &= o(x-a) \Leftrightarrow \\
 (x-a)^2 + (2a-c)(x-a) &= o(x-a) \Leftrightarrow \\
 \frac{(x-a)^2 + (2a-c)(x-a)}{x-a} &\rightarrow 0 \text{ as } x-a \rightarrow 0 \Leftrightarrow \\
 \frac{(x-a)^2}{x-a} + (2a-c)\frac{x-a}{x-a} &\rightarrow 0 \text{ as } x-a \rightarrow 0 \Leftrightarrow
 \end{aligned}$$

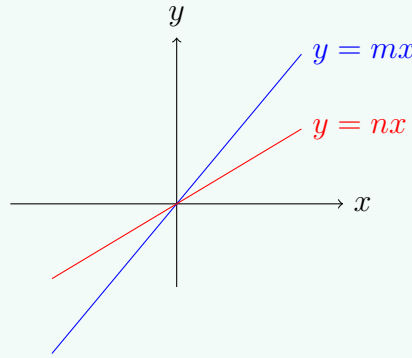
if $c = 2a$, $f'(a) = 2a$, if $c \neq 2a$ then it doesn't work.



Example 8.1.2

Linear case:

$$y = mx, \quad y = nx, \quad n \neq m$$



The vertical distance between the two lines at position x is

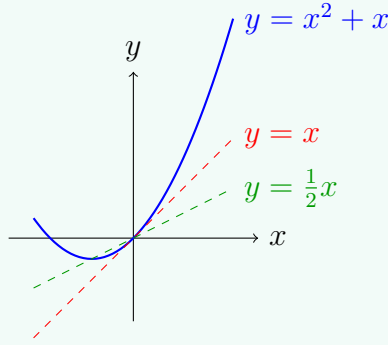
$$\text{distance} = |mx - nx| = |m - n| |x|.$$

As $x \rightarrow 0$, the distance tends to 0 *linearly* in x .

Nonlinear case: Consider $y = x^2 + x$.

$$\text{At } x = 0, \quad y = x^2 + x, \quad f'(0) = 1.$$

So the tangent line at $x = 0$ is $y = x$.



Let's compare distances:

- For tangent $y = \frac{1}{2}x$,

$$\text{distance} = (x^2 + x) - \frac{1}{2}x = x^2 + \frac{1}{2}x \rightarrow 0 \text{ linearly as } x \rightarrow 0.$$

- For tangent $y = x$,

$$\text{distance} = (x^2 + x) - x = x^2 \rightarrow 0 \text{ faster than linear.}$$

If the distance between $f(x)$ and its tangent line near $x = a$ goes to zero *faster than linearly* in $x - a$, that means

$$f(x) - [f(a) + f'(a)(x - a)] = o(|x - a|),$$

which is precisely the definition of differentiability:

$$f(x) = f(a) + f'(a)(x - a) + o(|x - a|).$$

Definition 8.1.5: Derivative def 4

f is differentiable at a if there exists $c \in \mathbb{R}$ and a function $r(x, a)$ so that

$$f(x) = f(a) + c(x - a) + r(x, a)$$

where

$$\frac{r(x, a)}{x - a} \rightarrow 0 \text{ as } x - a \rightarrow 0$$

If such c exists, we call $f'(a) = c$.

Claim 8.1.1

If $f(x) \equiv c$ for some $c \in \mathbb{R}$, then $f'(a) = 0$ for all $a \in \mathbb{R}$.

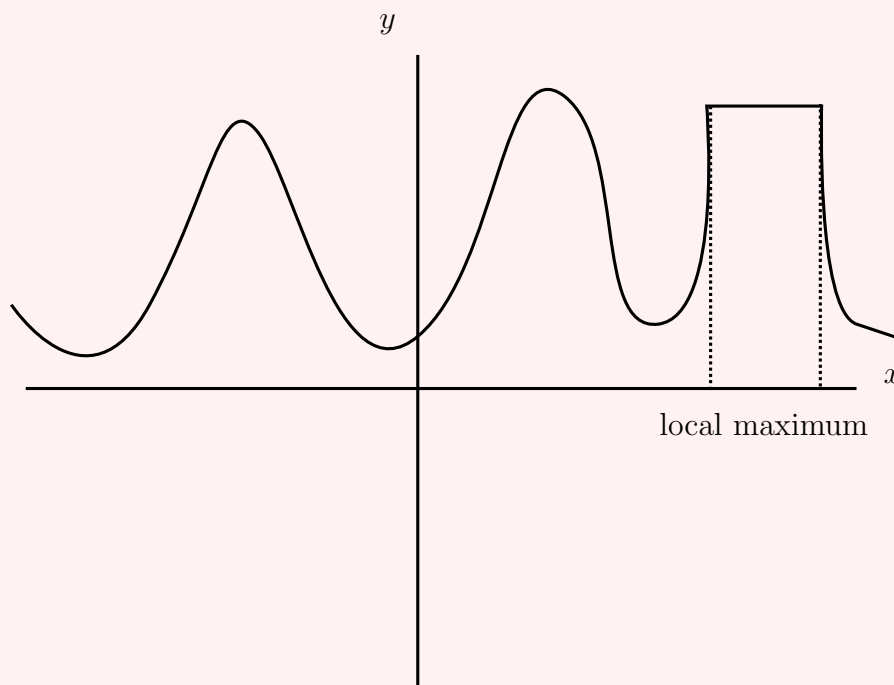
Proof: By the limit definition,

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = 0.$$



Definition 8.1.6: Local maximum

a is a local maximum of f if $\exists \delta > 0$ s.t. $(a - \delta, a + \delta) \subseteq \text{domain } f$ and $|x - a| < \delta \Rightarrow f(x) \leq f(a)$. There could be a plateau of local maximum, where the function has local maximum on a whole interval, for example:



Definition 8.1.7: Local minimum

a is a local minimum of f if a is a local maximum for $g = -f$

Theorem 8.1.3

If a is a local maximum of f and f is differentiable at a , then $f'(a) = 0$.

Sketch. Near a local maximum, a secant through a point to the *left* of a has positive slope, while a secant through a point to the *right* has negative slope. If the derivative exists, these one-sided slopes both converge to $f'(a)$, forcing $f'(a) = 0$.

Proof: Since a is a local maximum, there exists $\delta > 0$ such that

$$x \in (a - \delta, a + \delta) \implies f(x) \leq f(a).$$

Set $h = x - a$ and consider the difference quotient

$$Q(h) := \frac{f(a + h) - f(a)}{h} \quad \text{for } 0 < |h| < \delta.$$

Left side ($h < 0$). If $h < 0$ and $|h| < \delta$, then $a + h \in (a - \delta, a)$ and $f(a + h) \leq f(a)$. Because $h < 0$, dividing by the negative number h reverses the inequality:

$$\frac{f(a + h) - f(a)}{h} \geq 0 \quad (0 < |h| < \delta, h < 0).$$

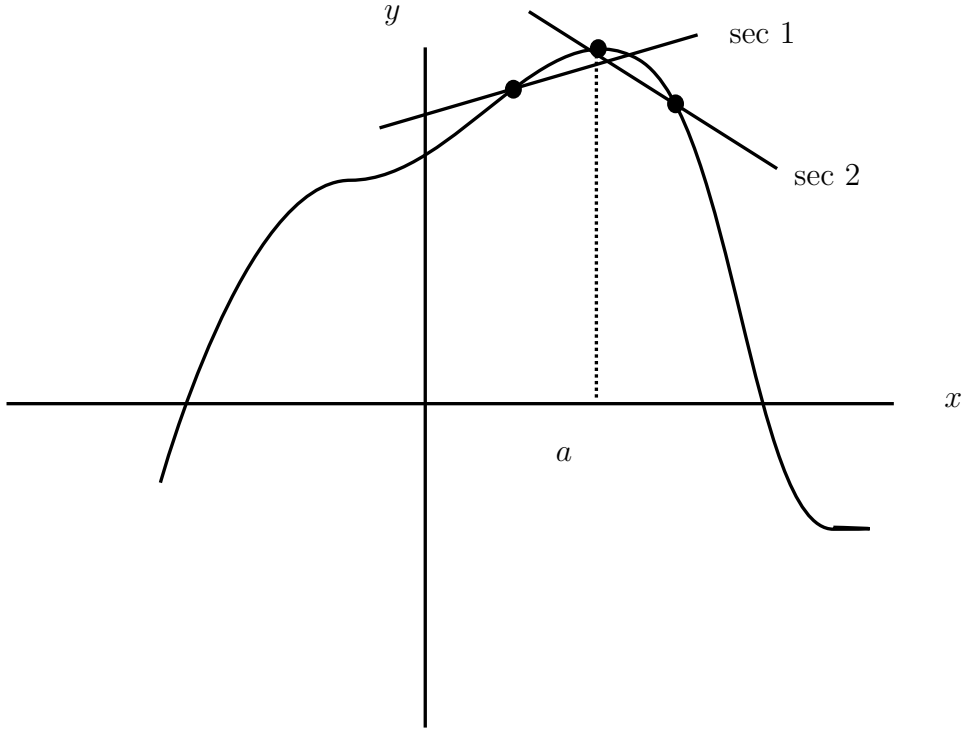


Figure 8.1: Two secants approaching the tangent at a local maximum a : the left secant has positive slope, the right secant has negative slope; their slopes converge to the slope of the tangent, which must be 0.

Hence every left-hand quotient $Q(h)$ is ≥ 0 , so

$$\lim_{h \rightarrow 0^-} Q(h) \geq 0.$$

Right side ($h > 0$). If $h > 0$ and $|h| < \delta$, then $a + h \in (a, a + \delta)$ and again $f(a + h) \leq f(a)$. Now $h > 0$, so dividing preserves the inequality:

$$\frac{f(a + h) - f(a)}{h} \leq 0 \quad (0 < |h| < \delta, h > 0).$$

Therefore every right-hand quotient $Q(h)$ is ≤ 0 , and

$$\lim_{h \rightarrow 0^+} Q(h) \leq 0.$$

Conclusion. Differentiability at a means the two one-sided limits exist and agree:

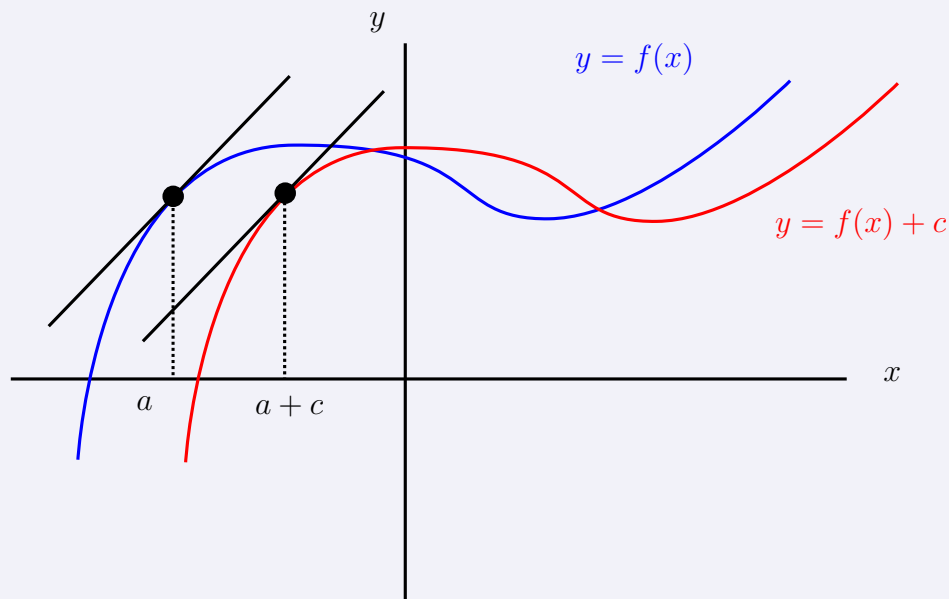
$$f'(a) = \lim_{h \rightarrow 0^-} Q(h) = \lim_{h \rightarrow 0^+} Q(h).$$

Combining with the inequalities above gives

$$f'(a) \geq 0 \quad \text{and} \quad f'(a) \leq 0,$$

hence $f'(a) = 0$.

☺

Theorem 8.1.4

Here are two curves: $y = f(x)$ and $y = f(x + c)$, $c < 0$. If $g(x) = f(x + c)$ for some $c \in \mathbb{R}$, then $g'(a) = f'(a + c)$. Formally, if f is differentiable at a then g is differentiable at $a + c$ and $g'(a) = f'(a + c)$

Proof:

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{g(a + h) - g(a)}{h} &= \lim_{h \rightarrow 0} \frac{f(a + c + h) - g(a + c)}{h} \quad (\text{by definition of } g) \\ &= \lim_{h \rightarrow 0} \frac{f((a + c) + h) - f(a + c)}{h} = f'(a + c) \end{aligned}$$

because f is differentiable at $a + c$.

**Theorem 8.1.5**

If f is differentiable at a , then f is continuous at a .

Proof:

$$\begin{aligned} \lim_{h \rightarrow 0} f(a + h) - f(a) &= \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h} \cdot h \\ &= \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h} \cdot \lim_{h \rightarrow 0} h \\ &= f'(a) \cdot 0 \\ &= 0 \end{aligned}$$

Since $\lim_{h \rightarrow 0} f(a+h) - f(a) = 0$ is equivalent to $\lim_{x \rightarrow a} f(x) = f(a)$, thus f is continuous at a . ☺

Note:-

The converse is not true, for example $f(x) = |x|$

8.1.1 Equivalence of the 4 definition for derivatives

Note:-

This part is from prof's notes

We'll show that if f is differentiable at a according to definition 1 then f is differentiable at a according to definition 2 and both definitions give the same value for $f'(a)$. We'll then show that if f is differentiable at a according to definition 2 then f is differentiable at a according to definition 3 and both definitions give the same value for $f'(a)$. Next, we'll show that if f is differentiable at a according to definition 3 then f is differentiable at a according to definition 4 and both definitions give the same value for $f'(a)$. And we'll close the loop by showing that if f is differentiable at a according to definition 4 then f is differentiable at a according to definition 1 and both definitions give the same value for $f'(a)$.

This proves that if f is differentiable at a according to one definition then it's differentiable at a according to the other three definitions and that all four definitions give the same value for $f'(a)$.

Claim 8.1.2

If f is differentiable at a according to definition 1 then f is differentiable at a according to definition 2 and the derivatives are equal.

Proof: By assumption, f is differentiable at a according to definition 1. That is, the limit

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

exists. Let's call the limit c_1 . By definition, given $\varepsilon > 0$ there exists $\delta > 0$ so that

$$|h| < \delta \implies \left| \frac{f(a+h) - f(a)}{h} - c_1 \right| < \varepsilon.$$

Let $x := a + h$. Then $h = x - a$ and, by definition,

$$\frac{f(a+h) - f(a)}{h} = \frac{f(x) - f(a)}{x - a}.$$

Therefore

$$|x - a| < \delta \implies \left| \frac{f(x) - f(a)}{x - a} - c_1 \right| < \varepsilon.$$

This proves

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = c_1.$$

Therefore f is differentiable at a according to definition 2 and the derivative computed using definition 2 equals c_1 , which is the derivative computed according to definition 1, as desired. ☺

Claim 8.1.3

If f is differentiable at a according to definition 2 then f is differentiable at a according to definition 3 and the derivatives are equal.

Proof: By assumption, f is differentiable at a according to definition 2. That is, the limit

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

exists. Let's call the limit c_2 . By definition, given $\varepsilon > 0$ there exists $\delta > 0$ so that

$$|x - a| < \delta \implies \left| \frac{f(x) - f(a)}{x - a} - c_2 \right| < \varepsilon.$$

Rearranging,

$$|x - a| < \delta \implies \left| \frac{f(x) - (f(a) + c_2(x - a))}{x - a} \right| < \varepsilon.$$

This proves

$$\lim_{x \rightarrow a} \frac{f(x) - (f(a) + c_2(x - a))}{x - a} = 0.$$

Which is the definition of

$$f(x) = f(a) + c_2(x - a) + o(x - a).$$

Therefore f is differentiable at a according to definition 3 and the derivative computed using definition 3 equals c_2 , which is the derivative computed according to definition 2, as desired. ☺

Claim 8.1.4

If f is differentiable at a according to definition 3 then f is differentiable at a according to definition 4 and the derivatives are equal.

Proof: By assumption, f is differentiable at a according to definition 3. That is, there exists $c_3 \in \mathbb{R}$ such that

$$f(x) = f(a) + c_3(x - a) + o(x - a).$$

From the definition of little o , this means

$$\lim_{x \rightarrow a} \frac{f(x) - (f(a) + c_3(x - a))}{x - a} = 0.$$

Let

$$r(x, a) := f(x) - f(a) - c_3(x - a).$$

Note that with this definition,

$$f(x) = f(a) + c_3(x - a) + r(x, a).$$

Also note that $\lim_{x \rightarrow a} \frac{r(x, a)}{x - a} = 0$. In this way we have found a constant c (specifically $c = c_3$) and a function e (specifically $r(x, a) = f(x) - f(a) - c_3(x - a)$) so that definition 4 holds.

Therefore f is differentiable at a according to definition 4 and the derivative computed using definition 4 equals c_3 , which is the derivative computed according to definition 3, as desired. ☺

Claim 8.1.5

If f is differentiable at a according to definition 4 then f is differentiable at a according to definition 1 and the derivatives are equal.

Proof: By assumption, f is differentiable at a according to definition 4. That is, there exists $c_4 \in \mathbb{R}$ and a function r such that

$$f(x) = f(a) + c_4(x - a) + r(x, a)$$

and

$$\lim_{x \rightarrow a} \frac{r(x, a)}{x - a} = 0.$$

Using the expression for $r(x, a)$,

$$\lim_{x \rightarrow a} \frac{f(x) - f(a) - c_4(x - a)}{x - a} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} - c_4 = 0.$$

We're done using the sum of limits law:

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a} \left(\left[\frac{f(x) - f(a)}{x - a} - c_4 \right] + c_4 \right) = \lim_{x \rightarrow a} \left[\frac{f(x) - f(a)}{x - a} - c_4 \right] + \lim_{x \rightarrow a} c_4 = 0 + c_4.$$

Therefore f is differentiable at a according to definition 1 and the derivative computed using definition 1 equals c_4 , which is the derivative computed according to definition 4, as desired. ☺

8.2 Rules for derivatives

8.2.1 Sum rule

Lemma 8.2.1

Let $w, v : \mathbb{R} \rightarrow \mathbb{R}$ be defined on a punctured neighborhood of $a \in \mathbb{R}$. If $\lim_{x \rightarrow a} w(x) = L$ and $\lim_{x \rightarrow a} v(x) = M$, then

$$\lim_{x \rightarrow a} (w(x) + v(x)) = L + M.$$

Proof: Fix $\varepsilon > 0$. Since $\lim_{x \rightarrow a} w(x) = L$, there exists $\delta_1 > 0$ such that $0 < |x - a| < \delta_1 \implies |w(x) - L| < \varepsilon/2$. Similarly, since $\lim_{x \rightarrow a} v(x) = M$, there exists $\delta_2 > 0$ such that $0 < |x - a| < \delta_2 \implies |v(x) - M| < \varepsilon/2$. Let $\delta = \min\{\delta_1, \delta_2\}$. Then for $0 < |x - a| < \delta$,

$$|(w(x) + v(x)) - (L + M)| \leq |w(x) - L| + |v(x) - M| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence $\lim_{x \rightarrow a} (w(x) + v(x)) = L + M$. ☺

Theorem 8.2.1

If f and g are differentiable at a , then $f + g$ is differentiable at a and $(f + g)'(a) = f'(a) + g'(a)$.

Proof: f differentiable at a implies

$$f(x) = f(a) + f'(a)(x - a) + r_f(x, a)$$

where $\lim_{x \rightarrow a} \frac{r_f(x, a)}{x - a} = 0$. r_f is the remainder term and goes to 0 faster than linear. Similarly

$$g(x) = g(a) + g'(a)(x - a) + r_g(x, a)$$

where $\lim_{x \rightarrow a} \frac{r_g(x, a)}{x - a} = 0$

$$f(x) + g(x) = f(a) + g(a) + [f'(a) + g'(a)](x - a) + r_f(x, a) + r_g(x, a)$$

By definition 4, if $\lim_{x \rightarrow a} \frac{r_f(x, a) + r_g(x, a)}{x - a} = 0$, then

$$(f + g)'(a) := f'(a) + g'(a)$$

$$\begin{aligned} \lim_{x \rightarrow a} \frac{r_f(x, a) + r_g(x, a)}{x - a} &= \lim_{x \rightarrow a} \frac{r_f(x, a)}{x - a} + \lim_{x \rightarrow a} \frac{r_g(x, a)}{x - a} \text{ by sum rule of limits} \\ &= 0 + 0 \\ &= 0 \end{aligned}$$

☺

Question 2: What does "goes faster than" mean?

- $r(x, a) \rightarrow 0$ is faster than $x - a$ iff $\lim_{x \rightarrow a} \frac{r(x, a)}{x - a} = 0$
- $r(x, a) \rightarrow 0$ is faster than $(x - a)^{1/3}$ iff $\lim_{x \rightarrow a} \frac{r(x, a)}{(x - a)^{1/3}} = 0$

Lemma 8.2.2

If $\tilde{r}(x, a)$ and $\hat{r}(x, a)$ are $o(x - a)$, and $\alpha \in \mathbb{R}, \beta \in \mathbb{R}$, then $\alpha\tilde{r}(x, a)$ and $\beta(x - a)\tilde{r}(x, a)$ and $\tilde{r}(x, a)\hat{r}(x, a)$ are all $o(x - a)$ and any linear combination of them is $o(x - a)$.

Proof:

$$\begin{aligned}
\lim_{x \rightarrow a} \frac{\alpha \tilde{r}(x, a)}{x - a} &= \lim_{x \rightarrow a} \alpha \frac{\tilde{r}(x - a)}{x - a} = \alpha \cdot 0 = 0 \\
\lim_{x \rightarrow a} \frac{\beta(x - a) \tilde{r}(x - a)}{x - a} &= \lim_{x \rightarrow a} \beta(x - a) \frac{\tilde{r}(x, a)}{x - a} = 0 \cdot 0 \\
\lim_{x \rightarrow a} \frac{\tilde{r}(x, a) \hat{r}(x, a)}{x - a} &= \lim_{x \rightarrow a} \frac{\tilde{r}(x, a)}{x - a} \frac{\hat{r}(x, a)}{x - a} = 0 \cdot 0 \cdot 0 \\
\lim_{x \rightarrow 0} (\text{linear combination}) &= \text{linear combination (limits)} = 0
\end{aligned}$$

by limit rule.

$$\lim_{x \rightarrow a} \frac{\tilde{r}(x, a) \hat{r}(x, a)}{x - a} = \lim_{x \rightarrow a} \tilde{r}(x, a) \frac{\hat{r}(x, a)}{x - a}$$

$\tilde{r}(x, a) \rightarrow 0$ because $\tilde{r} \rightarrow 0$ faster than $x - a$, which implies $\tilde{r} \rightarrow 0$ as $x \rightarrow 0$. $\frac{\hat{r}(x, a)}{x - a} \rightarrow 0$ because $\hat{r}$ is $o(x - a)$.

☺

8.2.2 Product rule

Theorem 8.2.2

If f and g are differentiable at a , then fg is differentiable at a and $(fg)'(a) = f'(a)g(a) + f(a)g'(a)$

Proof: We have:

$$f(x) = f(a) + f'(a)(x - a) + r_f(x, a)$$

where $\lim_{x \rightarrow a} \frac{r_f(x, a)}{x - a} = 0$ and

$$g(x) = g(a) + g'(a)(x - a) + r_g(x, a)$$

where $\lim_{x \rightarrow a} \frac{r_g(x, a)}{x - a} = 0$ Now,

$$\begin{aligned}
f(x)g(x) &= f(a)g(a) + f'(a)g(a)(x - a) + f(a)r_g(x, a) \\
&\quad + f'(a)g(a)(x - a) + f'(a)g'(a)(x - a)^2 + f'(a)(x - a)r_g(x, a) \\
&\quad + g(a)r_f(x, a) + g'(a)r_f(x, a)(x - a) + r_f(x, a)r_g(x, a)
\end{aligned}$$

By lemma,

$$\lim_{x \rightarrow a} \frac{f(a)r_g(x, a) + f'(a)g'(a)(x - a)^2 + \dots + r_f(x, a)r_g(x, a)}{x - a} = 0$$

Therefore, fg is differentiable and

$$(fg)'a = f(a)g'(a) + f'(a)g(a)$$

☺

8.2.3 Quotient rule

Theorem 8.2.3

If g is differentiable at a and $g(a) \neq 0$,
 then $\frac{1}{g}$ is differentiable at a and

$$\left(\frac{1}{g}\right)'(a) = -\frac{g'(a)}{[g(a)]^2}.$$

Corollary 8.2.1

If f and g are differentiable at a and $g(a) \neq 0$,
 then $\frac{f}{g}$ is differentiable at a and

$$\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - g'(a)f(a)}{[g(a)]^2}.$$

Proof:

$$\left(\frac{f}{g}\right) = f \left(\frac{1}{g}\right) \Rightarrow \left(\frac{f}{g}\right)' = f' \left(\frac{1}{g}\right) + f \left(\frac{1}{g}\right)'.$$

Substituting from the theorem gives

$$\left(\frac{f}{g}\right)' = \frac{f'g - g'f}{g^2}.$$

Sketch of Proof (for the Theorem).

Let

$$g(x) = g(a) + g'(a)(x - a) + r_g(x, a),$$

where $r_g(x, a)$ satisfies $r_g(x, a) = o(x - a)$ as $x \rightarrow a$.

Since g is differentiable at a , it is continuous at a . Hence, there exists $\delta_0 > 0$ such that if $|x - a| < \delta_0$, then $g(x) \neq 0$.

So, for $|x - a| < \delta_0$,

$$\frac{1}{g(x)} = \frac{1}{g(a) + g'(a)(x - a) + r_g(x, a)}.$$

We can factor out $g(a)$:

$$\frac{1}{g(x)} = \frac{1}{g(a)} \cdot \frac{1}{1 + \frac{g'(a)}{g(a)}(x - a) + \frac{r_g(x, a)}{g(a)}}.$$

Lemma 8.2.3

For small t ,

$$\frac{1}{1+t} = 1 - t + r(t), \quad \text{where } r(t) = o(t).$$

Applying the lemma with

$$t = \frac{g'(a)}{g(a)}(x - a) + \frac{r_g(x, a)}{g(a)},$$

we get

$$\frac{1}{g(x)} = \frac{1}{g(a)} \left[1 - \frac{g'(a)}{g(a)}(x - a) + r(x, a) \right],$$

where $r(x, a) = o(x - a)$.

Hence,

$$\frac{1}{g(x)} = \frac{1}{g(a)} - \frac{g'(a)}{[g(a)]^2}(x - a) + o(x - a),$$

$$\text{so } \left(\frac{1}{g} \right)'(a) = -\frac{g'(a)}{[g(a)]^2}.$$

☺

8.2.4 Chain rule

Theorem 8.2.4 Chain rule

If g is differentiable at a , and f is differentiable at $g(a)$

Proof: Let $f : A \rightarrow \mathbf{R}$ and $g : B \rightarrow \mathbf{R}$ satisfy $f(A) \subseteq B$ so that the composition $g \circ f$ is defined. If f is differentiable at $c \in A$ and if g is differentiable at $f(c) \in B$, then $g \circ f$ is differentiable at c with $(g \circ f)'(c) = g'(f(c)) \cdot f'(c)$.

Proof. Because g is differentiable at $f(c)$, we know that

$$g'(f(c)) = \lim_{y \rightarrow f(c)} \frac{g(y) - g(f(c))}{y - f(c)}. \quad (8.1)$$

Another way to assert this same fact is to let $d(y)$ be the difference quotient

$$d(y) = \frac{g(y) - g(f(c))}{y - f(c)}, \quad (8.2)$$

and observe that $\lim_{y \rightarrow f(c)} d(y) = g'(f(c))$. At the moment, $d(y)$ is not defined when $y = f(c)$, but it should seem natural to declare that $d(f(c)) = g'(f(c))$, so that d is continuous at $f(c)$.

Now, we come to the finesse. Equation (1) can be rewritten as

$$g(y) - g(f(c)) = d(y)(y - f(c)).$$

Observe that this equation holds for all $y \in B$ including $y = f(c)$. Thus, we are free to substitute $y = f(t)$ for any arbitrary $t \in A$. If $t \neq c$, we can divide equation (2) by $(t - c)$ to get

$$\frac{g(f(t)) - g(f(c))}{t - c} = d(f(t)) \frac{(f(t) - f(c))}{t - c}$$

for all $t \neq c$. Finally, taking the limit as $t \rightarrow c$ and applying the Algebraic Limit Theorem together with Theorem 4.3.9 yields the desired formula. ☺

8.3 Significance of the derivative

Definition 8.3.1: Global maximum

x_0 is a **global maximum** for f if $f(x_0) \geq f(x), \forall x \in A$.

Definition 8.3.2: Local maximum

x_0 is a **local maximum** for f if there exists $\delta > 0$ such that if $x \in A$ and $|x - x_0| < \delta$ then $f(x_0) \geq f(x)$.

Theorem 8.3.1

If x_0 is a local maximum or minimum, and $f'(x_0)$ exists, then $f'(x_0) = 0$

Proof: Suppose x_0 is a local maximum or minimum. Then there exists an interval $(x_0 - \delta, x_0 + \delta)$ such that

$$f(x) \leq f(x_0) \quad \text{for all } x \in (x_0 - \delta, x_0 + \delta)$$

in the case of a local maximum, and

$$f(x) \geq f(x_0) \quad \text{for all } x \in (x_0 - \delta, x_0 + \delta)$$

in the case of a local minimum.

Consider the difference quotient for $h > 0$: - If x_0 is a local maximum, then $f(x_0 + h) \leq f(x_0)$, so

$$\frac{f(x_0 + h) - f(x_0)}{h} \leq 0.$$

- If x_0 is a local minimum, then $f(x_0 + h) \geq f(x_0)$, so

$$\frac{f(x_0 + h) - f(x_0)}{h} \geq 0.$$

Similarly, for $h < 0$: - If x_0 is a local maximum, then $f(x_0 + h) \leq f(x_0)$, and since $h < 0$,

$$\frac{f(x_0 + h) - f(x_0)}{h} \geq 0.$$

- If x_0 is a local minimum, then $f(x_0 + h) \geq f(x_0)$, so

$$\frac{f(x_0 + h) - f(x_0)}{h} \leq 0.$$

Thus, in both the maximum and minimum cases, the right-hand limit of the difference quotient is ≤ 0 and the left-hand limit is ≥ 0 .

Since $f'(x_0)$ exists, the left and right limits must be equal. The only number that is both ≤ 0 and ≥ 0 is 0. Therefore,

$$f'(x_0) = 0.$$



Definition 8.3.3: Critical Point

A point x_0 in the domain of a function f is called a *critical point* if f is differentiable at this point and

$$f'(x_0) = 0,$$

Theorem 8.3.2 Rolle's Theorem

Let f be continuous on $[a, b]$, differentiable on (a, b) , and suppose $f(a) = f(b)$. Then there exists some $c \in (a, b)$ such that

$$f'(c) = 0.$$

Proof: Because f is continuous on $[a, b]$, it attains a maximum and a minimum on that interval.

If these extrema occur only at the endpoints and

$$f(a) = f(b),$$

then f must be constant on $[a, b]$, so $f'(x) = 0$ for all $x \in (a, b)$, and any c works.

Otherwise, at least one of the extrema occurs at some interior point

$$c \in (a, b).$$

At that interior point, f has either a local maximum or a local minimum. Since f is differentiable at c , the previous theorem implies

$$f'(c) = 0.$$

Thus, in all cases, there exists $c \in (a, b)$ with $f'(c) = 0$.



Theorem 8.3.3 Mean Value Theorem

Let f be continuous on $[a, b]$ and differentiable on (a, b) . Then there exists some $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof: Define a new function

$$g(x) = f(x) - \left(\frac{f(b) - f(a)}{b - a} \right) (x - a).$$

The line being subtracted is the secant line through $(a, f(a))$ and $(b, f(b))$.

Observe that

$$g(a) = f(a) - 0 = f(a), \quad g(b) = f(b) - (f(b) - f(a)) = f(a).$$

Hence

$$g(a) = g(b).$$

The function g is continuous on $[a, b]$ and differentiable on (a, b) , so by Rolle's Theorem, there exists $c \in (a, b)$ such that

$$g'(c) = 0.$$

But

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}.$$

Therefore,

$$0 = g'(c) = f'(c) - \frac{f(b) - f(a)}{b - a},$$

which implies

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

This proves the theorem.



Chapter 9

Week 9

Note:-

Additional practice on differentiation, the Mean Value Theorem, and optimization can be found in Chapter 23, Sections 23.5 and 23.6.

9.1 Optimization Problems

The Mean Value Theorem and the theory of critical points allow us to solve optimization problems rigorously.

Theorem 9.1.1 Global Extrema on a Closed Interval

Let f be continuous on $[a, b]$. Then f attains its maximum and minimum on $[a, b]$ (by the Extreme Value Theorem). Moreover, any global extremum of f on $[a, b]$ occurs either at an endpoint (a or b), or at a critical point $c \in (a, b)$ where $f'(c) = 0$ or $f'(c)$ does not exist.

Example 9.1.1 (Optimization on a closed interval)

Find the maximum of $f(x) = x^3 - 3x + 1$ on $[-2, 2]$.

We compute $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$, so the critical points are $x = \pm 1$. Evaluating:

$$f(-2) = -1, \quad f(-1) = 3, \quad f(1) = -1, \quad f(2) = 3.$$

Thus the maximum value is 3, attained at $x = -1$ and $x = 2$.

9.2 Second-Derivative Test

Theorem 9.2.1 Second-Derivative Test (Sufficient Condition)

Suppose f'' is continuous near a , $f'(a) = 0$, and $f''(a) \neq 0$.

1. If $f''(a) > 0$, then f has a local minimum at a .
2. If $f''(a) < 0$, then f has a local maximum at a .

Proof: Suppose $f'(a) = 0$ and $f''(a) > 0$. Since f'' is continuous and positive at a , there exists $\delta > 0$ such that $f''(x) > 0$ for all $x \in (a - \delta, a + \delta)$.

By the Mean Value Theorem, for $x \in (a - \delta, a + \delta)$ with $x \neq a$, there exists c between a and x with

$$f(x) - f(a) = f'(c)(x - a).$$

Since $f'(a) = 0$ and $f''(c) > 0$, we know f' is strictly increasing near a . Thus $f'(c) < 0$ for $c < a$ and $f'(c) > 0$ for $c > a$. In either case $f'(c)(x - a) > 0$, so $f(x) > f(a)$ for $x \neq a$ near a . ☺

Note:-

The second-derivative test gives a *sufficient* condition for local extrema. The converse fails: $f(x) = x^4$ has a local minimum at 0, but $f''(0) = 0$. If $f''(a) = 0$, the test is inconclusive.

9.3 Convexity

Definition 9.3.1: Convex and Concave Functions

A function f is **convex** on an interval I if for all $x, y \in I$ and all $\lambda \in [0, 1]$,

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

f is **concave** on I if $-f$ is convex on I .

Theorem 9.3.1 Second-Derivative Characterization of Convexity

If f is twice differentiable on an interval I , then:

1. f is convex on I if and only if $f''(x) \geq 0$ for all $x \in I$.
2. f is concave on I if and only if $f''(x) \leq 0$ for all $x \in I$.

Definition 9.3.2: Inflection Point

A point a is an **inflection point** of f if f changes concavity at a (i.e., f'' changes sign at a).

9.4 Graphing Differentiable Functions

To sketch the graph of a differentiable function f :

1. Find the domain and any symmetries (even/odd).
2. Compute $f'(x)$; find critical points where $f'(x) = 0$ or $f'(x)$ is undefined.
3. Determine intervals of increase ($f' > 0$) and decrease ($f' < 0$).
4. Compute $f''(x)$; find inflection points where f'' changes sign.
5. Determine intervals of convexity ($f'' > 0$) and concavity ($f'' < 0$).

6. Evaluate f at critical points, inflection points, and endpoints. Determine limits as $x \rightarrow \pm\infty$ if applicable.

Example 9.4.1 (Graphing)

Let $f(x) = x^4 - 4x^3$. Then $f'(x) = 4x^3 - 12x^2 = 4x^2(x - 3)$ and $f''(x) = 12x^2 - 24x = 12x(x - 2)$.

Critical points: $x = 0$ (neither max nor min since f' does not change sign) and $x = 3$ (local min since f' changes from $-$ to $+$).

Inflection points: $x = 0$ and $x = 2$ (where f'' changes sign).

f is convex on $(-\infty, 0)$ and $(2, \infty)$, concave on $(0, 2)$.

Chapter 10

Week 10

Note:-

Additional practice on differentiation rules and L'Hôpital's rule can be found in Chapter 23, Section 23.5.

Starter Examples

Example 10.0.1 (Example 1)

$$\ell = \lim_{x \rightarrow 0} \frac{5x + 4x^2}{x - 9x^3} = \lim_{x \rightarrow 0} \frac{x(5 + 4x)}{x(1 - 9x^2)} = \lim_{x \rightarrow 0} \frac{5 + 4x}{1 - 9x^2} = 5.$$

Example 10.0.2 (Example 2)

We compute

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}.$$

Using Taylor's theorem with remainder,

$$\sin(x) = \sin(0) + \cos(0)x + r(x, 0) = 0 + x + r(x, 0) = x + r(x, 0),$$

where the remainder satisfies

$$\frac{r(x, 0)}{x} \longrightarrow 0 \quad (x \rightarrow 0),$$

because $r(x, 0) = o(x)$.

Thus,

$$\frac{\sin x}{x} = \frac{x + r(x, 0)}{x} = \frac{x(1 + r(x, 0)/x)}{x} = 1 + \frac{r(x, 0)}{x}.$$

Taking the limit,

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \left(1 + \frac{r(x, 0)}{x} \right) = 1.$$

Example 10.0.3 (Taylor expansion limit)

Compute

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{\cos x - 1}.$$

Use second-order Taylor expansions at $x = 0$:

$$e^x = 1 + x + \frac{x^2}{2} + r_f(x), \quad \text{where } \frac{r_f(x)}{x^2} \rightarrow 0.$$

$$\cos x = 1 - \frac{x^2}{2} + r_g(x), \quad \text{where } \frac{r_g(x)}{x^2} \rightarrow 0.$$

Substitute:

$$e^x - 1 - x = \left(1 + x + \frac{x^2}{2} + r_f(x)\right) - 1 - x = \frac{x^2}{2} + r_f(x),$$

$$\cos x - 1 = \left(1 - \frac{x^2}{2} + r_g(x)\right) - 1 = -\frac{x^2}{2} + r_g(x).$$

Thus,

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{\cos x - 1} = \lim_{x \rightarrow 0} \frac{\frac{x^2}{2} + r_f(x)}{-\frac{x^2}{2} + r_g(x)}.$$

Factor out x^2 :

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{2} + \frac{r_f(x)}{x^2}}{-\frac{1}{2} + \frac{r_g(x)}{x^2}}.$$

Since $\frac{r_f(x)}{x^2} \rightarrow 0$ and $\frac{r_g(x)}{x^2} \rightarrow 0$,

$$= \frac{\frac{1}{2}}{-\frac{1}{2}} = -1.$$

10.1 L'Hôpital's rule

Theorem 10.1.1 L'Hôpital's Rule (the 0/0 case)

Let f and g be real-valued functions defined on an open interval containing a , except possibly at a itself. Assume:

1. $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$,
2. f and g are differentiable on $(a - \delta, a + \delta) \setminus \{a\}$ for some $\delta > 0$,
3. $g'(x) \neq 0$ for all $x \neq a$ sufficiently close to a ,
4. the limit $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists.

Then the limit $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ exists and

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

Proof: Define modified functions

$$\tilde{f}(x) = \begin{cases} f(x), & x \neq a, \\ 0, & x = a, \end{cases} \quad \tilde{g}(x) = \begin{cases} g(x), & x \neq a, \\ 0, & x = a. \end{cases}$$

Since $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$, both $\tilde{f}$ and $\tilde{g}$ are continuous at a , and coincide with f and g on $(a - \delta, a + \delta) \setminus \{a\}$. Thus $\tilde{f}(a) = \tilde{g}(a) = 0$.

Fix $x \neq a$ sufficiently close to a . Apply Cauchy's Mean Value Theorem to $\tilde{f}$ and $\tilde{g}$ on the interval with endpoints a and x . Both functions are continuous on the closed interval and differentiable on its interior, and $\tilde{g}'(t) = g'(t) \neq 0$ for $t \neq a$. Hence there exists c between a and x such that

$$\frac{\tilde{f}(x) - \tilde{f}(a)}{\tilde{g}(x) - \tilde{g}(a)} = \frac{\tilde{f}'(c)}{\tilde{g}'(c)}.$$

Because $\tilde{f}(a) = \tilde{g}(a) = 0$ and $\tilde{f}(x) = f(x)$, $\tilde{g}(x) = g(x)$ for $x \neq a$, we obtain

$$\frac{f(x)}{g(x)} = \frac{f'(c)}{g'(c)}.$$

Letting $x \rightarrow a$ forces $c \rightarrow a$ as well, and by assumption the limit $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists. Therefore,

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{c \rightarrow a} \frac{f'(c)}{g'(c)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

☺

Here's another proof

Proof: Since f and g are differentiable at a , write the Taylor expansions

$$f(x) = f(a) + f'(a)(x - a) + r_f(x, a), \quad g(x) = g(a) + g'(a)(x - a) + r_g(x, a),$$

where the remainder terms satisfy

$$\frac{r_f(x, a)}{x - a} \rightarrow 0, \quad \frac{r_g(x, a)}{x - a} \rightarrow 0 \quad (x \rightarrow a).$$

Because $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$, we obtain

$$f(a) = 0, \quad g(a) = 0.$$

because f and g are differentiable at a . Thus

$$\frac{f(x)}{g(x)} = \frac{f'(a)(x-a) + r_f(x,a)}{g'(a)(x-a) + r_g(x,a)}.$$

Divide numerator and denominator by $(x-a)$:

$$\frac{f(x)}{g(x)} = \frac{f'(a) + \frac{r_f(x,a)}{x-a}}{g'(a) + \frac{r_g(x,a)}{x-a}}.$$

Taking the limit as $x \rightarrow a$ gives

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{f'(a)}{g'(a)},$$

since both $\frac{r_f(x,a)}{x-a}$ and $\frac{r_g(x,a)}{x-a}$ tend to 0.



Chapter 11

Week 11

Note:-

Additional practice on the Cauchy Mean Value Theorem and related results can be found in Chapter 23, Section 23.5.

11.1 Cauchy MVT

Example 11.1.1 (Introduction to CMVT)

Let $f(x) = \sqrt{x}$ on $(0, 1]$ and $g(x) = x^2$ on $(0, 1]$.

Then

$$\frac{f(1) - f(0)}{1 - 0} = \frac{g(1) - g(0)}{1 - 0}.$$

Hence, by the Mean Value Theorem, there exists $c \in (0, 1)$ such that

$$f'(c) = g'(c).$$

Theorem 11.1.1 Cauchy Mean Value Theorem

Let f, g be continuous on $[a, b]$ and differentiable on (a, b) . If

$$f(b) - f(a) = g(b) - g(a),$$

then there exists $c \in (a, b)$ such that

$$f'(c) = g'(c).$$

Proof: Define

$$h(x) = f(x) - g(x).$$

Then h is continuous on $[a, b]$ and differentiable on (a, b) . Also,

$$h(a) = f(a) - g(a) = f(b) - g(b) = h(b).$$

By Rolle's Theorem, there exists $c \in (a, b)$ with

$$h'(c) = 0.$$

Thus

$$f'(c) - g'(c) = 0 \quad \Rightarrow \quad f'(c) = g'(c).$$



Example 11.1.2 (Another Example)

Let

$$f(x) = \frac{1}{2}x^3, \quad g(x) = x + x^2.$$

There is no $c \in (0, 1)$ such that $f'(c) = g'(c)$, because

$$f'(x) = \frac{3}{2}x^2, \quad g'(x) = 1 + 2x,$$

and these cannot be equal on $(0, 1)$.

Question 1: Question

How can we modify the function so that

$$\frac{f(b) - f(a)}{b - a} = \frac{g(b) - g(a)}{b - a}?$$

Solution: Use the Cauchy Mean Value Theorem in the following form.

Theorem 11.1.2 Cauchy Mean Value Theorem (Version 2)

Assume f, g are continuous on $[a, b]$ and differentiable on (a, b) with $g(b) \neq g(a)$. Then there exists $c \in (a, b)$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$

Proof: Define the auxiliary function

$$h(x) = f(x) - \frac{f(b) - f(a)}{g(b) - g(a)} g(x).$$

We first check that $h(a) = h(b)$:

$$h(a) = f(a) - \frac{f(b) - f(a)}{g(b) - g(a)} g(a),$$

$$h(b) = f(b) - \frac{f(b) - f(a)}{g(b) - g(a)} g(b).$$

Thus

$$\begin{aligned} h(b) - h(a) &= [f(b) - f(a)] - \frac{f(b) - f(a)}{g(b) - g(a)} [g(b) - g(a)] \\ &= (f(b) - f(a)) \left[1 - \frac{g(b) - g(a)}{g(b) - g(a)} \right] = 0. \end{aligned}$$

Hence $h(a) = h(b)$.

Since h is continuous on $[a, b]$ and differentiable on (a, b) , by Rolle's Theorem there exists $c \in (a, b)$ such that

$$h'(c) = 0.$$

Compute $h'(x)$:

$$h'(x) = f'(x) - \frac{f(b) - f(a)}{g(b) - g(a)} g'(x).$$

Thus,

$$0 = h'(c) = f'(c) - \frac{f(b) - f(a)}{g(b) - g(a)} g'(c),$$

and therefore

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$

This proves the theorem. ☺

Theorem 11.1.3 Cauchy Mean Value Theorem (Version 3)

Assume f, g are continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $c \in (a, b)$ such that

$$[g(b) - g(a)] f'(c) = [f(b) - f(a)] g'(c).$$

Note. This version allows the case $g(b) = g(a)$ or $f(b) = f(a)$. For example, if $g(b) = g(a)$ then by Rolle's Theorem there is $c \in (a, b)$ with $g'(c) = 0$. Similarly, if $f(b) = f(a)$ then there is $c \in (a, b)$ with $f'(c) = 0$.

Proof: Define the auxiliary function

$$F(x) = [g(b) - g(a)](f(x) - f(a)) - [f(b) - f(a)](g(x) - g(a)).$$

Since f and g are continuous on $[a, b]$ and differentiable on (a, b) , the same is true for F .

We compute the endpoint values:

$$F(a) = [g(b) - g(a)](f(a) - f(a)) - [f(b) - f(a)](g(a) - g(a)) = 0,$$

and similarly,

$$F(b) = [g(b) - g(a)](f(b) - f(a)) - [f(b) - f(a)](g(b) - g(a)) = 0.$$

Thus $F(a) = F(b)$.

Because F is continuous on $[a, b]$, differentiable on (a, b) , and satisfies $F(a) = F(b)$, Rolle's Theorem guarantees the existence of $c \in (a, b)$ such that

$$F'(c) = 0.$$

We now compute $F'(x)$:

$$F'(x) = [g(b) - g(a)]f'(x) - [f(b) - f(a)]g'(x).$$

Hence at $x = c$,

$$0 = F'(c) = [g(b) - g(a)]f'(c) - [f(b) - f(a)]g'(c).$$

Rearranging yields

$$[g(b) - g(a)]f'(c) = [f(b) - f(a)]g'(c),$$

which is the desired conclusion. $\ominus$

Note:-

Note that the cases " $\frac{\infty}{\infty}$ " for L'Hôpital's Rule are not included in the standard statement; read the professor's note of the proof.

Example 11.1.3 (Limit Example)

Consider

$$\lim_{x \rightarrow 0^+} x \ln(x).$$

This is not of type $\frac{0}{0}$ nor $\frac{\infty}{\infty}$, since

$$x \rightarrow 0^+, \quad \ln(x) \rightarrow -\infty, \quad x \ln(x) \rightarrow 0 \cdot (-\infty).$$

Rewrite it as

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} \quad \text{or} \quad \lim_{x \rightarrow 0^+} \frac{x}{1/\ln x}.$$

Using the second rewriting:

$$\lim_{x \rightarrow 0^+} \frac{(x)'}{(\frac{1}{\ln x})'} = \lim_{x \rightarrow 0^+} \frac{1}{-\frac{1}{(\ln x)^2} \cdot \frac{1}{x}} = \lim_{x \rightarrow 0^+} -x(\ln x)^2$$

which becomes worse.

Using the first rewriting:

$$\lim_{x \rightarrow 0^+} \frac{(\ln x)'}{(\frac{1}{x})'} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x \rightarrow 0^+} -x = 0$$

which works

11.2 Composite Functions

Example 11.2.1 (Epsilon–Delta Proof of $\lim_{x \rightarrow 0^+} x^2 \ln(x) = 0$)

Another method (using substitution):

Let

$$k = x^2.$$

Then as $x \rightarrow 0^+$, we also have $k \rightarrow 0^+$, and

$$x^2 \ln(x) = k \ln(k).$$

We know from L'Hôpital's rule that (proven above)

$$\lim_{k \rightarrow 0^+} k \ln(k) = 0.$$

So given any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < k < \delta \implies |k \ln k - 0| < \varepsilon.$$

Now we must rewrite this in terms of x .

Given this $\delta > 0$, choose $\widehat{\delta} > 0$ such that

$$|x - 0| < \widehat{\delta} \implies |x^2 - 0| < \delta,$$

that is,

$$0 < x < \widehat{\delta} \implies 0 < k = x^2 < \delta.$$

Thus for such x :

$$0 < x < \widehat{\delta} \implies |x^2 \ln(x)| = |k \ln(k)| < \varepsilon.$$

Therefore

$$\lim_{x \rightarrow 0^+} x^2 \ln(x) = 0.$$

Example 11.2.2 (Limit Example)

$$\lim_{x \rightarrow 1} (x^2 - 1) \ln |x^2 - 1|$$

Let

$$k = x^2 - 1.$$

Then as $x \rightarrow 1$, $k \rightarrow 0$, and

$$(x^2 - 1) \ln |x^2 - 1| = k \ln |k|.$$

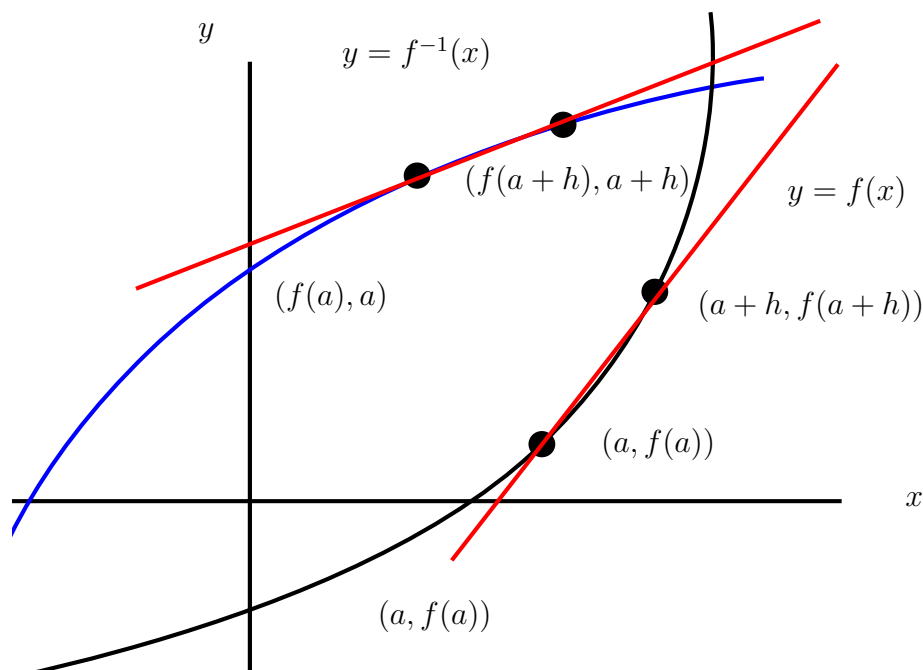
We know

$$\lim_{k \rightarrow 0} k \ln |k| = 0.$$

Hence

$$\lim_{x \rightarrow 1} (x^2 - 1) \ln |x^2 - 1| = 0.$$

11.3 Inverse function



Question 2

Is f^{-1} differentiable at $f(a)$?

Solution: We look at the secant slopes of the inverse function. For the inverse function:

$$(f^{-1})'(f(a)) = \lim_{h \rightarrow 0} \frac{(a+h) - a}{f(a+h) - f(a)}.$$

Rewrite:

$$(f^{-1})'(f(a)) = \lim_{h \rightarrow 0} \frac{h}{f(a+h) - f(a)} = \left(\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \right)^{-1}.$$

Since

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = f'(a)$$

and this limit exists, we obtain:

$$(f^{-1})'(f(a)) = \frac{1}{f'(a)}, \quad \text{provided } f'(a) \neq 0.$$

Thus the inverse function is differentiable at $f(a)$ whenever $f'(a) \neq 0$.



Example 11.3.1 (Derivative of the Square Root)

Prove that if

$$f(x) = \sqrt{x} \quad (x > 0),$$

then

$$f'(x) = \frac{1}{2\sqrt{x}}.$$

Incorrect Attempt (as often written):

Assume $f(x) = \sqrt{x}$. Then

$$(f(x))^2 = x.$$

Differentiate both sides:

$$2f(x)f'(x) = 1,$$

so

$$f'(x) = \frac{1}{2f(x)} = \frac{1}{2\sqrt{x}}.$$

Problem: This “proof” assumes *from the beginning* that f is differentiable at x , but differentiability is exactly what we are trying to prove. Thus the argument is circular: it differentiates $f(x)$ before proving that f has a derivative.

A correct proof must start from the definition

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h},$$

and evaluate this limit directly.

11.3.1 Inverse Function Theorem

Continuing from the previous discussion, we hope that if $f'(a) \neq 0$ then f^{-1} is differentiable at $f(a)$ and

$$(f^{-1})'(f(a)) = \frac{1}{f'(a)}.$$

More generally, if $b = f(a)$ then

$$(f^{-1})'(b) = \frac{1}{f'(f^{-1}(b))}.$$

Theorem 11.3.1 Inverse Function Theorem (Spivak)

Let f be a continuous 1–1 function on an open interval I . Assume $a \in I$ and f is differentiable at a with $f'(a) \neq 0$. Then f^{-1} is differentiable at $f(a)$ and

$$(f^{-1})'(f(a)) = \frac{1}{f'(a)}.$$

Proof: Since f is continuous and 1–1 on an open interval, there exists an open interval J such that $f : J \rightarrow f(J)$ is a bijection and $f^{-1} : f(J) \rightarrow J$ is continuous.

Let $b = f(a)$ and define

$$c = f^{-1}(b + k), \quad \text{so that} \quad b + k = f(c).$$

We want to know whether the limit

$$(f^{-1})'(b) = \lim_{k \rightarrow 0} \frac{f^{-1}(b + k) - f^{-1}(b)}{k}$$

exists and equals $\frac{1}{f'(a)}$.

If the above limit exists, then writing $c = f^{-1}(b + k)$ we have

$$\frac{f^{-1}(b + k) - f^{-1}(b)}{k} = \frac{c - a}{f(c) - f(a)}.$$

Let $h = c - a$, so $h \rightarrow 0$ as $k \rightarrow 0$. Then

$$\frac{c - a}{f(c) - f(a)} = \frac{h}{f(a + h) - f(a)}.$$

Thus

$$\lim_{k \rightarrow 0} \frac{f^{-1}(b + k) - f^{-1}(b)}{k} = \lim_{h \rightarrow 0} \frac{h}{f(a + h) - f(a)} = \frac{1}{f'(a)},$$

since the last limit is the reciprocal of the definition of $f'(a)$. ⊕

Theorem 11.3.2 Inverse Function Theorem (Stronger)

Assume $f'(a)$ exists and is nonzero. Assume further that f is continuous in a neighborhood of a . Then f^{-1} exists in a neighborhood of $f(a)$, is differentiable, and satisfies

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}$$

for all y in a neighborhood of $f(a)$.

Proof: Since $f'(a) \neq 0$ and f' is continuous at a , there exists $\delta > 0$ such that either

$$f'(x) > 0 \quad \text{for all } x \in (a - \delta, a + \delta),$$

or

$$f'(x) < 0 \quad \text{for all } x \in (a - \delta, a + \delta).$$

Thus f is strictly monotone on $(a - \delta, a + \delta)$, and hence injective on that interval. Because f is continuous, the restriction

$$f : (a - \delta, a + \delta) \longrightarrow J$$

is a bijection onto some open interval J containing $f(a)$, and its inverse $f^{-1} : J \rightarrow (a - \delta, a + \delta)$ is continuous.

To compute the derivative of the inverse, let $y \in J$ and write $x = f^{-1}(y)$. Then $f(x) = y$. Differentiating both sides with respect to y and applying the chain rule gives

$$f'(x) (f^{-1})'(y) = 1.$$

Hence

$$(f^{-1})'(y) = \frac{1}{f'(x)} = \frac{1}{f'(f^{-1}(y))},$$

which is well defined since $f'(x) \neq 0$ in the chosen neighborhood. ☺

Note:-

Difference From Spivak's Proof: Spivak assumes from the start that f is continuous and *one-to-one* on I , so monotonicity of f is already guaranteed and no appeal to continuity of f' is needed. In contrast, the more common textbook version assumes f' is continuous near a and uses that to deduce local monotonicity and the existence of the inverse.

Question 3: Implicit Differentiation (Chapter 14 of Spivak).

Prove there is a differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$(f(x))^5 + f(x) + x = 0.$$

Solution: We want to find a differentiable function f satisfying the equation. If $y = f(x)$ then $f^{-1}(y) = x$. (So we try to find a differentiable function f^{-1} such that)

$$y^5 + y + f^{-1}(y) = 0.$$

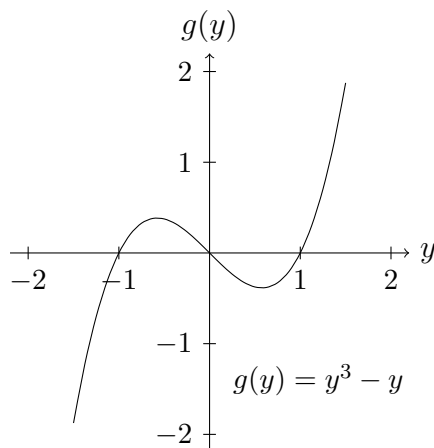
Thus

$$f^{-1}(y) = -y^5 - y.$$

This is differentiable, domain is $\mathbb{R}$, range is $\mathbb{R}$.

Is f^{-1} one-to-one? (This is the worry.)

Suppose $y = f(x)$.



The graph illustrates a potential case where f^{-1} is not invertible (cannot define $f(x) = (f^{-1})^{-1}(x)$).

But, this will not be the case since

$$f^{-1}(y) = -y(y^4 + 1).$$

Then

$$(f^{-1})'(y) = -5y^4 - 1 < 0 \quad \forall y,$$

so f^{-1} is strictly decreasing and $(f^{-1})'$ is never 0.

Therefore $f(x)$ exists and

$$f'(x) = \frac{1}{(f^{-1})'(f(x))} \quad \forall x.$$

A problematic solution:

$$(f(x))^5 + f(x) + x = 0.$$

Differentiate:

$$5(f(x))^4 f'(x) + f'(x) + 1 = 0.$$

So

$$f'(x) = \frac{-1}{5(f(x))^4 + 1}.$$

Need to prove f is differentiable and then differentiate.

Chapter 12

Week 1 (Winter 2026)

Note:-

Additional practice on sequences and series can be found in Chapter 23, Section 23.3.

12.1 Infinite sequences

Definition 12.1.1: Convergence of a Sequence

A sequence $(a_n)_{n \in \mathbb{N}}$ converges to a real number ℓ if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $n > N$,

$$|a_n - \ell| < \epsilon.$$

In this case, we write $\lim_{n \rightarrow \infty} a_n = \ell$.

Question 1

Question: What property of a sequence guarantees that it converges to some real number?

Intuitively, convergence means that if you wait long enough, all terms of the sequence stay arbitrarily close to the same real number.

Definition 12.1.2: Cauchy Sequence

A sequence $(a_n)_{n \in \mathbb{N}}$ is called a Cauchy sequence if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $m, n > N$,

$$|a_n - a_m| < \epsilon.$$

Intuitively, a sequence is Cauchy if, after some point, all its terms are close to each other.

Theorem 12.1.1

Every convergent sequence in $\mathbb{R}$ is a Cauchy sequence.

Proof: Let $\epsilon > 0$. Since (a_n) converges to ℓ , there exists $N \in \mathbb{N}$ such that for all $n > N$,

$$|a_n - \ell| < \frac{\epsilon}{2}.$$

If $m, n > N$, then by the triangle inequality,

$$|a_n - a_m| \leq |a_n - \ell| + |a_m - \ell| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Hence (a_n) is a Cauchy sequence. ☺

Theorem 12.1.2

If $(a_n)_{n \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}$, then it converges to some real number.

Note:-

This theorem expresses the completeness of $\mathbb{R}$. It is not true in all metric spaces.

Example 12.1.1 (Bounded but Non-Convergent Sequence)

Consider a bounded sequence (a_n) in $\mathbb{R}$. By the Bolzano–Weierstrass Theorem, (a_n) has a convergent subsequence. However, the sequence itself need not converge.

Note:-

Thus, boundedness alone does not guarantee convergence; the Cauchy property is the correct criterion in $\mathbb{R}$.

Theorem 12.1.3

Every Cauchy sequence in $\mathbb{R}$ converges.

Proof: We prove this in three steps.

Step 1: Cauchy $\implies$ bounded.

Let (a_n) be Cauchy. Choose $\epsilon = 1$. Then there exists $N_1 \in \mathbb{N}$ such that

$$n, m > N_1 \implies |a_n - a_m| < 1.$$

In particular, for $n > N_1$,

$$|a_n| = |a_n - a_{N_1+1} + a_{N_1+1}| \leq |a_n - a_{N_1+1}| + |a_{N_1+1}| < 1 + |a_{N_1+1}|.$$

Define

$$B = \max\{|a_1|, |a_2|, \dots, |a_{N_1+1}|, |a_{N_1+1}| + 1\}.$$

Then $|a_n| \leq B$ for all n , so (a_n) is bounded.

Step 2: Bounded $\implies$ existence of a convergent subsequence.

Since (a_n) is bounded, by the Bolzano–Weierstrass Theorem there exists a subsequence (a_{n_k}) that converges to some $L \in \mathbb{R}$.

Step 3: Cauchy + convergent subsequence $\implies$ convergence of the whole sequence.

Let $\epsilon > 0$. Since $(a_{n_k}) \rightarrow L$, there exists K such that

$$k > K \implies |a_{n_k} - L| < \epsilon/2.$$

Since (a_n) is Cauchy, there exists N_2 such that

$$n, m > N_2 \implies |a_n - a_m| < \epsilon/2.$$

Let

$$N = \max\{N_2, K\}.$$

Then for $n > N$,

$$\begin{aligned} |a_n - L| &= |a_n - a_{n_k} + a_{n_k} - L| \\ &\leq \underbrace{|a_n - a_{n_k}|}_{\text{Cauchy } (<\epsilon/2)} + \underbrace{|a_{n_k} - L|}_{\text{Subsequence convergence } (<\epsilon/2)} \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon. \end{aligned}$$

Hence $(a_n) \rightarrow L$.



With the two theorems, we now get that a sequence is Cauchy iff it's convergent

12.2 Infinite Series

Definition 12.2.1: Infinite Series

Let $(a_n)_{n=1}^{\infty}$ be a sequence in $\mathbb{R}$. The n th partial sum of the sequence is

$$s_n := a_1 + a_2 + \cdots + a_n.$$

Definition 12.2.2: Summable Sequence

A sequence $(a_n)_{n=1}^{\infty}$ is called *summable* if the sequence of partial sums (s_n) converges to a real number $L \in \mathbb{R}$, i.e.

$$\lim_{n \rightarrow \infty} s_n = L.$$

In this case, we write

$$\sum_{n=1}^{\infty} a_n := \lim_{n \rightarrow \infty} s_n.$$

Question 2

If we rearrange a summable sequence, is it still summable?

In general, the answer is no.

Example 12.2.1 (Non-summable Sequences)

1. Let $a_n = c$ for all n , where $c \neq 0$. The partial sums satisfy $s_n = nc \rightarrow \pm\infty$, so the sequence is not summable.
2. Let $a_n = (-1)^n$. The partial sums alternate between 0 and -1 , so $\lim s_n$ does not exist.

Example 12.2.2

Although $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$, the sequence $(\frac{1}{n})$ is not summable.

Indeed,

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

diverges. Grouping terms,

$$\left(\frac{1}{2} + \frac{1}{3}\right) + \left(\frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7}\right) + \left(\frac{1}{8} + \cdots + \frac{1}{15}\right) + \cdots$$

Each group has sum at least $\frac{1}{2}$, so the partial sums are unbounded.

Proposition 12.2.1

If $(a_n)_{n=1}^{\infty}$ is summable and $c \in \mathbb{R}$, then $(ca_n)_{n=1}^{\infty}$ is summable and

$$\sum_{n=1}^{\infty} ca_n = c \sum_{n=1}^{\infty} a_n.$$

Proof: Let $s_n = \sum_{k=1}^n a_k$. Then the partial sums of (ca_n) are

$$\tilde{s}_n = \sum_{k=1}^n ca_k = cs_n.$$

Since $s_n \rightarrow L$, we have $\tilde{s}_n \rightarrow cL$, so (ca_n) is summable. ☺

Proposition 12.2.2

If (a_n) and (b_n) are summable, then $(a_n + b_n)$ is summable and

$$\sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n.$$

Theorem 12.2.1 Convergent geometric series

Let $|r| < 1$. Then the geometric series

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}.$$

Proof:

$$\begin{aligned}s_n &= 1 + r + \cdots + r^n \\ r s_n &= r + r^2 + \cdots + r^n + r^{n+1} \\ s_n(1 - r) &= 1 - r^{n+1} \\ s_n &= \frac{1 - r^{n+1}}{1 - r}\end{aligned}$$



Question 3

Find

$$\sum_{n=3}^{\infty} 7 \left(\frac{1}{x} \right)^n.$$

Solution: Method 1:

$$7 + \frac{7}{x} + \frac{7}{x^2} + \sum_{n=3}^{\infty} 7 \left(\frac{1}{x} \right)^n = \sum_{n=0}^{\infty} 7 \left(\frac{1}{x} \right)^n = \frac{7}{1 - \frac{1}{x}} = \frac{7x}{x - 1}.$$

Hence,

$$\sum_{n=3}^{\infty} 7 \left(\frac{1}{x} \right)^n = \frac{7x}{x - 1} - 7 - \frac{7}{x} - \frac{7}{x^2}.$$

Method 2:

$$\sum_{n=3}^{\infty} 7 \left(\frac{1}{x} \right)^n = 7 \left(\frac{1}{x} \right)^3 \sum_{m=0}^{\infty} \left(\frac{1}{x} \right)^m = \frac{7}{x^3} \cdot \frac{1}{1 - \frac{1}{x}} = \frac{7}{x^2(x - 1)}.$$

Theorem 12.2.2 Cauchy Criterion for Series

A sequence $(a_n)_{n=1}^{\infty}$ is summable if and only if

$$\lim_{n, m \rightarrow \infty} (a_{n+1} + a_{n+2} + \cdots + a_m) = 0.$$

Proof: $(\Rightarrow)$ If (a_n) is summable, then its partial sums (s_n) converge, hence are Cauchy:

$$\forall \epsilon > 0, \exists N \text{ s.t. } m, n > N \Rightarrow |s_m - s_n| < \epsilon.$$

But

$$s_m - s_n = a_{n+1} + \cdots + a_m.$$

$(\Leftarrow)$ If the tail sums converge to 0, then (s_n) is Cauchy, hence convergent in $\mathbb{R}$.



Corollary 12.2.1

If $(a_n)_{n=1}^{\infty}$ is summable, then

$$\lim_{n \rightarrow \infty} a_n = 0.$$

Theorem 12.2.3 The vanishing condition

If $\{a_n\}$ is summable, then

$$\lim_{n \rightarrow \infty} a_n = 0$$

This condition follows from Cauchy criterion by taking $m = n + 1$. An alternative proof:

Proof: If $\lim_{n \rightarrow \infty} s_n = l$, then

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (s_n - s_{n-1}) = \lim_{n \rightarrow \infty} s_n - \lim_{n \rightarrow \infty} s_{n-1} = l - l = 0$$

☺

However the condition is not sufficient, for example $\lim_{n \rightarrow \infty} 1/n = 0$, but the sequence $\{1/n\}$ is not summable, as shown previously.

Theorem 12.2.4 The boundedness criterion

A nonnegative sequence $\{a_n\}$ is summable iff the set of partial sums s_n is bounded.

Proof: Assume throughout that $a_n \geq 0$ for all n . Then the partial sums satisfy

$$s_{n+1} - s_n = a_{n+1} \geq 0,$$

so $\{s_n\}$ is increasing.

($\Rightarrow$) If $\{a_n\}$ is summable, then by definition the sequence of partial sums $\{s_n\}$ converges to some real number $L \in \mathbb{R}$. Every convergent sequence is bounded, hence $\{s_n\}$ is bounded.

($\Leftarrow$) Conversely, suppose the set $\{s_n : n \in \mathbb{N}\}$ is bounded above. Since $\{s_n\}$ is increasing, it has a least upper bound

$$L := \sup\{s_n : n \in \mathbb{N}\} \in \mathbb{R}.$$

We claim that $s_n \rightarrow L$. Let $\epsilon > 0$. By the definition of supremum, $L - \epsilon$ is not an upper bound for $\{s_n\}$, so there exists $N \in \mathbb{N}$ such that

$$s_N > L - \epsilon.$$

For all $n \geq N$, monotonicity gives $s_n \geq s_N > L - \epsilon$, and also $s_n \leq L$ since L is an upper bound. Therefore, for all $n \geq N$,

$$|s_n - L| = L - s_n < \epsilon.$$

Hence $s_n \rightarrow L$, so the partial sums converge and $\{a_n\}$ is summable.

☺

Theorem 12.2.5 Comparison test

Suppose

$$0 \leq a_n \leq b_n \quad \forall n$$

then if $\sum_{n=1}^{\infty} b_n$ converges, so does $\sum_{n=1}^{\infty} a_n$.

Proof: If

$$s_n = a_1 + \dots + a_n$$

$$t_n = b_1 + \dots + b_n$$

then

$$0 \leq s_n \leq t_n \quad \forall n$$

Now $\{t_n\}$ is bounded, since $\sum_{n=1}^{\infty} b_n$ converges. Therefore $\{s_n\}$ is bounded; consequently, by boundedness criterion, $\sum_{n=1}^{\infty} a_n$ converges. $\odot$

Theorem 12.2.6 Limit Comparison Test

If $a_n, b_n > 0$ and $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c \neq 0$, then $\sum_{n=1}^{\infty} a_n$ converges iff $\sum_{n=1}^{\infty} b_n$ converges.

Proof: Suppose $\sum_{n=1}^{\infty} b_n$ converges. Since $\lim_{n \rightarrow \infty} a_n/b_n = c$, there is some N such that

$$a_n \leq 2cb_n \quad \text{for } n \geq N$$

But the sequence $2c \sum_{n=N}^{\infty} b_n$ converges. Then the comparison test shows that $\sum_{n=N}^{\infty} a_n$ converges, and this implies convergence of the whole series $\sum_{n=1}^{\infty} a_n$, which has only finitely many additional terms. The converse follows immediately, since we also have $\lim_{n \rightarrow \infty} b_n/a_n = 1/c \neq 0$. $\odot$

Theorem 12.2.7 The Ratio Test

Let $a_n > 0$ for all n , and suppose that

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = r$$

Then $\sum_{n=1}^{\infty} a_n$ converges if $r < 1$. On the other hand, if $r > 1$, then the term a_n are unbounded, so $\sum_{n=1}^{\infty} a_n$ diverges.

Proof: Suppose first that $r < 1$, choose any number s with $r < s < 1$. The hypothesis

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = r < 1$$

implies that there is some N such that

$$\frac{a_{n+1}}{a_n} \leq s \quad \text{for } n \geq N$$

This can be written

$$a_{n+1} \leq sa_n \quad \text{for } n \geq N$$

Thus

$$\begin{aligned} a_{N+1} &\leq s a_N \\ a_{N+2} &\leq s a_{N+1} \leq s^2 a_N \\ &\vdots \\ a_{N+k} &\leq s^k a_N \end{aligned}$$

Since $\sum_{k=0}^{\infty} a_N s^k = a_N \sum_{k=0}^{\infty} s^k$ converges, the comparison test shows that

$$\sum_{n=N}^{\infty} a_n = \sum_{k=0}^{\infty} a_{N+k}$$

converges. This implies the convergence of the whole series $\sum_{n=1}^{\infty} a_n$.
The case $r > 1$ is easier. If $1 < s < r$, then there is number N such that

$$\frac{a_{n+1}}{a_n} \geq s \quad \text{for } n \geq N$$

which means that

$$a_{N+k} \geq a_N s^k \quad k = 0, 1, \dots$$

so that the terms are unbounded



Definition 12.2.3: Absolutely convergent

The series $\sum_{n=1}^{\infty} a_n$ is **absolutely convergent** if the series $\sum_{n=1}^{\infty} |a_n|$ is convergent. (The sequence $\{a_n\}$ is **absolutely summable** if the sequence $\{|a_n|\}$ is summable)

Theorem 12.2.8 Absolute convergence implies convergence

If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

Proof: Since $\sum |a_n|$ converges, for every $\varepsilon > 0$ there exists N such that for all $m > n \geq N$,

$$\sum_{k=n+1}^m |a_k| < \varepsilon.$$

By the triangle inequality,

$$\left| \sum_{k=n+1}^m a_k \right| \leq \sum_{k=n+1}^m |a_k| < \varepsilon.$$

By the Cauchy criterion, $\sum a_n$ converges.



Definition 12.2.4: Conditional convergence

A series $\sum_{n=1}^{\infty} a_n$ is **conditionally convergent** if it converges but does not converge absolutely (i.e., $\sum a_n$ converges but $\sum |a_n|$ diverges).

Example 12.2.3 (The harmonic series diverges)

The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges. To see this, group terms:

$$1 + \frac{1}{2} + \underbrace{\frac{1}{3} + \frac{1}{4}}_{\geq 1/2} + \underbrace{\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}}_{\geq 1/2} + \cdots$$

Each group of 2^{k-1} terms (from $\frac{1}{2^{k-1}+1}$ to $\frac{1}{2^k}$) has sum $\geq \frac{1}{2}$, so the partial sums are unbounded.

Theorem 12.2.9 Alternating Series Test (Leibniz)

If (a_n) is a decreasing sequence of positive numbers with $\lim_{n \rightarrow \infty} a_n = 0$, then the alternating series

$$\sum_{n=1}^{\infty} (-1)^{n+1} a_n = a_1 - a_2 + a_3 - a_4 + \cdots$$

converges.

Proof: The even partial sums s_{2n} are increasing: $s_{2n+2} = s_{2n} + (a_{2n+1} - a_{2n+2}) \geq s_{2n}$ since $a_{2n+1} \geq a_{2n+2}$. They are also bounded above: $s_{2n} = a_1 - (a_2 - a_3) - (a_4 - a_5) - \cdots \leq a_1$. So s_{2n} converges to some limit s . Since $s_{2n+1} = s_{2n} + a_{2n+1}$ and $a_{2n+1} \rightarrow 0$, we get $s_{2n+1} \rightarrow s$ as well. Hence $s_n \rightarrow s$. ☺

Example 12.2.4 (The alternating harmonic series converges conditionally)

The series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$ converges by the alternating series test (since $1/n$ is decreasing to 0). But the harmonic series $\sum \frac{1}{n}$ diverges, so the convergence is conditional.

Chapter 13

Week 2

Note:-

Additional practice on Riemann integration and the Fundamental Theorem of Calculus can be found in Chapter 23, Section 23.7.

13.1 Riemann Sums

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function.

Partitions

A *partition* of $[a, b]$ is a finite set of points

$$P = \{t_0, t_1, \dots, t_n\}$$

such that

$$a = t_0 < t_1 < \dots < t_n = b.$$

Infimum and Supremum on Subintervals

For each subinterval $[t_{i-1}, t_i]$, define

$$m_i := \inf\{f(x) \mid t_{i-1} \leq x \leq t_i\}, \quad M_i := \sup\{f(x) \mid t_{i-1} \leq x \leq t_i\}.$$

Geometrically, the rectangle of height m_i over $[t_{i-1}, t_i]$ has area

$$m_i(t_i - t_{i-1}),$$

and the rectangle of height M_i over $[t_{i-1}, t_i]$ has area

$$M_i(t_i - t_{i-1}).$$

By definition,

$$m_i(t_i - t_{i-1}) \leq f(x)(t_i - t_{i-1}) \leq M_i(t_i - t_{i-1}) \quad \text{for all } x \in [t_{i-1}, t_i].$$

Lower and Upper Sums

For this partition P , define the *lower sum* of f with respect to P by

$$L(f, P) := \sum_{i=1}^n m_i(t_i - t_{i-1}),$$

and the *upper sum* of f with respect to P by

$$U(f, P) := \sum_{i=1}^n M_i(t_i - t_{i-1}).$$

Clearly,

$$L(f, P) \leq U(f, P).$$

Extremal Values over All Partitions

Consider all possible partitions P of $[a, b]$. Then

$$\inf\{U(f, P) \mid P \text{ is a partition of } [a, b]\}$$

and

$$\sup\{L(f, P) \mid P \text{ is a partition of } [a, b]\}$$

are well-defined real numbers.

Definition 13.1.1: Formal Definition

Assume f is bounded on $[a, b]$ and let

$$P = \{t_0, t_1, \dots, t_n\}$$

be a partition of $[a, b]$ with

$$a = t_0 < t_1 < \dots < t_n = b.$$

Define

$$m_i := \inf\{f(x) \mid x \in [t_{i-1}, t_i]\}, \quad M_i := \sup\{f(x) \mid x \in [t_{i-1}, t_i]\}.$$

Then the lower sum of f for P is

$$L(f, P) := \sum_{i=1}^n m_i(t_i - t_{i-1}),$$

and the upper sum of f for P is

$$U(f, P) := \sum_{i=1}^n M_i(t_i - t_{i-1}).$$

Proposition 13.1.1

For any bounded function f and any partition P of $[a, b]$,

$$L(f, P) \leq U(f, P).$$

Proof: By definition, for all $x \in [t_{i-1}, t_i]$,

$$m_i \leq f(x) \leq M_i.$$

Multiplying by $(t_i - t_{i-1})$ and summing over $i = 1, \dots, n$ gives

$$\sum_{i=1}^n m_i(t_i - t_{i-1}) \leq \sum_{i=1}^n M_i(t_i - t_{i-1}),$$

which is exactly $L(f, P) \leq U(f, P)$. ☺

Definition 13.1.2: Lower and Upper Integrals

The **lower integral** of f over $[a, b]$ is defined by

$$\underline{\int_a^b} f := \sup_P L(f, P),$$

and the **upper integral** of f over $[a, b]$ is defined by

$$\overline{\int_a^b} f := \inf_P U(f, P),$$

where the supremum and infimum are taken over all partitions P of $[a, b]$.

Definition 13.1.3: Riemann Integrability

A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is called **Riemann integrable** if

$$\underline{\int_a^b} f = \overline{\int_a^b} f.$$

In this case, their common value is denoted by

$$\int_a^b f(x) dx.$$

Example 13.1.1 (Constant Function)

Let $f(x) = c$ on $[a, b]$, where $c > 0$. For any partition P ,

$$m_i = M_i = c.$$

Hence

$$L(f, P) = U(f, P) = \sum_{i=1}^n c(t_i - t_{i-1}) = c(b - a).$$

Therefore,

$$\int_a^b f(x) dx = c(b - a).$$

Example 13.1.2 (Dirichlet Function)

Define

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

On every subinterval $[t_{i-1}, t_i]$,

$$m_i = 0, \quad M_i = 1.$$

Thus for every partition P ,

$$L(f, P) = 0, \quad U(f, P) = b - a.$$

Since the lower and upper integrals are not equal, f is not Riemann integrable.

Example 13.1.3 (Linear Function)

Let $f(x) = x$ on $[a, b]$ with $a < b$. Let $P = \{t_0, \dots, t_n\}$ be a partition of $[a, b]$. Since f is increasing,

$$m_i = t_{i-1}, \quad M_i = t_i.$$

Thus

$$L(f, P) = \sum_{i=1}^n t_{i-1}(t_i - t_{i-1}), \quad U(f, P) = \sum_{i=1}^n t_i(t_i - t_{i-1}).$$

For a uniform partition with $\Delta t = \frac{b-a}{n}$ and $t_i = a + i\Delta t$, one obtains

$$\lim_{n \rightarrow \infty} L(f, P) = \lim_{n \rightarrow \infty} U(f, P) = \frac{1}{2}(b^2 - a^2).$$

Hence

$$\int_a^b x dx = \frac{1}{2}(b^2 - a^2).$$

Theorem 13.1.1

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. Then f is Riemann integrable on $[a, b]$ iff

$$\forall \varepsilon > 0, \exists \text{ a partition } P_\varepsilon \text{ of } [a, b] \text{ such that } U(f, P_\varepsilon) - L(f, P_\varepsilon) < \varepsilon.$$

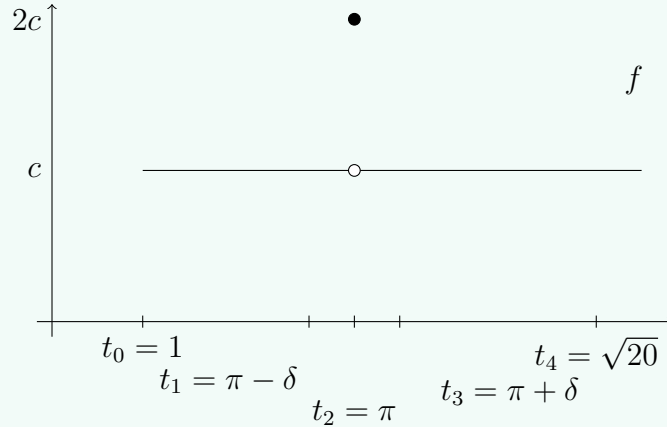
Since $U(f, P) \geq L(f, P)$ for every partition P , this is equivalent to

$$|U(f, P_\varepsilon) - L(f, P_\varepsilon)| < \varepsilon.$$

Example 13.1.4 (A Function with One “Bad” Point is Integrable)

Fix $c > 0$. Define $f : [1, \sqrt{20}] \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} c, & x \neq \pi, \\ 2c, & x = \pi. \end{cases}$$



Claim: f is Riemann integrable and

$$\int_1^{\sqrt{20}} f = c(\sqrt{20} - 1).$$

Let $\delta > 0$ be such that $\pi - \delta \geq 1$ and $\pi + \delta \leq \sqrt{20}$, and take the partition

$$P_\delta = \{t_0, t_1, t_2, t_3, t_4\} = \{1, \pi - \delta, \pi, \pi + \delta, \sqrt{20}\}.$$

On each subinterval not containing π , we have $m_i = M_i = c$. On the two subintervals containing π , we have $m_i = c$ and $M_i = 2c$.

Thus

$$L(f, P_\delta) = c((\pi - \delta) - 1) + c(\pi - (\pi - \delta)) + c((\pi + \delta) - \pi) + c(\sqrt{20} - (\pi + \delta)),$$

so

$$L(f, P_\delta) = c(\sqrt{20} - 1).$$

Also

$$U(f, P_\delta) = c((\pi - \delta) - 1) + 2c(\pi - (\pi - \delta)) + 2c((\pi + \delta) - \pi) + c(\sqrt{20} - (\pi + \delta)).$$

Therefore,

$$U(f, P_\delta) - L(f, P_\delta) = c\delta + c\delta = 2c\delta.$$

Given $\varepsilon > 0$, choose $\delta < \frac{\varepsilon}{2c}$. Then

$$U(f, P_\delta) - L(f, P_\delta) = 2c\delta < \varepsilon,$$

so f is Riemann integrable. Since $L(f, P_\delta) = c(\sqrt{20} - 1)$ for this partition, we get

$$\int_1^{\sqrt{20}} f = c(\sqrt{20} - 1).$$

Chapter 14

Week 3

Note:-

Additional practice on Riemann integration and integrability criteria can be found in Chapter 23, Section 23.7.

14.1 A Criterion for Riemann Integrability

Theorem 14.1.1

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. Then f is Riemann integrable on $[a, b]$ if and only if

$$\forall \varepsilon > 0, \exists \text{ a partition } P_\varepsilon \text{ of } [a, b]$$

such that

$$U(f, P_\varepsilon) - L(f, P_\varepsilon) < \varepsilon.$$

Proof: ($\Leftarrow$) Suppose that for every $\varepsilon > 0$ there exists a partition P such that

$$U(f, P) - L(f, P) < \varepsilon.$$

Then

$$\inf\{U(f, P') \mid P' \text{ a partition}\} - \sup\{L(f, P') \mid P' \text{ a partition}\} \leq U(f, P) - L(f, P) < \varepsilon.$$

Since this holds for all $\varepsilon > 0$, we obtain

$$\inf\{U(f, P)\} = \sup\{L(f, P)\}.$$

By definition, f is Riemann integrable.

($\Rightarrow$) Now suppose f is Riemann integrable, so that

$$\inf\{U(f, P)\} = \sup\{L(f, P)\}.$$

Let $\varepsilon > 0$. Then there exist partitions P' and P'' such that

$$U(f, P'') - \sup\{L(f, P)\} < \frac{\varepsilon}{2}, \quad \inf\{U(f, P)\} - L(f, P') < \frac{\varepsilon}{2}.$$

Let $P = P' \cup P''$. By the lemma below,

$$U(f, P) \leq U(f, P''), \quad L(f, P) \geq L(f, P').$$

Hence

$$U(f, P) - L(f, P) \leq U(f, P'') - L(f, P') < \varepsilon.$$

This completes the proof. ☺

Lemma 14.1.1

Let P and Q be partitions of $[a, b]$ with Q containing P . Then

$$L(f, P) \leq L(f, Q) \quad \text{and} \quad U(f, P) \geq U(f, Q).$$

Proof: It suffices to consider the case where Q is obtained from P by adding a single point.

Let

$$P = \{t_0, t_1, \dots, t_n\}, \quad a = t_0 < \dots < t_n = b,$$

and let $u \in (t_{k-1}, t_k)$. Define

$$Q = \{t_0, \dots, t_{k-1}, u, t_k, \dots, t_n\}.$$

Lower sums. Let

$$m_k = \inf\{f(x) \mid t_{k-1} \leq x \leq t_k\},$$

$$m'_k = \inf\{f(x) \mid t_{k-1} \leq x \leq u\}, \quad m''_k = \inf\{f(x) \mid u \leq x \leq t_k\}.$$

Then

$$L(f, P) = \sum_{i=1}^n m_i(t_i - t_{i-1}),$$

and

$$L(f, Q) = \sum_{i=1}^{k-1} m_i(t_i - t_{i-1}) + m'_k(u - t_{k-1}) + m''_k(t_k - u) + \sum_{i=k+1}^n m_i(t_i - t_{i-1}).$$

Since

$$\{f(x) \mid t_{k-1} \leq x \leq t_k\} \supset \{f(x) \mid t_{k-1} \leq x \leq u\}, \quad \{f(x) \mid t_{k-1} \leq x \leq t_k\} \supset \{f(x) \mid u \leq x \leq t_k\},$$

we have

$$m_k \leq m'_k, \quad m_k \leq m''_k.$$

Therefore,

$$m_k(t_k - t_{k-1}) = m_k(u - t_{k-1}) + m_k(t_k - u) \leq m'_k(u - t_{k-1}) + m''_k(t_k - u),$$

which implies

$$L(f, P) \leq L(f, Q).$$

Upper sums. Let

$$M_k = \sup\{f(x) \mid t_{k-1} \leq x \leq t_k\},$$

$$M'_k = \sup\{f(x) \mid t_{k-1} \leq x \leq u\}, \quad M''_k = \sup\{f(x) \mid u \leq x \leq t_k\}.$$

Then

$$U(f, P) = \sum_{i=1}^n M_i(t_i - t_{i-1}),$$

and

$$U(f, Q) = \sum_{i=1}^{k-1} M_i(t_i - t_{i-1}) + M'_k(u - t_{k-1}) + M''_k(t_k - u) + \sum_{i=k+1}^n M_i(t_i - t_{i-1}).$$

Since

$$\{f(x) \mid t_{k-1} \leq x \leq u\} \subset \{f(x) \mid t_{k-1} \leq x \leq t_k\}, \quad \{f(x) \mid u \leq x \leq t_k\} \subset \{f(x) \mid t_{k-1} \leq x \leq t_k\},$$

we have

$$M'_k \leq M_k, \quad M''_k \leq M_k.$$

Therefore,

$$U(f, Q) \leq \sum_{i=1}^{k-1} M_i(t_i - t_{i-1}) + M_k(u - t_{k-1}) + M_k(t_k - u) + \sum_{i=k+1}^n M_i(t_i - t_{i-1}) = U(f, P),$$

so

$$U(f, P) \geq U(f, Q).$$

Thus, adding one point increases lower sums and decreases upper sums. Since any refinement can be obtained by adding points one at a time, the result holds for general $Q \supset P$. ☺

Theorem 14.1.2 Additivity of the Riemann Integral

Let $a < c < b$.

1. If f is integrable on $[a, b]$, then f is integrable on $[a, c]$ and on $[c, b]$.
2. If f is integrable on $[a, c]$ and on $[c, b]$, then f is integrable on $[a, b]$.
3. If f is integrable on $[a, b]$, then

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

Proof: First suppose that f is integrable on $[a, b]$. Let $\varepsilon > 0$. Then there exists a partition

$$P = \{t_0, \dots, t_n\}$$

of $[a, b]$ such that

$$U(f, P) - L(f, P) < \varepsilon.$$

We may assume that $c = t_j$ for some j . Otherwise, let Q be the partition obtained by adding c to P . Since Q contains P , we have

$$U(f, Q) - L(f, Q) \leq U(f, P) - L(f, P) < \varepsilon.$$

Define

$$P' = \{t_0, \dots, t_j\}, \quad P'' = \{t_j, \dots, t_n\}.$$

Then P' is a partition of $[a, c]$ and P'' is a partition of $[c, b]$. Moreover,

$$L(f, P) = L(f, P') + L(f, P''), \quad U(f, P) = U(f, P') + U(f, P'').$$

Hence

$$[U(f, P') - L(f, P')] + [U(f, P'') - L(f, P'')] = U(f, P) - L(f, P) < \varepsilon.$$

Since both terms are nonnegative, each is less than ε . This shows that f is integrable on $[a, c]$ and on $[c, b]$.

Note also that

$$L(f, P') \leq \int_a^c f \leq U(f, P'), \quad L(f, P'') \leq \int_c^b f \leq U(f, P'').$$

Adding these inequalities gives

$$L(f, P) \leq \int_a^c f + \int_c^b f \leq U(f, P).$$

Since this holds for all partitions P , we conclude that

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

Now suppose that f is integrable on $[a, c]$ and on $[c, b]$. Let $\varepsilon > 0$. Then there exist partitions P' of $[a, c]$ and P'' of $[c, b]$ such that

$$U(f, P') - L(f, P') < \frac{\varepsilon}{2}, \quad U(f, P'') - L(f, P'') < \frac{\varepsilon}{2}.$$

Let P be the partition of $[a, b]$ consisting of all points of P' and P'' . Then

$$L(f, P) = L(f, P') + L(f, P''), \quad U(f, P) = U(f, P') + U(f, P'').$$

Therefore,

$$U(f, P) - L(f, P) = [U(f, P') - L(f, P')] + [U(f, P'') - L(f, P'')] < \varepsilon.$$

Hence f is integrable on $[a, b]$. ☺

Note:-

Using this theorem, we extend the definition of the integral by setting

$$\int_a^a f = 0, \quad \int_a^b f = -\int_b^a f \quad \text{if } a > b.$$

With these conventions,

$$\int_a^c f + \int_c^b f = \int_a^b f$$

holds for all real numbers a, b, c .

The following are notes during lecture:

14.2 Properties of the Integral

Theorem 14.2.1 Positivity

If f is integrable on $[a, b]$ and $f(x) \geq 0$ for all $x \in [a, b]$, then

$$\int_a^b f \geq 0.$$

Proof: Since $f(x) \geq 0$, we have $m_i = \inf\{f(x) \mid x \in [t_{i-1}, t_i]\} \geq 0$ for every subinterval. Thus $L(f, P) \geq 0$ for every partition P . Since $\int_a^b f \geq L(f, P)$ for all P , we get $\int_a^b f \geq 0$. $\odot$

Corollary 14.2.1

If f and g are integrable on $[a, b]$ and $f(x) \leq g(x)$ for all $x \in [a, b]$, then

$$\int_a^b f \leq \int_a^b g.$$

Theorem 14.2.2 Linearity of the Integral

If f and g are integrable on $[a, b]$ and $c \in \mathbb{R}$, then $f + g$ and cf are integrable on $[a, b]$, and

$$\int_a^b (f + g) = \int_a^b f + \int_a^b g, \quad \int_a^b cf = c \int_a^b f.$$

Proof: For any partition P ,

$$\inf_{[t_{i-1}, t_i]} f + \inf_{[t_{i-1}, t_i]} g \leq \inf_{[t_{i-1}, t_i]} (f + g),$$

so $L(f, P) + L(g, P) \leq L(f + g, P)$. Similarly, $U(f + g, P) \leq U(f, P) + U(g, P)$. Given $\varepsilon > 0$, choose partitions P_1, P_2 with $U(f, P_1) - L(f, P_1) < \varepsilon/2$ and $U(g, P_2) - L(g, P_2) < \varepsilon/2$. Let $P = P_1 \cup P_2$. Then

$$U(f + g, P) - L(f + g, P) \leq [U(f, P) - L(f, P)] + [U(g, P) - L(g, P)] < \varepsilon,$$

so $f + g$ is integrable. The equality follows since for all P :

$$L(f, P) + L(g, P) \leq \int_a^b f + \int_a^b g \leq U(f, P) + U(g, P),$$

and $\int_a^b (f + g)$ is the unique number squeezed between $L(f + g, P)$ and $U(f + g, P)$. The proof for cf is similar. $\odot$

Theorem 14.2.3 Products of Integrable Functions are Integrable

If f and g are integrable on $[a, b]$, then fg is integrable on $[a, b]$.

Proof: First note that if f is integrable, then f^2 is integrable. Since f is bounded, say $|f(x)| \leq M$, we have

$$\sup_{[t_{i-1}, t_i]} f^2 - \inf_{[t_{i-1}, t_i]} f^2 \leq 2M \left(\sup_{[t_{i-1}, t_i]} f - \inf_{[t_{i-1}, t_i]} f \right).$$

So $U(f^2, P) - L(f^2, P) \leq 2M[U(f, P) - L(f, P)]$, and integrability of f^2 follows.

Now use the identity

$$fg = \frac{1}{4} [(f + g)^2 - (f - g)^2].$$

Since $f + g$ and $f - g$ are integrable (by linearity), $(f + g)^2$ and $(f - g)^2$ are integrable, and hence fg is integrable. $\odot$

Corollary 14.2.2

If f is bounded on $[a, b]$ and has only finitely many jump discontinuities and removable discontinuities, then f is integrable on $[a, b]$.

Proof: We know that f is integrable on $[a, b]$ if and only if for every $\varepsilon > 0$ there exists a partition P_ε such that

$$U(f, P_\varepsilon) - L(f, P_\varepsilon) < \varepsilon.$$

Let

$$P_\varepsilon = \{t_0, t_1, \dots, t_n\}$$

be a partition of $[a, b]$. Then

$$U(f, P_\varepsilon) - L(f, P_\varepsilon) = \sum_{i=1}^n M_i(t_i - t_{i-1}) - \sum_{i=1}^n m_i(t_i - t_{i-1}) = \sum_{i=1}^n (M_i - m_i)(t_i - t_{i-1}),$$

where

$$M_i = \sup\{f(x) \mid x \in [t_{i-1}, t_i]\}, \quad m_i = \inf\{f(x) \mid x \in [t_{i-1}, t_i]\}.$$

$\odot$

Theorem 14.2.4 Continuous implies integrable

Prove that if f is continuous on $[a, b]$, then f is integrable on $[a, b]$.

Note:-

Wrong Proof Attempt

Fix $\varepsilon > 0$ and construct a partition P as follows.

Let $t_0 = a$. Since f is continuous at t_0 , there exists $\delta > 0$ such that

$$x \in (t_0, t_0 + \delta) \implies |f(x) - f(t_0)| < \frac{\varepsilon}{2}.$$

Define

$$t_1 = t_0 + \frac{1}{2}\delta.$$

Since f is continuous on $[t_0, t_1]$, there exist points

$$x_{\max}, x_{\min} \in [t_0, t_1]$$

such that

$$M_1 = f(x_{\max}), \quad m_1 = f(x_{\min}).$$

Then

$$|M_1 - m_1| = |f(x_{\max}) - f(x_{\min})| \leq |f(x_{\max}) - f(t_0)| + |f(x_{\min}) - f(t_0)| < \varepsilon$$

by the triangle inequality.

Repeat this construction inductively. After n steps, we obtain a partition

$$P = \{t_0, t_1, \dots, t_n\}$$

such that

$$[t_n - \delta, t_n + \delta] \supset [t_n, b],$$

and hence b is added to the partition.

Then

$$U(f, P) - L(f, P) = \sum_{i=1}^{n+1} (M_i - m_i)(t_i - t_{i-1}) < \sum_{i=1}^{n+1} \varepsilon \delta = (n+1)\varepsilon\delta.$$

Note:-

Why the Above Proof Is Wrong

The mistake occurs in the final estimate

$$U(f, P) - L(f, P) < (n+1)\varepsilon\delta.$$

The error is *not* algebraic, but conceptual.

1. **The number of subintervals is not controlled.** Although each interval has length less than δ , the number of intervals satisfies

$$n \approx \frac{b-a}{\delta}.$$

Thus the bound

$$(n+1)\varepsilon\delta$$

does not necessarily tend to 0 as $\delta \rightarrow 0$.

2. **Small terms do not imply a small sum.** The argument assumes that making each factor $(M_i - m_i)$ small is sufficient. However, the sum involves *many* such terms, and their number grows as the partition is refined.
3. **Continuity at points is insufficient.** The construction uses continuity at individual points to choose δ , but does not produce a *single uniform bound* on $M_i - m_i$ that works across all subintervals.

The correct proof avoids this problem by using *uniform continuity* of f on $[a, b]$, which guarantees the existence of $\delta > 0$ such that

$$M_i - m_i < \frac{\varepsilon}{b-a} \quad \text{for all } i.$$

Then

$$U(f, P) - L(f, P) < \frac{\varepsilon}{b-a} \sum_{i=1}^n (t_i - t_{i-1}) = \varepsilon,$$

since $\sum_{i=1}^n (t_i - t_{i-1}) = b - a$.

14.3 Uniform Continuity

Definition 14.3.1: Uniform Continuity

A function $f : A \rightarrow \mathbb{R}$ is said to be *uniformly continuous* on a set A if

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } x, y \in A \text{ and } |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

Note:-

Translation

There exists a single $\delta > 0$ such that for every $a \in A$,

$$x \in A \text{ and } |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

The same δ works for all points $a \in A$.

Example 14.3.1

The function $f(x) = x^2$ is continuous on $\mathbb{R}$, but not uniformly continuous on $\mathbb{R}$.

Indeed,

$$|f(x) - f(a)| = |x^2 - a^2| = |x - a||x + a|.$$

If $|x - a| < 1$, then

$$|f(x) - f(a)| < |x - a|(2|a| + 1).$$

Thus to guarantee $|f(x) - f(a)| < \varepsilon$, we must choose

$$\delta = \frac{\varepsilon}{2|a| + 1},$$

which depends on a . Hence f is not uniformly continuous on $\mathbb{R}$.

Example 14.3.2

The function $f(x) = \frac{1}{x}$ is continuous on $(0, 1]$, but not uniformly continuous on $(0, 1]$.

14.4 Uniform Continuity Implies Integrability

Theorem 14.4.1

If f is continuous on $[a, b]$, then f is Riemann integrable on $[a, b]$.

Proof: Let $\varepsilon > 0$. Since f is continuous on $[a, b]$, it is uniformly continuous on $[a, b]$. Thus there exists $\delta > 0$ such that

$$x, y \in [a, b], |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

Let P_ε be a *uniform partition* of $[a, b]$ with

$$\Delta t = \frac{b - a}{n} < \delta, \quad t_i = a + i\Delta t, \quad i = 0, \dots, n.$$

For each subinterval $[t_{i-1}, t_i]$, let

$$M_i = f(x_i^{\max}), \quad m_i = f(x_i^{\min}),$$

where $x_i^{\max}, x_i^{\min} \in [t_{i-1}, t_i]$.

Since

$$|x_i^{\max} - x_i^{\min}| \leq \Delta t < \delta,$$

we have

$$|f(x_i^{\max}) - f(x_i^{\min})| < \varepsilon,$$

and hence

$$M_i - m_i < \varepsilon \quad \text{for all } i.$$

Therefore,

$$\begin{aligned}
 U(f, P_\varepsilon) - L(f, P_\varepsilon) &= \sum_{i=1}^n (M_i - m_i)(t_i - t_{i-1}) \\
 &< \sum_{i=1}^n \varepsilon \Delta t \\
 &= n\varepsilon \Delta t \\
 &= \varepsilon(b - a).
 \end{aligned}$$

Since $\varepsilon > 0$ was arbitrary, this shows that f is Riemann integrable on $[a, b]$. $\odot$

Note:-

Why This Proof Works

The key point is the use of a *uniform partition*. Since all subintervals have the same length Δt , we can factor Δt out of the sum:

$$\sum_{i=1}^n (M_i - m_i)(t_i - t_{i-1}) = \Delta t \sum_{i=1}^n (M_i - m_i).$$

This avoids the error made in the earlier incorrect proof, where the number of subintervals was not controlled. Uniform continuity ensures that $M_i - m_i$ is uniformly small, and the fixed total length $b - a$ ensures the sum is small.

Theorem 14.4.2 (Spivak proof) Continuity Implies Integrability

If f is continuous on $[a, b]$, then f is Riemann integrable on $[a, b]$.

Proof: First observe that f is bounded on $[a, b]$, since it is continuous on the compact interval $[a, b]$.

To prove that f is integrable on $[a, b]$, we use the integrability criterion and show that for every $\varepsilon > 0$ there exists a partition P of $[a, b]$ such that

$$U(f, P) - L(f, P) < \varepsilon.$$

Since f is continuous on $[a, b]$, it is uniformly continuous on $[a, b]$. Therefore, there exists $\delta > 0$ such that for all $x, y \in [a, b]$,

$$|x - y| < \delta \implies |f(x) - f(y)| < \frac{\varepsilon}{2(b - a)}.$$

Choose a partition

$$P = \{t_0, t_1, \dots, t_n\}$$

of $[a, b]$ such that

$$|t_i - t_{i-1}| < \delta \quad \text{for all } i.$$

For each i , and for all $x, y \in [t_{i-1}, t_i]$, we then have

$$|f(x) - f(y)| < \frac{\varepsilon}{2(b-a)}.$$

Hence,

$$M_i - m_i \leq \frac{\varepsilon}{2(b-a)} < \frac{\varepsilon}{b-a},$$

where

$$M_i = \sup\{f(x) \mid x \in [t_{i-1}, t_i]\}, \quad m_i = \inf\{f(x) \mid x \in [t_{i-1}, t_i]\}.$$

Therefore,

$$U(f, P) - L(f, P) = \sum_{i=1}^n (M_i - m_i)(t_i - t_{i-1}) < \frac{\varepsilon}{b-a} \sum_{i=1}^n (t_i - t_{i-1}).$$

Since

$$\sum_{i=1}^n (t_i - t_{i-1}) = b - a,$$

we obtain

$$U(f, P) - L(f, P) < \varepsilon.$$

This proves that f is Riemann integrable on $[a, b]$.

☺

Chapter 15

Week 4

Note:-

Additional practice on the Fundamental Theorem of Calculus and integration techniques can be found in Chapter 23, Section 23.7.

15.1 The Fundamental Theorem of Calculus

Theorem 15.1.1 First Fundamental Theorem of Calculus

Let f be integrable on $[a, b]$. Define $F : [a, b] \rightarrow \mathbb{R}$ by

$$F(x) = \int_a^x f(t) dt.$$

Then F is continuous on $[a, b]$. Moreover, if f is continuous at $c \in (a, b)$, then F is differentiable at c and

$$F'(c) = f(c).$$

Proof: **Continuity of F :** Since f is integrable on $[a, b]$, it is bounded: $|f(t)| \leq M$ for all $t \in [a, b]$. For $x, y \in [a, b]$,

$$|F(x) - F(y)| = \left| \int_y^x f(t) dt \right| \leq M|x - y|.$$

So F is Lipschitz, hence continuous.

Differentiability at c : Suppose f is continuous at c . Let $\varepsilon > 0$. Choose $\delta > 0$ such that $|t - c| < \delta \Rightarrow |f(t) - f(c)| < \varepsilon$. For $0 < |h| < \delta$,

$$\frac{F(c+h) - F(c)}{h} - f(c) = \frac{1}{h} \int_c^{c+h} [f(t) - f(c)] dt.$$

Taking absolute values (assuming $h > 0$; the case $h < 0$ is similar):

$$\left| \frac{F(c+h) - F(c)}{h} - f(c) \right| \leq \frac{1}{h} \int_c^{c+h} |f(t) - f(c)| dt < \frac{1}{h} \cdot \varepsilon \cdot h = \varepsilon.$$

Hence $F'(c) = f(c)$. ☺

Theorem 15.1.2 Second Fundamental Theorem of Calculus

Let f be integrable on $[a, b]$, and suppose there exists a function G that is continuous on $[a, b]$ and differentiable on (a, b) with $G'(x) = f(x)$ for all $x \in (a, b)$. Then

$$\int_a^b f = G(b) - G(a).$$

Proof: Let $P = \{t_0, \dots, t_n\}$ be any partition of $[a, b]$. By the Mean Value Theorem, for each i there exists $c_i \in (t_{i-1}, t_i)$ with

$$G(t_i) - G(t_{i-1}) = G'(c_i)(t_i - t_{i-1}) = f(c_i)(t_i - t_{i-1}).$$

Summing over i gives a telescoping sum:

$$G(b) - G(a) = \sum_{i=1}^n f(c_i)(t_i - t_{i-1}).$$

Since $m_i \leq f(c_i) \leq M_i$, we have

$$L(f, P) \leq G(b) - G(a) \leq U(f, P)$$

for every partition P . Since f is integrable, the only number with this property is $\int_a^b f$. ☺

Note:-

Together, the two parts say: (1) every continuous function has an antiderivative (given by $F(x) = \int_a^x f$), and (2) any antiderivative can be used to evaluate definite integrals. In practice, to compute $\int_a^b f$, find any G with $G' = f$ and compute $G(b) - G(a)$.

15.2 Integration Techniques

Theorem 15.2.1 Integration by Substitution

If g is differentiable on $[a, b]$ with g' integrable, and f is continuous on the range of g , then

$$\int_a^b f(g(t)) g'(t) dt = \int_{g(a)}^{g(b)} f(u) du.$$

Proof: Let F be an antiderivative of f (exists by FTC I since f is continuous). Then $(F \circ g)'(t) = F'(g(t)) \cdot g'(t) = f(g(t)) \cdot g'(t)$ by the chain rule. By FTC II:

$$\int_a^b f(g(t)) g'(t) dt = F(g(b)) - F(g(a)) = \int_{g(a)}^{g(b)} f(u) du.$$

☺

Theorem 15.2.2 Integration by Parts

If f and g are differentiable on $[a, b]$ with f' and g' integrable, then

$$\int_a^b f(x) g'(x) dx = f(b)g(b) - f(a)g(a) - \int_a^b f'(x) g(x) dx.$$

Proof: By the product rule, $(fg)' = f'g + fg'$. Since $f'g$ and fg' are integrable (products of integrable functions), we apply FTC II to fg :

$$f(b)g(b) - f(a)g(a) = \int_a^b (fg)' = \int_a^b f'g + \int_a^b fg'.$$

Rearranging gives the result.



Chapter 16

Week 5

Note:-

Additional practice on integration techniques, the logarithm, and the exponential function can be found in Chapter 23, Section 23.7.

Example 16.0.1

$$\int \frac{x^2 + x + 1}{(x-1)^2(x+1)} dx$$

For this, partial fraction doesn't work, for example:

$$\frac{a}{(x-1)} + \frac{b}{(x+1)}$$

Should try

$$\frac{Ax+B}{(x-1)^2} + \frac{C}{(x+1)}$$

or

$$\frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$$

and we have

$$\int \frac{x^2 + x + 1}{(x-1)^2(x+1)} dx = \frac{A(x-1)^2 + B(x-1)(x+1) + C(x+1)}{(x-1)^2(x+1)}$$

Example 16.0.2

$$\int \frac{2x^2 + 1}{(x-1)(x^2 - 2x + 4)} dx = \int \frac{2x^2 + 1}{(x-1)((x-1)^2 + 3)} dx$$

Note that $x^2 - 2x + 4$ has no real roots. Using

$$\frac{A}{(x-1)} + \frac{Bx+C}{((x-1)^2 + 3)}$$

then,

$$\begin{aligned} &= \int \frac{1}{x-1} + \frac{x-3}{(x-1)^2+3} dx \\ &= \log|x-1| + \frac{(x-1)-2}{(x-1)^2+3} dx \end{aligned}$$

16.1 The Logarithm and Exponential Functions

Theorem 16.1.1

- Define $\log : (0, \infty) \rightarrow \mathbb{R}$, show it's strictly increasing, $\log(x) \rightarrow \infty$ as $x \rightarrow \infty$, $\log(xy) = \log(x) + \log(y)$ for all $x, y > 0$.
- Define $\exp : \mathbb{R} \rightarrow (0, \infty)$ as $\log^{-1}$. Show it has nice properties.
- Define $e := \exp(1)$, define e^x and show it equals $\exp(x)$
- Define a^x and $\log_a(x)$ for $a > 0$.

Proof: 1)

$$\log(x) := \int_1^x \frac{1}{t} dt$$

$$\log'(x) = \frac{1}{x} \quad \text{strictly increasing}$$

$\log < 0$ on $(0, 1)$, $\log > 0$ on $(1, \infty)$, $\log(1) = 0$.

$$\log(xy) = \log(x) + \log(y)$$

Prove by induction

$$\log(x^n) = n \log(x) \quad n \in \mathbb{N}$$

☺

Claim 16.1.1

$\log(x) \rightarrow \infty$ as $x \rightarrow \infty$, $\log(x) \rightarrow -\infty$ as $x \rightarrow 0^+$

Proof: Pick a number > 1 , 1.1 for example show $\log(1.1^n) \rightarrow \infty$ as $n \rightarrow \infty$ gives a sequence x_n with $\log(x_n) \rightarrow \infty$, use that $\log$ is nondecreasing to prove $\lim_{x \rightarrow \infty} \log(x) = \infty$

$$\log(1.1^n) = n \log(1.1)$$

Similarly, pick a number in $(0, 1)$ to show $\log(x) \rightarrow -\infty$.

☺

Definition 16.1.1: Exp

$$\exp : \mathbb{R} \rightarrow (0, \infty)$$

is by definition

$$\log^{-1}$$

Note:-

$\exp$ is a bijection because $\log$ is. $\exp$ is strictly increasing because $\log$ is.

Theorem 16.1.2

$\exp$ is differentiable because $\log$ is differentiable and $\log'(x) = \frac{1}{x} \neq 0$

$$\exp'(x) = \frac{1}{\log'(\exp(x))} = \frac{1}{1/\exp(x)} = \exp(x)$$

Theorem 16.1.3

$$\exp(x + y) = \exp(x)\exp(y) \quad \forall x, y \in \mathbb{R}$$

Proof:

$$\begin{aligned} \exp(x + y) &= \exp(x)\exp(y) \\ \Leftrightarrow \log(\exp(x + y)) &= \log(\exp(x)\exp(y)) \quad \log \text{ is a bijection} \\ \Leftrightarrow x + y &= \log(\exp(x)) + \log(\exp(y)) \quad \exp := \log^{-1} \\ &= x + y = x + y \end{aligned}$$



Chapter 17

Week 6

Note:-

Additional practice on the logarithm, exponential, and trigonometric functions can be found in Chapter 23, Section 23.7.

17.1 The logarithm and exponential functions

Definition 17.1.1

If $x > 0$, then

$$\log x = \int_1^x \frac{1}{t} dt$$

If $x \leq 0$, a number $\log x$ cannot be defined, since $f(t) = \frac{1}{t} \rightarrow \infty$ as $t \rightarrow 0^+$, so it's not bounded on $[x, 1]$.

Theorem 17.1.1

If $x, y > 0$, then

$$\log xy = \log x + \log y$$

Proof: Notice that $\log'(x) = \frac{1}{x}$, by FTC, choose number $y > 0$, let

$$f(x) = \log(xy)$$

then

$$f'(x) = \log'(xy) \cdot y = \frac{1}{xy} \cdot y = \frac{1}{x}$$

Thus $f' = \log'$, meaning that there is a number c such that

$$f(x) = \log x + c \quad \forall x > 0$$

that is,

$$\log xy = \log x + c \quad \forall x > 0$$

The c can be evaluated by noting that when $x = 1$,

$$\log 1 \cdot y = \log 1 + c = c$$

Thus

$$\log xy = \log x + \log y \quad \forall x$$

Since this is true for all $y > 0$, the theorem is proved. 

Corollary 17.1.1

If n is natural number and $x > 0$, then

$$\log x^n = n \log x$$

Corollary 17.1.2

If $x, y > 0$, then

$$\log \frac{x}{y} = \log x - \log y$$

Proof:

$$\log x = \log \frac{x}{y} \cdot y = \log \frac{x}{y} + \log y$$

so

$$\log x - \log y = \log \frac{x}{y}$$



Definition 17.1.2: Exponential function

The "exponential function" $\exp : \mathbb{R} \rightarrow (0, \infty)$ is defined by

$$\exp = \log^{-1}$$

Theorem 17.1.2

For all numbers x ,

$$\begin{aligned} \exp'(x) &= (\log^{-1})(x) = \frac{1}{\log'(\log^{-1}(x))} \\ &= \frac{1}{\frac{1}{\log^{-1}(x)}} = \log^{-1}(x) = \exp(x) \end{aligned}$$

Theorem 17.1.3

If x, y are any two numbers, then

$$\exp(x + y) = \exp(x) \cdot \exp(y)$$

Proof: Let $x' = \exp(x)$, $y' = \exp(y)$, so that

$$x = \log x', \quad y = \log y'.$$

then,

$$x + y = \log x' + \log y' = \log x'y'$$

meaning that

$$\exp(x + y) = x'y' = \exp(x) \cdot \exp(y)$$



Definition 17.1.3

$$e = \exp(1)$$

$$1 = \log e = \int_1^e \frac{1}{t} dt$$

Definition 17.1.4

For any number x ,

$$e^x = \exp(x)$$

Definition 17.1.5

If $a > 0$, then, for any real number x ,

$$a^x = e^{x \log a}$$

Theorem 17.1.4

If $a > 0$, then

1. $(a^b)^c = a^{bc} \forall b, c$
2. $a^1 = a$ and $a^{x+y} = a^x \cdot a^y, \forall x, y$

Proof: 1. $(a^b)^c = (e^{b \log a})^c = e^{cb \log a} = a^{bc}$

2. $a^1 = e^{1 \cdot \log a} = e^{\log a} = a$, and

$$a^{x+y} = e^{(x+y) \log a} = e^{x \log a} \cdot e^{y \log a} = a^x \cdot a^y$$



Theorem 17.1.5

If f is differentiable and

$$f'(x) = f(x) \quad \forall x$$

then there exists a number c such that

$$f(x) = ce^x \quad \forall x$$

Proof: Let $g(x) = \frac{f(x)}{e^x}$, then

$$g'(x) = \frac{e^x f'(x) - f(x)e^x}{(e^x)^2} = 0$$

(which is allowed, since $e^x \neq 0$ for all x). Then

$$g'(x) = \frac{e^x f'(x) - f(x)e^x}{(e^x)^2} = 0$$

Therefore $\exists c$ such that

$$g(x) = \frac{f(x)}{e^x} = c \quad \forall x$$

☺

Theorem 17.1.6

For any natural number n ,

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^n} = \infty$$

Proof: **Step 1** $e^x > x$ for all x , and consequently $\lim_{x \rightarrow \infty} e^x = \infty$ (this may be considered to be the base case $n = 0$). To prove this statement (which is clear for $x \leq 0$), it suffices to show that

$$x > \log x \quad \forall x > 0$$

This is because $e^{\log x} = x$, so $e^x > e^{\log x}$ iff $x > \log x$.

If $x < 1$, this is true, since $\log x < 0$. If $x > 1$, then $x - 1$ is an upper sum for $f(t) = 1/t$ on $[1, x]$, so $\log x < x - 1 < x$.

Step 2 $\lim_{x \rightarrow \infty} \frac{e^x}{x} = \infty$. To prove this, note that

$$\frac{e^x}{x} = \frac{e^{x/2} \cdot e^{x/2}}{\frac{x}{2} \cdot 2} = \frac{1}{2} \left(\frac{e^{x/2}}{x/2} \right) \cdot e^{x/2}$$

By step 1, the expression in parentheses is greater than 1, and $\lim_{x \rightarrow \infty} e^{x/2} = \infty$, this shows that $\lim_{x \rightarrow \infty} \frac{e^x}{x} = \infty$.

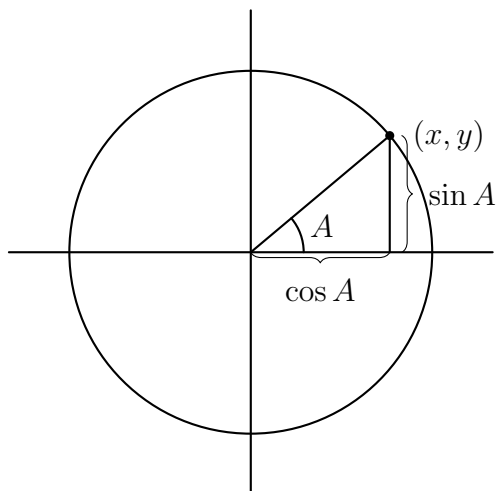
Step 3 $\lim_{x \rightarrow \infty} \frac{e^x}{x^n} = \infty$. Note that

$$\frac{e^x}{x^n} = \frac{(e^{x/n})^n}{(x/n)^n \cdot n^n} = \frac{1}{n^n} \cdot \left(\frac{e^{x/n}}{x/n} \right)^n$$

By step 2, the expression in parentheses tends to infinity as $x \rightarrow \infty$, so the whole expression tends to infinity. ☺

17.2 The trigonometric functions

The following unit circle defines sine and cosine functions.



Definition 17.2.1: Pi

$$\pi = 2 \cdot \int_{-1}^1 \sqrt{1-x^2} dx$$

Definition 17.2.2

If $-1 \leq x \leq 1$, then

$$A(x) = \frac{x\sqrt{1-x^2}}{2} + \int_x^1 \sqrt{1-t^2} dt$$

Definition 17.2.3

If $0 \leq x \leq \pi$, then $\cos x$ is the unique number in $[-1, 1]$ such that

$$A(\cos x) = \frac{x}{2}$$

and

$$\sin x = \sqrt{1 - (\cos x)^2}$$

Theorem 17.2.1

If $0 < x < \pi$, then

$$\begin{aligned}\cos'(x) &= -\sin x, \\ \sin'(x) &= \cos x.\end{aligned}$$

Proof: If $B = 2A$, then the definition $A(\cos x) = x/2$ can be written

$$B(\cos x) = x;$$

in other words, $\cos$ is just the inverse of B . We have already computed that

$$A'(x) = -\frac{1}{2\sqrt{1-x^2}},$$

from which we conclude that

$$B'(x) = -\frac{1}{\sqrt{1-x^2}}.$$

Consequently,

$$\begin{aligned}\cos'(x) &= (B^{-1})'(x) \\ &= \frac{1}{B'(B^{-1}(x))} \\ &= \frac{1}{-\frac{1}{\sqrt{1-[B^{-1}(x)]^2}}} \\ &= -\sqrt{1-(\cos x)^2} \\ &= -\sin x.\end{aligned}$$

Since

$$\sin x = \sqrt{1-(\cos x)^2},$$

we also obtain

$$\begin{aligned}\sin'(x) &= \frac{1}{2} \cdot \frac{-2 \cos x \cdot \cos'(x)}{\sqrt{1-(\cos x)^2}} \\ &= \frac{\cos x \sin x}{\sin x} \\ &= \cos x.\end{aligned}$$



Definition 17.2.4: Other Trigonometric Functions

$$\left. \begin{aligned}\sec x &= \frac{1}{\cos x} \\ \tan x &= \frac{\sin x}{\cos x}\end{aligned}\right\} x \neq k\pi + \pi/2,$$

$$\left. \begin{aligned}\csc x &= \frac{1}{\sin x} \\ \cot x &= \frac{\cos x}{\sin x}\end{aligned}\right\} x \neq k\pi.$$

Theorem 17.2.2

If $x \neq k\pi + \pi/2$, then

$$\begin{aligned}\sec'(x) &= \sec x \tan x, \\ \tan'(x) &= \sec^2 x.\end{aligned}$$

If $x \neq k\pi$, then

$$\begin{aligned}\csc'(x) &= -\csc x \cot x, \\ \cot'(x) &= -\csc^2 x.\end{aligned}$$

Proof: We apply the quotient rule and the results from Theorem 1.

1. For tangent:

$$\begin{aligned}\tan'(x) &= \left(\frac{\sin x}{\cos x} \right)' = \frac{\cos x(\sin x)' - \sin x(\cos x)'}{\cos^2 x} \\ &= \frac{\cos x(\cos x) - \sin x(-\sin x)}{\cos^2 x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x.\end{aligned}$$

2. For secant:

$$\begin{aligned}\sec'(x) &= \left(\frac{1}{\cos x} \right)' = \frac{\cos x(0) - 1(\cos x)'}{\cos^2 x} \\ &= \frac{-(-\sin x)}{\cos^2 x} = \frac{\sin x}{\cos^2 x} = \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x} = \sec x \tan x.\end{aligned}$$

3. For cotangent:

$$\begin{aligned}\cot'(x) &= \left(\frac{\cos x}{\sin x} \right)' = \frac{\sin x(\cos x)' - \cos x(\sin x)'}{\sin^2 x} \\ &= \frac{\sin x(-\sin x) - \cos x(\cos x)}{\sin^2 x} \\ &= \frac{-(\sin^2 x + \cos^2 x)}{\sin^2 x} = -\frac{1}{\sin^2 x} = -\csc^2 x.\end{aligned}$$

4. For cosecant:

$$\begin{aligned}\csc'(x) &= \left(\frac{1}{\sin x} \right)' = \frac{\sin x(0) - 1(\sin x)'}{\sin^2 x} \\ &= \frac{-\cos x}{\sin^2 x} = -\frac{1}{\sin x} \cdot \frac{\cos x}{\sin x} = -\csc x \cot x.\end{aligned}$$



The inverses of the trigonometric functions are also easily differentiated. The trigonometric functions are not one-one, so it is first necessary to restrict them to suitable intervals; the largest possible length obtainable is π , and the intervals usually chosen are

$$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \text{ for } \sin, \quad [0, \pi] \text{ for } \cos, \quad \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \text{ for } \tan.$$

(The inverses of the other trigonometric functions are so rarely used that they will not even be discussed here.)

The inverse of the function

$$f(x) = \sin x, \quad -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

is denoted by $\arcsin$; the domain of $\arcsin$ is $[-1, 1]$. The notation $\sin^{-1}$ has been avoided because $\arcsin$ is not the inverse of $\sin$ (which is not one-one), but of the restricted function f ; sometimes this function f is denoted by Sin , and $\arcsin$ by Sin^{-1} .

The inverse of the function

$$g(x) = \cos x, \quad 0 \leq x \leq \pi$$

is denoted by $\arccos$; the domain of $\arccos$ is $[-1, 1]$. Sometimes g is denoted by Cos , and $\arccos$ by Cos^{-1} .

The inverse of the function

$$h(x) = \tan x, \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$$

is denoted by $\arctan$; $\arctan$ is one of the simplest examples of a differentiable function which is bounded even though it is one-one on all of $\mathbb{R}$. Sometimes the function h is denoted by Tan , and $\arctan$ by Tan^{-1} .

The derivatives of the inverse trigonometric functions are surprisingly simple, and do not involve trigonometric functions at all. Finding the derivatives is a simple matter, but to express them in a suitable form we will have to simplify expressions like

$$\cos(\arcsin x), \quad \sec(\arctan x).$$

Theorem 17.2.3

If $-1 < x < 1$, then

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}, \quad (\arccos x)' = \frac{-1}{\sqrt{1-x^2}}.$$

Moreover, for all x we have

$$(\arctan x)' = \frac{1}{1+x^2}.$$

Proof: We prove the first formula.

$$(\arcsin x)' = (f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))} = \frac{1}{\sin'(\arcsin x)} = \frac{1}{\cos(\arcsin x)}.$$

Now

$$[\sin(\arcsin x)]^2 + [\cos(\arcsin x)]^2 = 1,$$

that is,

$$x^2 + [\cos(\arcsin x)]^2 = 1;$$

therefore,

$$\cos(\arcsin x) = \sqrt{1 - x^2}.$$

(The positive square root is to be taken because $\arcsin x$ is in $(-\pi/2, \pi/2)$, so $\cos(\arcsin x) > 0$.) This proves the first formula.

The second formula has already been established (in the proof of Theorem 1). It is also possible to imitate the proof for the first formula, a valuable exercise if that proof presented any difficulties. The third formula is proved as follows.

$$(\arctan x)' = (h^{-1})'(x) = \frac{1}{h'(h^{-1}(x))} = \frac{1}{\tan'(\arctan x)} = \frac{1}{\sec^2(\arctan x)}.$$

Dividing both sides of the identity

$$\sin^2 a + \cos^2 a = 1$$

by $\cos^2 a$ yields

$$\tan^2 a + 1 = \sec^2 a.$$

It follows that

$$[\tan(\arctan x)]^2 + 1 = \sec^2(\arctan x),$$

or

$$x^2 + 1 = \sec^2(\arctan x),$$

which proves the third formula. ☺

Lemma 17.2.1

Suppose f has a second derivative everywhere and that

$$f'' + f = 0, \quad f(0) = 0, \quad f'(0) = 0.$$

Then $f = 0$.

Proof: Multiplying both sides of the first equation by f' yields

$$f' f'' + f f' = 0.$$

Thus

$$[(f')^2 + f^2]' = 2(f' f'' + f f') = 0,$$

so $(f')^2 + f^2$ is a constant function. From $f(0) = 0$ and $f'(0) = 0$ it follows that the constant is 0; thus

$$[f'(x)]^2 + [f(x)]^2 = 0 \quad \text{for all } x.$$

This implies that

$$f(x) = 0 \quad \text{for all } x.$$



Theorem 17.2.4

If f has a second derivative everywhere and

$$f'' + f = 0, \quad f(0) = a, \quad f'(0) = b,$$

then

$$f = b \cdot \sin + a \cdot \cos.$$

(In particular, if $f(0) = 0$ and $f'(0) = 1$, then $f = \sin$; if $f(0) = 1$ and $f'(0) = 0$, then $f = \cos$.)

Proof: Let

$$g(x) = f(x) - b \sin x - a \cos x.$$

Then

$$g'(x) = f'(x) - b \cos x + a \sin x, \quad g''(x) = f''(x) + b \sin x + a \cos x.$$

Consequently,

$$g'' + g = 0, \quad g(0) = 0, \quad g'(0) = 0,$$

which shows that

$$0 = g(x) = f(x) - b \sin x - a \cos x, \quad \text{for all } x.$$



Theorem 17.2.5

If x and y are any two numbers, then

$$\sin(x + y) = \sin x \cos y + \cos x \sin y, \quad \cos(x + y) = \cos x \cos y - \sin x \sin y.$$

Proof: For any particular number y we can define a function f by

$$f(x) = \sin(x + y).$$

Then

$$f'(x) = \cos(x + y), \quad f''(x) = -\sin(x + y).$$

Consequently,

$$f'' + f = 0, \quad f(0) = \sin y, \quad f'(0) = \cos y.$$

It follows from Theorem 4 that

$$f = (\cos y) \cdot \sin + (\sin y) \cdot \cos;$$

that is,

$$\sin(x + y) = \cos y \sin x + \sin y \cos x, \quad \text{for all } x.$$

Since any number y could have been chosen to begin with, this proves the first formula for all x and y . The second formula is proved similarly. ☺

Theorem 17.2.6

For all real numbers x ,

$$\sin^2 x + \cos^2 x = 1.$$

Proof: Define a function F by

$$F(x) = \sin^2 x + \cos^2 x.$$

Then F has a derivative everywhere, and

$$F'(x) = 2 \sin x \cos x + 2 \cos x(-\sin x) = 0.$$

Hence F is constant. Evaluating at $x = 0$ gives

$$F(0) = \sin^2 0 + \cos^2 0 = 0^2 + 1^2 = 1.$$

Therefore $F(x) = 1$ for all x , i.e.

$$\sin^2 x + \cos^2 x = 1$$

for all real x . ☺

Example 17.2.1 (Integrals Involving arctan and Improper Integrals)

Recall that

$$\frac{d}{dx} \arctan(x) = \frac{1}{1 + x^2}.$$

Also,

$$\tan : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R} \quad \text{is bijective, and} \quad \arctan : \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \quad \text{is its inverse.}$$

1. A definite integral

$$\int_0^1 \frac{1}{1 + t^2} dt = \arctan(t) \Big|_0^1 = \arctan(1) - \arctan(0) = \frac{\pi}{4} - 0 = \frac{\pi}{4}.$$

2. An improper integral on $[0, \infty)$

$$\int_0^\infty \frac{1}{1 + t^2} dt := \lim_{R \rightarrow \infty} \int_0^R \frac{1}{1 + t^2} dt \quad (\text{if the limit exists}).$$

Compute:

$$\begin{aligned}\lim_{R \rightarrow \infty} \int_0^R \frac{1}{1+t^2} dt &= \lim_{R \rightarrow \infty} \left[\arctan(t) \right]_0^R \\ &= \lim_{R \rightarrow \infty} (\arctan(R) - \arctan(0)) \\ &= \lim_{R \rightarrow \infty} \arctan(R) = \frac{\pi}{2}.\end{aligned}$$

Hence

$$\int_0^\infty \frac{1}{1+t^2} dt = \frac{\pi}{2}.$$

3. An improper integral on $(-\infty, \infty)$

$$\int_{-\infty}^\infty \frac{1}{1+t^2} dt := \lim_{R \rightarrow \infty} \int_{-R}^R \frac{1}{1+t^2} dt.$$

Then

$$\begin{aligned}\lim_{R \rightarrow \infty} \int_{-R}^R \frac{1}{1+t^2} dt &= \lim_{R \rightarrow \infty} \left[\arctan(t) \right]_{-R}^R \\ &= \lim_{R \rightarrow \infty} (\arctan(R) - \arctan(-R)) \\ &= \lim_{R \rightarrow \infty} \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right) = \pi.\end{aligned}$$

Therefore,

$$\int_{-\infty}^\infty \frac{1}{1+t^2} dt = \pi.$$

Example 17.2.2 (Computing $\int \frac{1}{4x^2 - 4x + 4} dx$)

Step 1: Complete the square.

$$\begin{aligned}4x^2 - 4x + 4 &= 4(x^2 - x + 1) \\ &= 4 \left(x^2 - x + \frac{1}{4} + \frac{3}{4} \right) \\ &= 4 \left(\left(x - \frac{1}{2} \right)^2 + \frac{3}{4} \right).\end{aligned}$$

Factor out $\frac{3}{4}$:

$$\begin{aligned}4 \left(\left(x - \frac{1}{2} \right)^2 + \frac{3}{4} \right) &= 4 \cdot \frac{3}{4} \left[\frac{4}{3} \left(x - \frac{1}{2} \right)^2 + 1 \right] \\ &= 3 \left[\frac{4}{3} \left(x - \frac{1}{2} \right)^2 + 1 \right].\end{aligned}$$

Rewrite:

$$4x^2 - 4x + 4 = 3 \left[\left(\frac{2}{\sqrt{3}} \left(x - \frac{1}{2} \right) \right)^2 + 1 \right].$$

Step 2: Substitute into the integral.

$$\int \frac{1}{4x^2 - 4x + 4} dx = \int \frac{1}{3 \left[\left(\frac{2}{\sqrt{3}} \left(x - \frac{1}{2} \right) \right)^2 + 1 \right]} dx.$$

Factor out $\frac{1}{3}$:

$$= \frac{1}{3} \int \frac{1}{1 + \left(\frac{2}{\sqrt{3}} \left(x - \frac{1}{2} \right) \right)^2} dx.$$

Let

$$u = \frac{2}{\sqrt{3}} \left(x - \frac{1}{2} \right).$$

Then

$$du = \frac{2}{\sqrt{3}} dx \quad \implies \quad dx = \frac{\sqrt{3}}{2} du.$$

Substitute:

$$\frac{1}{3} \int \frac{1}{1 + u^2} \frac{\sqrt{3}}{2} du = \frac{\sqrt{3}}{6} \int \frac{1}{1 + u^2} du.$$

Step 3: Use the arctangent formula.

Since

$$\int \frac{1}{1 + u^2} du = \arctan(u),$$

we obtain

$$\frac{\sqrt{3}}{6} \arctan(u) + C.$$

Substitute back:

$$\boxed{\int \frac{1}{4x^2 - 4x + 4} dx = \frac{\sqrt{3}}{6} \arctan \left(\frac{2}{\sqrt{3}} \left(x - \frac{1}{2} \right) \right) + C.}$$

Chapter 18

Week 7

Note:-

Additional practice on Taylor polynomials, Taylor series, and power series can be found in Chapter 23, Section 23.8.

18.1 Taylor polynomial and Taylor series

To begin with, look at the polynomial function p :

$$p(x) = a_0 + a_1x + \dots + a_nx^n$$

then $p(x)$ can be computed by simply plugging in x . However, this does not work for every function like sin, log and so on. Taylor polynomial is a great tool to approximate many functions.

We can see that

$$p(0) = a_0$$

differentiating:

$$p'(x) = a_1 + 2a_2x + \dots + na_nx^{n-1}$$

thus,

$$p'(0) = p^{(1)}(0) = a_1$$

differentiating, we can get

$$p''(x) = 2a_2 + 3 \cdot 2 \cdot a_3x + \dots + n(n-1)a_nx^{n-2}$$

and we get

$$p''(0) = 2a_2$$

So in general, we have

$$p^{(k)}(0) = k!a_k \Rightarrow a_k = \frac{p^{(k)}(0)}{k!}$$

Now, if we define p as

$$p(x) = a_0 + a_1(x-a) + \dots + a_n(x-a)^n$$

then we get

$$a_k = \frac{p^{(k)}(a)}{k!}$$

Suppose f is a function where it's first to n th derivative exist, then

$$a_k = \frac{p^{(k)}(a)}{k!}, \quad 0 \leq k \leq n$$

and here we get the definition of **Taylor Polynomial**:

Definition 18.1.1: Taylor polynomial

The Taylor polynomial of degree n for f at a is:

$$P_{n,a} = a_0 + a_1(x - a) + \dots + a_n(x - a)^n$$

Example 18.1.1 (Taylor polynomial various functions)

The Taylor polynomial for $P_{2n+1,0}$ for sine:

$$P_{2n+1,0}(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots + (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

The Taylor polynomial $P_{2n,0}$ of degree $2n$ for cos at 0 is (the computations are left to you)

$$P_{2n,0}(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots + (-1)^n \frac{x^{2n}}{(2n)!}.$$

The Taylor polynomial for exp is especially easy to compute. Since $\exp^{(k)}(0) = \exp(0) = 1$ for all k , the Taylor polynomial of degree n at 0 is

$$P_{n,0}(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots + \frac{x^n}{n!}.$$

Theorem 18.1.1

suppose f is a function for which

$$f'(a), \dots, f^{(n)}(a)$$

all exist. Let

$$a_k = \frac{f^{(k)}(a)}{k!}, \quad 0 \leq k \leq n$$

and define

$$P_{n,a}(x) = a_0 + a_1(x - a) + \dots + a_n(x - a)^n$$

then

$$\lim_{x \rightarrow a} \frac{f(x) - P_{n,a}(x)}{(x - a)^n} = 0$$

Theorem 18.1.2 Peano form of remainder

suppose f is a function for which

$$f'(a), \dots, f^{(n)}(a)$$

all exist. Let

$$a_k = \frac{f^{(k)}(a)}{k!}, \quad 0 \leq k \leq n$$

and define

$$P_{n,a}(x) = a_0 + a_1(x-a) + \dots + a_n(x-a)^n$$

then

$$R_{n,a}(x) = f(x) - P_{n,a}(x) = h(x)(x-a)^n$$

where h is continuous at $x = a$ and $h(a) = 0$.

Proof: Writing out $P_{n,a}(x)$ explicitly, we obtain

$$\frac{f(x) - P_{n,a}(x)}{(x-a)^n} = \frac{f(x) - \sum_{i=0}^{n-1} \frac{f^{(i)}(a)}{i!}(x-a)^i}{(x-a)^n} - \frac{f^{(n)}(a)}{n!}.$$

It will help to introduce the new functions

$$Q(x) = \sum_{i=0}^{n-1} \frac{f^{(i)}(a)}{i!}(x-a)^i \quad \text{and} \quad g(x) = (x-a)^n;$$

now we must prove that

$$\lim_{x \rightarrow a} \frac{f(x) - Q(x)}{g(x)} = \frac{f^{(n)}(a)}{n!}.$$

Notice that

$$\begin{aligned} Q^{(k)}(a) &= f^{(k)}(a), \quad k \leq n-1, \\ g^{(k)}(x) &= n!(x-a)^{n-k}/(n-k)!. \end{aligned}$$

Thus

$$\begin{aligned} \lim_{x \rightarrow a} [f(x) - Q(x)] &= f(a) - Q(a) = 0, \\ \lim_{x \rightarrow a} [f'(x) - Q'(x)] &= f'(a) - Q'(a) = 0, \\ &\vdots \\ \lim_{x \rightarrow a} [f^{(n-2)}(x) - Q^{(n-2)}(x)] &= f^{(n-2)}(a) - Q^{(n-2)}(a) = 0. \end{aligned}$$

and

$$\lim_{x \rightarrow a} g(x) = \lim_{x \rightarrow a} g'(x) = \dots = \lim_{x \rightarrow a} g^{(n-2)}(x) = 0.$$

We may therefore apply l'Hôpital's Rule $n - 1$ times to obtain

$$\lim_{x \rightarrow a} \frac{f(x) - Q(x)}{(x - a)^n} = \lim_{x \rightarrow a} \frac{f^{(n-1)}(x) - Q^{(n-1)}(x)}{n!(x - a)}.$$

Since Q is a polynomial of degree $n - 1$, its $(n - 1)$ st derivative is a constant; in fact, $Q^{(n-1)}(x) = f^{(n-1)}(a)$. Thus

$$\lim_{x \rightarrow a} \frac{f(x) - Q(x)}{(x - a)^n} = \lim_{x \rightarrow a} \frac{f^{(n-1)}(x) - f^{(n-1)}(a)}{n!(x - a)}$$

and this last limit is $f^{(n)}(a)/n!$ by definition of $f^{(n)}(a)$. ☺

Theorem 18.1.3

Assume $f'(a) = \dots = f^{(n-1)}(a) = 0$, $f^{(n)} \neq 0$,

1. If n is even and $f^{(n)}(a) > 0$, then f has a local minimum at a .
2. If n is odd and $f^{(n)}(a) < 0$, then f has a local maximum at a .
3. If n is odd, then f has neither a local maximum nor a local minimum at a .

Proof from the book:

Proof: There is clearly no loss of generality in assuming that $f(a) = 0$, since neither the hypotheses nor the conclusion are affected if f is replaced by $f - f(a)$. Then, since the first $n - 1$ derivatives of f at a are 0, the Taylor polynomial $P_{n,a}$ of f is

$$\begin{aligned} P_{n,a}(x) &= f(a) + \frac{f'(a)}{1!}(x - a) + \dots + \frac{f^{(n)}(a)}{n!}(x - a)^n \\ &= \frac{f^{(n)}(a)}{n!}(x - a)^n. \end{aligned}$$

Thus, Theorem 1 states that

$$0 = \lim_{x \rightarrow a} \frac{f(x) - P_{n,a}(x)}{(x - a)^n} = \lim_{x \rightarrow a} \left[\frac{f(x)}{(x - a)^n} - \frac{f^{(n)}(a)}{n!} \right].$$

Consequently, if x is sufficiently close to a , then

$$\frac{f(x)}{(x - a)^n} \quad \text{has the same sign as} \quad \frac{f^{(n)}(a)}{n!}.$$

Suppose now that n is even. In this case $(x - a)^n > 0$ for all $x \neq a$. Since $f(x)/(x - a)^n$ has the same sign as $f^{(n)}(a)/n!$ for x sufficiently close to a , it follows

that $f(x)$ itself has the same sign as $f^{(n)}(a)/n!$ for x sufficiently close to a . If $f^{(n)}(a) > 0$, this means that

$$f(x) > 0 = f(a)$$

for x close to a . Consequently, f has a local minimum at a . A similar proof works for the case $f^{(n)}(a) < 0$.

Now suppose that n is odd. The same argument as before shows that if x is sufficiently close to a , then

$$\frac{f(x)}{(x-a)^n} \text{ always has the same sign.}$$

But $(x-a)^n > 0$ for $x > a$ and $(x-a)^n < 0$ for $x < a$. Therefore $f(x)$ has *different* signs for $x > a$ and $x < a$. This proves that f has neither a local maximum nor a local minimum at a . $\ominus$

An alternative proof from the book:

Proof: We know $P_{n,a}(x) = f(a) + \frac{f^{(n)}(a)}{n!}(x-a)^n$. By theorem,

$$f(x) - P_{n,a}(x) = f(x) - \left[f(a) + \frac{f^{(n)}(a)}{n!}(x-a)^n \right] = h(x)(x-a)^n$$

where $h(x) \rightarrow 0$ as $x \rightarrow a$.

$$\frac{f(x) - f(a)}{(x-a)^n} - \frac{f^{(n)}(a)}{n!} = h(x) \rightarrow 0$$

Case 1

Case 1: Assume n is even. There is a $\delta > 0$ so that $0 < |x-a| < \delta$ then

$$\frac{3}{2} \frac{f^{(n)}(a)}{n!} > \frac{f(x) - f(a)}{(x-a)^n} = h(x) + \frac{f^{(n)}(a)}{n!} > \frac{1}{2} \frac{f^{(n)}(a)}{n!}$$

(took $\epsilon = \frac{1}{2} \frac{f^{(n)}(a)}{n!}$ in scratch work). If n even, multiply by $(x-a)^n$ and add $f(a)$ everywhere:

$$0 < |x-a| < \delta \implies f(a) + \frac{3}{2} \frac{f^{(n)}(a)}{n!}(x-a)^n > f(x) > f(a) + \frac{1}{2} \frac{f^{(n)}(a)}{n!}(x-a)^n$$

If $f^{(n)}(a) > 0$, then $f(x) > f(a) + \frac{1}{2} \frac{f^{(n)}(a)}{n!}(x-a)^n$ (positive because $0 < |x-a| < \delta$) $\implies$ local min. If $f^{(n)}(a) < 0$, then $f(a) + \frac{3}{2} \frac{f^{(n)}(a)}{n!}(x-a)^n$ (negative) $> f(x) \implies$ local max.

Case 2

Assume n is even and $f^{(n)}(a) < 0$. Following the same argument as in Case 1, we choose a $\delta > 0$ such that for $0 < |x-a| < \delta$:

$$\frac{f^{(n)}(a)}{2} < \frac{f(x) - f(a)}{(x-a)^n} - \frac{f^{(n)}(a)}{n!} < -\frac{f^{(n)}(a)}{2}$$

As shown previously, when n is even, $(x-a)^n > 0$. Multiplying through and rearranging lead to the conclusion that $f(x) < f(a)$ for all x in the neighborhood, which implies a **local max**.

Case 3

Assume n is odd. We have the same initial relationship:

$$\frac{f(x) - f(a)}{(x - a)^n} - \frac{f^{(n)}(a)}{n!} = h(x)$$

There is a $\delta > 0$ so that if $0 < |x - a| < \delta$, then $\frac{f(x) - f(a)}{(x - a)^n}$ has the same sign as $\frac{f^{(n)}(a)}{n!}$.

However, because n is odd, the term $(x - a)^n$ changes sign depending on whether $x > a$ or $x < a$:

- If $x > a$, then $(x - a)^n > 0$.
- If $x < a$, then $(x - a)^n < 0$.

This means that $f(x) - f(a)$ will have different signs on opposite sides of a . Therefore, f has **neither a local max nor a local min** at a . ☹

Geometric series. Let

$$f(x) = \frac{1}{1 - x}.$$

Its n th Taylor polynomial about 0 is

$$P_{n,0}(x) = 1 + x + x^2 + x^3 + \cdots + x^n.$$

Question 1

Is

$$P(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots + \frac{x^{n+1}}{n+1}$$

the $(n + 1)$ st Taylor polynomial of $-\log(1 - x)$?

Solution: Yes. Since

$$-\log(1 - x) = \int_0^x \frac{1}{1 - t} dt,$$

and

$$\frac{1}{1 - t} = 1 + t + t^2 + \cdots,$$

integrating term-by-term gives the Maclaurin series

$$-\log(1 - x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots.$$

Hence the $(n + 1)$ st Taylor polynomial is exactly

$$x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots + \frac{x^{n+1}}{n+1}.$$

Question 2

Is

$$\tilde{P}(x) = 1 - x^2 + x^4 - x^6 + \cdots + (-1)^n x^{2n}$$

the $(2n)$ th Taylor polynomial of $\frac{1}{1+x^2}$?

Solution: Yes. Note that

$$\frac{1}{1+x^2} = \frac{1}{1-(-x^2)}.$$

Using the geometric series with $u = -x^2$,

$$\frac{1}{1-u} = 1 + u + u^2 + \cdots,$$

we get

$$\frac{1}{1+x^2} = 1 + (-x^2) + (-x^2)^2 + \cdots = 1 - x^2 + x^4 - x^6 + \cdots.$$

Therefore the $(2n)$ th Taylor polynomial is

$$1 - x^2 + x^4 - x^6 + \cdots + (-1)^n x^{2n},$$

which can be viewed as substituting $x \mapsto -x^2$ into the geometric series polynomial.

Theorem 18.1.4 Uniqueness of a Polynomial Determined by Its Order at a

Let P and Q be two polynomials in $(x-a)$ of degree $\leq n$, and suppose that P and Q are equal up to order n at a . Then $P = Q$.

Proof: Let $R = P - Q$. Then R is a polynomial in $(x-a)$ of degree $\leq n$. Write

$$R(x) = b_0 + b_1(x-a) + \cdots + b_n(x-a)^n.$$

Since P and Q are equal up to order n at a , we have

$$\lim_{x \rightarrow a} \frac{R(x)}{(x-a)^n} = 0.$$

This hypothesis implies in particular that for each $0 \leq i \leq n$,

$$\lim_{x \rightarrow a} \frac{R(x)}{(x-a)^i} = 0.$$

For $i = 0$ this gives $\lim_{x \rightarrow a} R(x) = 0$. But

$$\lim_{x \rightarrow a} R(x) = \lim_{x \rightarrow a} (b_0 + b_1(x-a) + \cdots + b_n(x-a)^n) = b_0,$$

so $b_0 = 0$. Hence

$$R(x) = b_1(x-a) + \cdots + b_n(x-a)^n.$$

Divide by $(x - a)$:

$$\frac{R(x)}{x - a} = b_1 + b_2(x - a) + \cdots + b_n(x - a)^{n-1},$$

and taking $x \rightarrow a$ yields

$$\lim_{x \rightarrow a} \frac{R(x)}{x - a} = b_1.$$

But the left-hand limit is 0, so $b_1 = 0$. Thus

$$R(x) = b_2(x - a)^2 + \cdots + b_n(x - a)^n.$$

Continuing in this way, we obtain successively

$$b_0 = b_1 = \cdots = b_n = 0,$$

so $R \equiv 0$. Therefore $P - Q = 0$, i.e. $P = Q$. ☺

Corollary 18.1.1

Let f be n times differentiable at a , and suppose that P is a polynomial in $(x - a)$ of degree $\leq n$, which equals f up to order n at a , then $P = P_{n,a,f}$.

Proof: Since P and $P_{n,a,f}$ both equal f up to order n at a , it is easy to see that P equals $P_{n,a,f}$ up to order n at a . Consequently, $P = P_{n,a,f}$ by Theorem. ☺

18.2 Integral Form of the Remainder

Theorem 18.2.1 Taylor's Theorem with Integral Remainder

Suppose $f^{(n+1)}$ is continuous on $[a, x]$ (or $[x, a]$). Then

$$f(x) = P_{n,a}(x) + R_{n,a}(x),$$

where

$$R_{n,a}(x) = \frac{1}{n!} \int_a^x (x - t)^n f^{(n+1)}(t) dt.$$

Proof: By integration by parts with $u = f^{(n+1)}(t)$ and $dv = -\frac{(x-t)^n}{n!} dt$ (so $v = \frac{(x-t)^{n+1}}{(n+1)!}$, but we use the negative sign convention). More directly, proceed by induction on n .

Base case ($n = 0$): By FTC,

$$f(x) = f(a) + \int_a^x f'(t) dt,$$

which is $P_{0,a}(x) + R_{0,a}(x)$.

Inductive step: Assume the result for $n - 1$:

$$f(x) = P_{n-1,a}(x) + \frac{1}{(n-1)!} \int_a^x (x - t)^{n-1} f^{(n)}(t) dt.$$

Apply integration by parts to the remainder with $u = f^{(n)}(t)$, $dv = \frac{(x-t)^{n-1}}{(n-1)!} dt$:

$$\frac{1}{(n-1)!} \int_a^x (x-t)^{n-1} f^{(n)}(t) dt = \frac{f^{(n)}(a)}{n!} (x-a)^n + \frac{1}{n!} \int_a^x (x-t)^n f^{(n+1)}(t) dt.$$

Adding $P_{n-1,a}(x)$ to the first term gives $P_{n,a}(x)$, completing the induction. $\ominus$

Note:-

The integral remainder is often more useful than the Peano form because it gives an *explicit bound* on $|R_{n,a}(x)|$: if $|f^{(n+1)}(t)| \leq M$ for t between a and x , then

$$|R_{n,a}(x)| \leq \frac{M|x-a|^{n+1}}{(n+1)!}.$$

18.3 Taylor Series

Definition 18.3.1: Taylor Series

If f is infinitely differentiable at a , the **Taylor series** of f at a is the formal series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k.$$

When $a = 0$, this is called the **Maclaurin series**.

Note:-

A Taylor series may or may not converge to $f(x)$. It converges to $f(x)$ if and only if $R_{n,a}(x) \rightarrow 0$ as $n \rightarrow \infty$. A classical counterexample: the function

$$f(x) = \begin{cases} e^{-1/x^2} & x \neq 0, \\ 0 & x = 0 \end{cases}$$

has $f^{(k)}(0) = 0$ for all k , so its Maclaurin series is identically 0, but $f(x) \neq 0$ for $x \neq 0$.

Example 18.3.1 (Important Taylor series)

Using the integral remainder to verify convergence:

1. $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$ for all $x \in \mathbb{R}$. (Since $|R_{n,0}(x)| \leq \frac{e^{|x|}|x|^{n+1}}{(n+1)!} \rightarrow 0$.)
2. $\sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}$ for all $x \in \mathbb{R}$.
3. $\cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}$ for all $x \in \mathbb{R}$.

Chapter 19

Week 8

Note:-

Additional practice on sequences of functions, uniform convergence, and power series can be found in Chapter 23, Section 23.8.

19.1 Sequences of Functions and Uniform Convergence

Definition 19.1.1: Pointwise Convergence

A sequence of functions (f_n) defined on a set A **converges pointwise** to f on A if for every $x \in A$,

$$\lim_{n \rightarrow \infty} f_n(x) = f(x).$$

Note:-

Pointwise convergence can behave badly: the limit of continuous functions need not be continuous, and

$$\lim_{n \rightarrow \infty} \int_a^b f_n \neq \int_a^b \lim_{n \rightarrow \infty} f_n$$

in general.

Example 19.1.1 (Pointwise but not uniform)

Let $f_n(x) = x^n$ on $[0, 1]$. Then $f_n \rightarrow f$ pointwise, where $f(x) = 0$ for $x \in [0, 1)$ and $f(1) = 1$. Each f_n is continuous, but f is not.

Definition 19.1.2: Uniform Convergence

(f_n) **converges uniformly** to f on A if for every $\varepsilon > 0$ there exists N such that for all $n \geq N$ and all $x \in A$,

$$|f_n(x) - f(x)| < \varepsilon.$$

Equivalently, $\sup_{x \in A} |f_n(x) - f(x)| \rightarrow 0$ as $n \rightarrow \infty$.

Note:-

The key difference: in pointwise convergence, N may depend on both ε and x . In uniform convergence, N depends only on ε .

Theorem 19.1.1 Uniform Limit and Integration

Suppose that $\{f_n\}$ is a sequence of functions which are integrable on $[a, b]$, and that $\{f_n\}$ converges uniformly on $[a, b]$ to a function f which is integrable on $[a, b]$. Then

$$\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n.$$

Proof: Let $\varepsilon > 0$. There is some N such that for all $n > N$ we have

$$|f(x) - f_n(x)| < \varepsilon \quad \text{for all } x \in [a, b].$$

Thus, if $n > N$ we have

$$\begin{aligned} \left| \int_a^b f(x) dx - \int_a^b f_n(x) dx \right| &= \left| \int_a^b [f(x) - f_n(x)] dx \right| \\ &\leq \int_a^b |f(x) - f_n(x)| dx \\ &\leq \int_a^b \varepsilon dx \\ &= \varepsilon(b - a). \end{aligned}$$

Since this is true for any $\varepsilon > 0$, it follows that

$$\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n. \quad \odot$$

Theorem 19.1.2 Uniform Limit of Continuous Functions is Continuous

Suppose that $\{f_n\}$ is a sequence of functions which are continuous on $[a, b]$, and that $\{f_n\}$ converges uniformly on $[a, b]$ to f . Then f is also continuous on $[a, b]$.

Proof: For each x in $[a, b]$ we must prove that f is continuous at x . We will deal only with x in (a, b) ; the cases $x = a$ and $x = b$ require the usual simple modifications.

Let $\varepsilon > 0$. Since $\{f_n\}$ converges uniformly to f on $[a, b]$, there is some n such that

$$|f(y) - f_n(y)| < \frac{\varepsilon}{3} \quad \text{for all } y \in [a, b].$$

In particular, for all h such that $x + h$ is in $[a, b]$, we have

1. $|f(x) - f_n(x)| < \frac{\varepsilon}{3},$
2. $|f(x + h) - f_n(x + h)| < \frac{\varepsilon}{3}.$

Now f_n is continuous, so there is some $\delta > 0$ such that for $|h| < \delta$ we have

$$3. |f_n(x) - f_n(x+h)| < \frac{\varepsilon}{3}.$$

Thus, if $|h| < \delta$, then

$$\begin{aligned} |f(x+h) - f(x)| &= |f(x+h) - f_n(x+h) + f_n(x+h) - f_n(x) + f_n(x) - f(x)| \\ &\leq |f(x+h) - f_n(x+h)| + |f_n(x+h) - f_n(x)| + |f_n(x) - f(x)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\ &= \varepsilon. \end{aligned}$$

This proves that f is continuous at x . ☺

Theorem 19.1.3 Uniform Limit and Differentiation

Suppose that $\{f_n\}$ is a sequence of functions which are differentiable on $[a, b]$, with integrable derivatives f'_n , and that $\{f_n\}$ converges (pointwise) to f . Suppose, moreover, that $\{f'_n\}$ converges uniformly on $[a, b]$ to some continuous function g . Then f is differentiable and

$$f'(x) = \lim_{n \rightarrow \infty} f'_n(x).$$

Proof: Applying Theorem 19.1.1 to the interval $[a, x]$, we see that for each x we have

$$\begin{aligned} \int_a^x g &= \lim_{n \rightarrow \infty} \int_a^x f'_n \\ &= \lim_{n \rightarrow \infty} [f_n(x) - f_n(a)] \\ &= f(x) - f(a). \end{aligned}$$

Since g is continuous, it follows that $f'(x) = g(x) = \lim_{n \rightarrow \infty} f'_n(x)$ for all x in the interval $[a, b]$. ☺

Uniform Convergence of Series

Definition 19.1.3: Uniform Convergence of Series

The series $\sum_{n=1}^{\infty} f_n$ **converges uniformly** (more formally: the sequence $\{f_n\}$ is **uniformly summable**) **to f on A** , if the sequence

$$f_1, \quad f_1 + f_2, \quad f_1 + f_2 + f_3, \quad \dots$$

converges uniformly to f on A .

Corollary 19.1.1 Uniform Convergence of Series

Let $\sum_{n=1}^{\infty} f_n$ converge uniformly to f on $[a, b]$.

1. If each f_n is continuous on $[a, b]$, then f is continuous on $[a, b]$.
2. If f and each f_n is integrable on $[a, b]$, then

$$\int_a^b f = \sum_{n=1}^{\infty} \int_a^b f_n.$$

Moreover, if $\sum_{n=1}^{\infty} f_n$ converges (pointwise) to f on $[a, b]$, each f_n has an integrable derivative

f'_n and $\sum_{n=1}^{\infty} f'_n$ converges uniformly on $[a, b]$ to some continuous function, then

$$3. f'(x) = \sum_{n=1}^{\infty} f'_n(x) \quad \text{for all } x \text{ in } [a, b].$$

Proof: (1) If each f_n is continuous, then so is each $f_1 + \cdots + f_n$, and f is the uniform limit of the sequence $f_1, f_1 + f_2, f_1 + f_2 + f_3, \dots$, so f is continuous by Theorem 19.1.2.

(2) Since $f_1, f_1 + f_2, f_1 + f_2 + f_3, \dots$ converges uniformly to f , it follows from Theorem 19.1.1 that

$$\begin{aligned} \int_a^b f &= \lim_{n \rightarrow \infty} \int_a^b (f_1 + \cdots + f_n) \\ &= \lim_{n \rightarrow \infty} \left(\int_a^b f_1 + \cdots + \int_a^b f_n \right) \\ &= \sum_{n=1}^{\infty} \int_a^b f_n. \end{aligned}$$

(3) Each function $f_1 + \cdots + f_n$ is differentiable, with derivative $f'_1 + \cdots + f'_n$, and $f'_1, f'_1 + f'_2, f'_1 + f'_2 + f'_3, \dots$ converges uniformly to a continuous function, by hypothesis. It follows from Theorem 19.1.3 that

$$\begin{aligned} f'(x) &= \lim_{n \rightarrow \infty} [f'_1(x) + \cdots + f'_n(x)] \\ &= \sum_{n=1}^{\infty} f'_n(x). \end{aligned}$$

☺

Theorem 19.1.4 The Weierstrass M-Test

Let $\{f_n\}$ be a sequence of functions defined on A , and suppose that $\{M_n\}$ is a sequence of numbers such that

$$|f_n(x)| \leq M_n \quad \text{for all } x \in A.$$

Suppose moreover that $\sum_{n=1}^{\infty} M_n$ converges. Then for each x in A the series $\sum_{n=1}^{\infty} f_n(x)$ converges (in fact, it converges absolutely), and $\sum_{n=1}^{\infty} f_n$ converges uniformly on A to the function

$$f(x) = \sum_{n=1}^{\infty} f_n(x).$$

Proof: For each x in A the series $\sum_{n=1}^{\infty} |f_n(x)|$ converges, by the comparison test; consequently $\sum_{n=1}^{\infty} f_n(x)$ converges (absolutely). Moreover, for all x in A we have

$$\begin{aligned} |f(x) - [f_1(x) + \cdots + f_N(x)]| &= \left| \sum_{n=N+1}^{\infty} f_n(x) \right| \\ &\leq \sum_{n=N+1}^{\infty} |f_n(x)| \\ &\leq \sum_{n=N+1}^{\infty} M_n. \end{aligned}$$

Since $\sum_{n=1}^{\infty} M_n$ converges, the number $\sum_{n=N+1}^{\infty} M_n$ can be made as small as desired, by choosing N sufficiently large. ☺

Theorem 19.1.5 A Continuous Nowhere Differentiable Function

The function

$$f(x) = \sum_{n=1}^{\infty} \frac{1}{10^n} \{10^n x\}$$

is continuous everywhere and differentiable nowhere!

Here $\{y\} = y - [y]$ denotes the fractional part of y .

Proof: We have just shown that f is continuous; this is the only part of the proof which uses uniform convergence. We will prove that f is not differentiable at a , for any a , by the straightforward method of exhibiting a particular sequence $\{h_m\}$ approaching 0 for which

$$\lim_{m \rightarrow \infty} \frac{f(a + h_m) - f(a)}{h_m}$$

does not exist. It obviously suffices to consider only those numbers a satisfying $0 < a \leq 1$.

Suppose that the decimal expansion of a is

$$a = 0.a_1a_2a_3a_4 \dots$$

Let $h_m = 10^{-m}$ if $a_m \neq 4$ or 9 , but let $h_m = -10^{-m}$ if $a_m = 4$ or 9 (the reason for these two exceptions will appear soon). Then

$$\begin{aligned}\frac{f(a + h_m) - f(a)}{h_m} &= \sum_{n=1}^{\infty} \frac{1}{10^n} \cdot \frac{\{10^n(a + h_m)\} - \{10^n a\}}{\pm 10^{-m}} \\ &= \sum_{n=1}^{\infty} \pm 10^{m-n} [\{10^n(a + h_m)\} - \{10^n a\}].\end{aligned}$$

This infinite series is really a finite sum, because if $n \geq m$, then $10^n h_m$ is an integer, so

$$\{10^n(a + h_m)\} - \{10^n a\} = 0.$$

On the other hand, for $n < m$ we can write

$$\begin{aligned}10^n a &= \text{integer} + 0.a_{n+1}a_{n+2}a_{n+3} \dots a_m \dots \\ 10^n(a + h_m) &= \text{integer} + 0.a_{n+1}a_{n+2}a_{n+3} \dots (a_m \pm 1) \dots\end{aligned}$$

(in order for the second equation to be true it is essential that we choose $h_m = -10^{-m}$ when $a_m = 9$). Now suppose that

$$0.a_{n+1}a_{n+2}a_{n+3} \dots a_m \dots \leq \frac{1}{2}.$$

Then we also have

$$0.a_{n+1}a_{n+2}a_{n+3} \dots (a_m \pm 1) \dots \leq \frac{1}{2}$$

(in the special case $m = n + 1$ the second equation is true because we chose $h_m = -10^{-m}$ when $a_m = 4$). This means that

$$\{10^n(a + h_m)\} - \{10^n a\} = \pm 10^{n-m},$$

and exactly the same equation can be derived when $0.a_{n+1}a_{n+2}a_{n+3} \dots > \frac{1}{2}$. Thus, for $n < m$ we have

$$10^{m-n} [\{10^n(a + h_m)\} - \{10^n a\}] = \pm 1.$$

In other words,

$$\frac{f(a + h_m) - f(a)}{h_m}$$

is the sum of $m - 1$ numbers, each of which is ± 1 . Now adding $+1$ or -1 to a number changes it from odd to even, and vice versa. The sum of $m - 1$ numbers each ± 1 is therefore an *even integer* if m is odd, and an *odd integer* if m is even. Consequently the sequence of ratios

$$\frac{f(a + h_m) - f(a)}{h_m}$$

cannot possibly converge, since it is a sequence of integers which are alternately odd and even. ☺

19.2 Power Series

Definition 19.2.1: Power Series

A **power series** centered at a is a series of the form

$$\sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots.$$

Theorem 19.2.1 Radius of Convergence

For every power series $\sum c_n(x-a)^n$, there exists $R \in [0, \infty]$ (the **radius of convergence**) such that:

1. The series converges absolutely for $|x-a| < R$.
2. The series diverges for $|x-a| > R$.
3. At $|x-a| = R$, the series may converge or diverge.

Note:-

The radius of convergence can be computed by

$$\frac{1}{R} = \limsup_{n \rightarrow \infty} |c_n|^{1/n},$$

or, when the limit exists, by the ratio test:

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right|.$$

Theorem 19.2.2 Uniform Convergence and Differentiability of Power Series

Suppose that the series

$$f(x_0) = \sum_{n=0}^{\infty} a_n x_0^n$$

converges, and let a be any number with $0 < a < |x_0|$. Then on $[-a, a]$ the series

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

converges uniformly (and absolutely). Moreover, the same is true for the series

$$g(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}.$$

Finally, f is differentiable and

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

for all x with $|x| < |x_0|$.

Proof: Since $\sum_{n=0}^{\infty} a_n x_0^n$ converges, the terms $a_n x_0^n$ approach 0. Hence they are surely bounded: there is some number M such that

$$|a_n x_0^n| = |a_n| \cdot |x_0|^n \leq M \quad \text{for all } n.$$

Now if x is in $[-a, a]$, then $|x| \leq |a|$, so

$$\begin{aligned} |a_n x^n| &= |a_n| \cdot |x|^n \\ &\leq |a_n| \cdot |a|^n \\ &= |a_n| \cdot |x_0|^n \cdot \left| \frac{a}{x_0} \right|^n \\ &\leq M \left| \frac{a}{x_0} \right|^n. \end{aligned}$$

But $|a/x_0| < 1$, so the (geometric) series

$$\sum_{n=0}^{\infty} M \left| \frac{a}{x_0} \right|^n = M \sum_{n=0}^{\infty} \left| \frac{a}{x_0} \right|^n$$

converges. Choosing $M \cdot |a/x_0|^n$ as the number M_n in the Weierstrass M -test, it follows that $\sum_{n=0}^{\infty} a_n x^n$ converges uniformly on $[-a, a]$.

To prove the same assertion for $g(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$ notice that

$$\begin{aligned} |n a_n x^{n-1}| &= n |a_n| \cdot |x|^{n-1} \\ &\leq n |a_n| \cdot |a|^{n-1} \\ &= \frac{|a_n|}{|a|} \cdot |x_0|^n n \left| \frac{a}{x_0} \right|^n \\ &\leq \frac{M}{|a|} n \left| \frac{a}{x_0} \right|^n. \end{aligned}$$

Since $|a/x_0| < 1$, the series

$$\sum_{n=1}^{\infty} \frac{M}{|a|} n \left| \frac{a}{x_0} \right|^n = \frac{M}{|a|} \sum_{n=1}^{\infty} n \left| \frac{a}{x_0} \right|^n$$

converges (this fact was proved in Chapter 23 as an application of the ratio test). Another appeal to the Weierstrass M -test proves that $\sum_{n=1}^{\infty} n a_n x^{n-1}$ converges uniformly on $[-a, a]$.

Finally, our corollary proves, first that g is continuous, and then that

$$f'(x) = g(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} \quad \text{for } x \text{ in } [-a, a].$$

Since we could have chosen any number a with $0 < a < |x_0|$, this result holds for all x with $|x| < |x_0|$. ⊖

Chapter 20

Week 9 (Winter)

Note:-

Additional practice on parametrized curves, arc length, and improper integrals can be found in Chapter 23, Section 23.9.

20.1 Parametrized Curves

Definition 20.1.1: Parametrized Curve

A **parametrized curve** in $\mathbb{R}^2$ is a pair of continuous functions $\gamma(t) = (x(t), y(t))$ defined on an interval $[a, b]$.

Definition 20.1.2: Polar Coordinates

A point in $\mathbb{R}^2$ can be described by **polar coordinates** (r, θ) , where

$$x = r \cos \theta, \quad y = r \sin \theta.$$

A curve given in polar form $r = f(\theta)$ can be parametrized as $\gamma(\theta) = (f(\theta) \cos \theta, f(\theta) \sin \theta)$.

Definition 20.1.3: Velocity and Speed

If $\gamma(t) = (x(t), y(t))$ is differentiable, the **velocity** is $\gamma'(t) = (x'(t), y'(t))$ and the **speed** is

$$|\gamma'(t)| = \sqrt{x'(t)^2 + y'(t)^2}.$$

20.2 Arc Length

Definition 20.2.1: Arc Length

Let $\gamma(t) = (x(t), y(t))$ be a curve on $[a, b]$. The **length** of γ is

$$L(\gamma) = \sup_P \sum_{i=1}^n |\gamma(t_i) - \gamma(t_{i-1})|,$$

where the supremum is over all partitions $P = \{t_0, \dots, t_n\}$ of $[a, b]$, and $|\cdot|$ denotes the Euclidean distance. If $L(\gamma) < \infty$, the curve is called **rectifiable**.

Theorem 20.2.1 Arc Length Integral Formula

If x' and y' are continuous on $[a, b]$, then γ is rectifiable and

$$L(\gamma) = \int_a^b \sqrt{x'(t)^2 + y'(t)^2} dt.$$

Proof: For any partition $P = \{t_0, \dots, t_n\}$, by the Mean Value Theorem applied to each component, there exist $c_i, d_i \in (t_{i-1}, t_i)$ with

$$x(t_i) - x(t_{i-1}) = x'(c_i)(t_i - t_{i-1}), \quad y(t_i) - y(t_{i-1}) = y'(d_i)(t_i - t_{i-1}).$$

Thus

$$\sum_{i=1}^n |\gamma(t_i) - \gamma(t_{i-1})| = \sum_{i=1}^n \sqrt{x'(c_i)^2 + y'(d_i)^2} (t_i - t_{i-1}).$$

This is close to a Riemann sum for $\int_a^b \sqrt{x'^2 + y'^2} dt$. A careful argument using uniform continuity of x' and y' shows that the supremum over all partitions equals the integral. $\odot$

Example 20.2.1 (Arc length of a circle)

Parametrize the unit circle by $\gamma(t) = (\cos t, \sin t)$ for $t \in [0, 2\pi]$. Then $\gamma'(t) = (-\sin t, \cos t)$ and $|\gamma'(t)| = 1$, so

$$L = \int_0^{2\pi} 1 dt = 2\pi.$$

Chapter 21

Week 10 (Winter)

Note:-

Additional practice on applications of the integral can be found in Chapter 23, Section 23.9.

21.1 Applications of the Integral

Cavalieri's Principle

Note:-

Cavalieri's Principle: If two solids have equal cross-sectional areas at every height, then they have equal volumes. More precisely, if $A(x)$ is the cross-sectional area of a solid at height x , then

$$V = \int_a^b A(x) dx.$$

Volumes of Solids of Revolution

Theorem 21.1.1 Disk Method

If the region under $y = f(x) \geq 0$ on $[a, b]$ is revolved about the x -axis, the volume of the resulting solid is

$$V = \pi \int_a^b [f(x)]^2 dx.$$

Proof: At position x , the cross-section is a disk of radius $f(x)$, so $A(x) = \pi[f(x)]^2$. By Cavalieri's principle, $V = \int_a^b A(x) dx$. ☺

Example 21.1.1 (Volume of a sphere)

Revolve $f(x) = \sqrt{r^2 - x^2}$ on $[-r, r]$ about the x -axis:

$$V = \pi \int_{-r}^r (r^2 - x^2) dx = \pi \left[r^2x - \frac{x^3}{3} \right]_{-r}^r = \frac{4}{3}\pi r^3.$$

Surface Area of Solids of Revolution

Theorem 21.1.2 Surface Area Formula

If f is continuously differentiable on $[a, b]$ with $f(x) \geq 0$, the surface area of the solid obtained by revolving $y = f(x)$ about the x -axis is

$$S = 2\pi \int_a^b f(x) \sqrt{1 + [f'(x)]^2} dx.$$

Example 21.1.2 (Surface area of a sphere)

With $f(x) = \sqrt{r^2 - x^2}$ on $[-r, r]$, we have $f'(x) = \frac{-x}{\sqrt{r^2 - x^2}}$, so

$$\sqrt{1 + [f'(x)]^2} = \frac{r}{\sqrt{r^2 - x^2}}.$$

Thus

$$S = 2\pi \int_{-r}^r \sqrt{r^2 - x^2} \cdot \frac{r}{\sqrt{r^2 - x^2}} dx = 2\pi r \int_{-r}^r dx = 4\pi r^2.$$

Chapter 22

Week 11 (Winter)

Note:-

Additional practice on Dedekind cuts and foundational constructions can be found in Chapter 23, Section 23.1.

22.1 The Construction of $\mathbb{R}$ by Dedekind Cuts

Definition 22.1.1: Real Number (Dedekind Cut)

A **real number** is a set α , of rational numbers, with the following four properties:

1. If x is in α and y is a rational number with $y < x$, then y is also in α .
2. $\alpha \neq \emptyset$.
3. $\alpha \neq \mathbb{Q}$.
4. There is no greatest element in α ; in other words, if x is in α , then there is some y in α with $y > x$.

The set of all real numbers is denoted by $\mathbb{R}$.

Definition 22.1.2: Order on Cuts

For cuts α, β , define $\alpha < \beta$ if and only if $\alpha \subsetneq \beta$.

Definition 22.1.3: Addition of Cuts

If α and β are real numbers, then

$$\alpha + \beta = \{x : x = y + z \text{ for some } y \text{ in } \alpha \text{ and some } z \text{ in } \beta\}.$$

Theorem 22.1.1 Closure of Addition

If α and β are real numbers, then $\alpha + \beta$ is a real number.

Proof: Once again four facts must be verified.

1. Suppose $w < x$ for some x in $\alpha + \beta$. Then $x = y + z$ for some y in α and some z in β , which means that $w < y + z$, and consequently, $w - y < z$. This shows that $w - y$ is in β (since z is in β , and β is a real number). Since $w = y + (w - y)$, it follows that w is in $\alpha + \beta$.
2. It is clear that $\alpha + \beta \neq \emptyset$, since $\alpha \neq \emptyset$ and $\beta \neq \emptyset$.
3. Since $\alpha \neq \mathbb{Q}$ and $\beta \neq \mathbb{Q}$, there are rational numbers a not in α and b not in β . Any x in α satisfies $x < a$ (for if $a < x$, then condition (1) for a real number would imply that a is in α); similarly any y in β satisfies $y < b$. Thus $x + y < a + b$ for any x in α and y in β . This shows that $a + b$ is not in $\alpha + \beta$, so $\alpha + \beta \neq \mathbb{Q}$.
4. If x is in $\alpha + \beta$, then $x = y + z$ for y in α and z in β . There are y' in α and z' in β with $y < y'$ and $z < z'$; then $x < y' + z'$ and $y' + z'$ is in $\alpha + \beta$. Thus $\alpha + \beta$ has no greatest member.

☺

Theorem 22.1.2 Associativity of Addition

If α , β , and γ are real numbers, then $(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$.

Proof: Since $(x + y) + z = x + (y + z)$ for all rational numbers x , y , and z , every member of $(\alpha + \beta) + \gamma$ is also a member of $\alpha + (\beta + \gamma)$, and vice versa.

☺

Theorem 22.1.3 Commutativity of Addition

If α and β are real numbers, then $\alpha + \beta = \beta + \alpha$.

Proof: Left to you (even easier).

☺

To prove the other properties of $+$ we first define $\mathbf{0}$.

Definition 22.1.4: The Real Number $\mathbf{0}$

$$\mathbf{0} = \{x \text{ in } \mathbb{Q} : x < 0\}.$$

Note:-

It is obvious that $\mathbf{0}$ is a real number.

Theorem 22.1.4 Additive Identity

If α is a real number, then $\alpha + \mathbf{0} = \alpha$.

Proof: If x is in α and y is in $\mathbf{0}$, then $y < 0$, so $x + y < x$. This implies that $x + y$ is in α . Thus every member of $\alpha + \mathbf{0}$ is also a member of α .

On the other hand, if x is in α , then there is a rational number y in α such that $y > x$. Since $x = y + (x - y)$, where y is in α , and $x - y < 0$ (so that $x - y$ is in $\mathbf{0}$), this shows that x is in $\alpha + \mathbf{0}$. Thus every member of α is also a member of $\alpha + \mathbf{0}$.

☺

The reasonable candidate for $-\alpha$ would seem to be the set

$$\{x \text{ in } \mathbb{Q} : -x \text{ is not in } \alpha\}$$

(since $-x$ not in α means, intuitively, that $-x > \alpha$, so that $x < -\alpha$). But in certain cases this set will not even be a real number. Although a real number α does not have a greatest member, the set

$$\mathbb{Q} - \alpha = \{x \text{ in } \mathbb{Q} : x \text{ is not in } \alpha\}$$

may have a *least* element x_0 ; when α is a real number of this kind, the set $\{x : -x \text{ is not in } \alpha\}$ will have a greatest element $-x_0$. It is therefore necessary to introduce a slight modification into the definition of $-\alpha$, which comes equipped with a theorem.

Definition 22.1.5: Additive Inverse

If α is a real number, then

$$-\alpha = \{x \text{ in } \mathbb{Q} : -x \text{ is not in } \alpha, \text{ but } -x \text{ is not the least element of } \mathbb{Q} - \alpha\}.$$

Theorem 22.1.5 $-\alpha$ is a Real Number

If α is a real number, then $-\alpha$ is a real number.

Proof:

1. Suppose that x is in $-\alpha$ and $y < x$. Then $-y > -x$. Since $-x$ is not in α , it is also true that $-y$ is not in α . Moreover, it is clear that $-y$ is not the smallest element of $\mathbb{Q} - \alpha$, since $-x$ is a smaller element. This shows that y is in $-\alpha$.
2. Since $\alpha \neq \mathbb{Q}$, there is some rational number y which is not in α . We can assume that y is not the smallest rational number in $\mathbb{Q} - \alpha$ (since y can always be replaced by any $y' > y$). Then $-y$ is in $-\alpha$. Thus $-\alpha \neq \emptyset$.
3. Since $\alpha \neq \emptyset$, there is some x in α . Then $-x$ cannot possibly be in $-\alpha$, so $-\alpha \neq \mathbb{Q}$.
4. If x is in $-\alpha$, then $-x$ is not in α , and there is a rational number $y < -x$ which is also not in α . Let z be a rational number with $y < z < -x$. Then z is also not in α , and z is clearly not the smallest element of $\mathbb{Q} - \alpha$. So $-z$ is in $-\alpha$. Since $-z > x$, this shows that $-\alpha$ does not have a greatest element.

☺

Lemma 22.1.1 Key Lemma for Additive Inverse

Let α be a real number, and z a positive rational number. Then there are rational numbers x in α , and y not in α , such that $y - x = z$. Moreover, we may assume that y is not the smallest element of $\mathbb{Q} - \alpha$.

Proof: Suppose first that z is in α . If the numbers

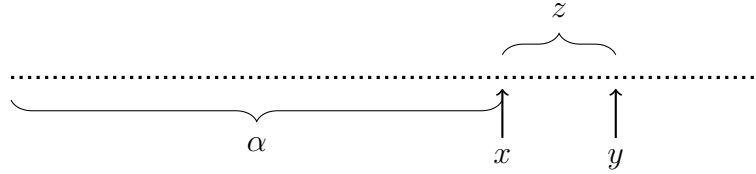
$$z, 2z, 3z, \dots$$

were *all* in α , then *every* rational number would be in α , since every rational number w satisfies $w < nz$ for some n , by the Archimedean property. This contradicts the fact that α is a real

number, so there is some k such that $x = kz$ is in α and $y = (k+1)z$ is not in α . Clearly $y - x = z$.

Moreover, if y happens to be the smallest element of $\mathbb{Q} - \alpha$, let $x' > x$ be an element of α , and replace x by x' , and y by $y + (x' - x)$.

If z is not in α , there is a similar proof, based on the fact that the numbers $(-n)z$ cannot all fail to be in α . ☺



Theorem 22.1.6 Additive Inverse Property

If α is a real number, then

$$\alpha + (-\alpha) = \mathbf{0}.$$

Proof: Suppose x is in α and y is in $-\alpha$. Then $-y$ is not in α , so $-y > x$. Hence $x + y < 0$, so $x + y$ is in $\mathbf{0}$. Thus every member of $\alpha + (-\alpha)$ is in $\mathbf{0}$.

It is a little more difficult to go in the other direction. If z is in $\mathbf{0}$, then $-z > 0$. According to the lemma, there is some x in α , and some y not in α , with y not the smallest element of $\mathbb{Q} - \alpha$, such that $y - x = -z$. This equation can be written $x + (-y) = z$. Since x is in α , and $-y$ is in $-\alpha$, this proves that z is in $\alpha + (-\alpha)$. ☺

Before proceeding with multiplication, we define the “positive elements” and prove a basic property:

Definition 22.1.6: Positive Real Numbers

$$\mathbf{P} = \{\alpha \text{ in } \mathbb{R} : \alpha > \mathbf{0}\}.$$

Note:-

Notice that $\alpha + \beta$ is clearly in $\mathbf{P}$ if α and β are.

Theorem 22.1.7 Trichotomy

If α is a real number, then one and only one of the following conditions holds:

- (i) $\alpha = \mathbf{0}$,
- (ii) α is in $\mathbf{P}$,
- (iii) $-\alpha$ is in $\mathbf{P}$.

Proof: If α contains any positive rational number, then α certainly contains all negative rational numbers, so α contains $\mathbf{0}$ and $\alpha \neq \mathbf{0}$, i.e., α is in $\mathbf{P}$. If α contains no positive rational numbers, then one of two possibilities must hold:

1. α contains all negative rational numbers; then $\alpha = \mathbf{0}$.

2. there is some negative rational number x which is not in α ; it can be assumed that x is not the least element of $\mathbb{Q} - \alpha$ (since x could be replaced by $x/2 > x$); then $-\alpha$ contains the positive rational number $-x$, so, as we have just proved, $-\alpha$ is in $\mathbf{P}$.

This shows that *at least one* of (i)–(iii) must hold. If $\alpha = \mathbf{0}$, it is clearly impossible for condition (ii) or (iii) to hold. Moreover, it is impossible that $\alpha > \mathbf{0}$ and $-\alpha > \mathbf{0}$ both hold, since this would imply that $\mathbf{0} = \alpha + (-\alpha) > \mathbf{0}$. $\odot$

Theorem 22.1.8 Addition Preserves Order

If α , β , and γ are real numbers and $\alpha > \beta$, then $\alpha + \gamma > \beta + \gamma$.

Proof: The hypothesis $\alpha > \beta$ implies that β is contained in α ; it follows immediately from the definition of $+$ that $\beta + \gamma$ is contained in $\alpha + \gamma$. This shows that $\alpha + \gamma \geq \beta + \gamma$. We can easily rule out the possibility of equality, for if

$$\alpha + \gamma = \beta + \gamma,$$

then

$$\alpha = (\alpha + \gamma) + (-\gamma) = (\beta + \gamma) + (-\gamma) = \beta,$$

which is false. Thus $\alpha + \gamma > \beta + \gamma$. $\odot$

Multiplication presents difficulties of its own. If $\alpha, \beta > \mathbf{0}$, then $\alpha \cdot \beta$ can be defined as follows.

Definition 22.1.7: Multiplication of Positive Cuts

If α and β are real numbers and $\alpha, \beta > \mathbf{0}$, then

$$\alpha \cdot \beta = \{z : z \leq 0 \text{ or } z = x \cdot y \text{ for some } x \text{ in } \alpha \text{ and } y \text{ in } \beta \text{ with } x, y > 0\}.$$

Theorem 22.1.9 Closure of Multiplication (Positive Case)

If α and β are real numbers with $\alpha, \beta > \mathbf{0}$, then $\alpha \cdot \beta$ is a real number.

Proof: As usual, we must check four conditions.

1. Suppose $w < z$, where z is in $\alpha \cdot \beta$. If $w \leq 0$, then w is automatically in $\alpha \cdot \beta$. Suppose that $w > 0$. Then $z > 0$, so $z = x \cdot y$ for some positive x in α and positive y in β . Now

$$w = \frac{wz}{z} = \frac{wxy}{z} = \left(\frac{w}{z} \cdot x\right) \cdot y.$$

Since $0 < w < z$, we have $w/z < 1$, so $(w/z) \cdot x$ is in α . Thus w is in $\alpha \cdot \beta$.

2. Clearly $\alpha \cdot \beta \neq \emptyset$.
3. If x is not in α , and y is not in β , then $x > x'$ for all x' in α , and $y > y'$ for all y' in β . Hence $xy > x'y'$ for all such positive x' and y' . So xy is not in $\alpha \cdot \beta$; thus $\alpha \cdot \beta \neq \mathbb{Q}$.

4. Suppose w is in $\alpha \cdot \beta$, and $w \leq 0$. There is some x in α with $x > 0$ and some y in β with $y > 0$. Then $z = xy$ is in $\alpha \cdot \beta$ and $z > w$. Now suppose $w > 0$. Then $w = xy$ for some positive x in α and some positive y in β . Moreover, α contains some $x' > x$; if $z = x'y$, then $z > xy = w$, and z is in $\alpha \cdot \beta$. Thus $\alpha \cdot \beta$ does not have a greatest element.

☺

Note:-

Notice that $\alpha \cdot \beta$ is clearly in $\mathbf{P}$ if α and β are. This completes the verification of all properties of $\mathbf{P}$.

To complete the definition of $\cdot$ we first define $|\alpha|$.

Definition 22.1.8: Absolute Value of a Cut

If α is a real number, then

$$|\alpha| = \begin{cases} \alpha, & \text{if } \alpha \geq 0, \\ -\alpha, & \text{if } \alpha \leq 0. \end{cases}$$

Definition 22.1.9: Multiplication of Cuts (General)

If α and β are real numbers, then

$$\alpha \cdot \beta = \begin{cases} 0, & \text{if } \alpha = 0 \text{ or } \beta = 0, \\ |\alpha| \cdot |\beta|, & \text{if } \alpha > 0, \beta > 0 \text{ or } \alpha < 0, \beta < 0, \\ -(|\alpha| \cdot |\beta|), & \text{if } \alpha > 0, \beta < 0 \text{ or } \alpha < 0, \beta > 0. \end{cases}$$

Note:-

As one might suspect, the proofs of the properties of multiplication usually involve reduction to the case of positive numbers.

Theorem 22.1.10 Associativity of Multiplication

If α , β , and γ are real numbers, then $\alpha \cdot (\beta \cdot \gamma) = (\alpha \cdot \beta) \cdot \gamma$.

Proof: This is clear if $\alpha, \beta, \gamma > 0$. The proof for the general case requires considering separate cases (and is simplified slightly if one uses the following theorem). ☺

Theorem 22.1.11 Commutativity of Multiplication

If α and β are real numbers, then $\alpha \cdot \beta = \beta \cdot \alpha$.

Proof: This is clear if $\alpha, \beta > 0$, and the other cases are easily checked. ☺

Definition 22.1.10: The Real Number 1

$$1 = \{x \text{ in } \mathbb{Q} : x < 1\}.$$

(It is clear that **1** is a real number.)

Theorem 22.1.12 Multiplicative Identity

If α is a real number, then $\alpha \cdot \mathbf{1} = \alpha$.

Proof: Let $\alpha > \mathbf{0}$. It is easy to see that every member of $\alpha \cdot \mathbf{1}$ is also a member of α . On the other hand, suppose x is in α . If $x \leq 0$, then x is automatically in $\alpha \cdot \mathbf{1}$. If $x > 0$, then there is some rational number y in α such that $x < y$. Then $x = y \cdot (x/y)$, and x/y is in $\mathbf{1}$, so x is in $\alpha \cdot \mathbf{1}$. This proves that $\alpha \cdot \mathbf{1} = \alpha$ if $\alpha > \mathbf{0}$.

If $\alpha < \mathbf{0}$, then, applying the result just proved, we have

$$\alpha \cdot \mathbf{1} = -(|\alpha| \cdot |\mathbf{1}|) = -(|\alpha|) = \alpha.$$

Finally, the theorem is obvious when $\alpha = \mathbf{0}$. ☺

Definition 22.1.11: Multiplicative Inverse

If α is a real number and $\alpha > \mathbf{0}$, then

$$\begin{aligned} \alpha^{-1} = \{x \text{ in } \mathbb{Q} : & x \leq 0, \text{ or} \\ & x > 0 \text{ and } 1/x \text{ is not in } \alpha, \\ & \text{but } 1/x \text{ is not the smallest member of } \mathbb{Q} - \alpha\}; \end{aligned}$$

if $\alpha < \mathbf{0}$, then $\alpha^{-1} = -(|\alpha|)^{-1}$.

Theorem 22.1.13 α^{-1} is a Real Number

If α is a real number unequal to $\mathbf{0}$, then α^{-1} is a real number.

Proof: Clearly it suffices to consider only $\alpha > \mathbf{0}$. Four conditions must be checked.

1. Suppose $y < x$, and x is in α^{-1} . If $y \leq 0$, then y is in α^{-1} . If $y > 0$, then $x > 0$, so $1/x$ is not in α . Since $1/y > 1/x$, it follows that $1/y$ is not in α , and $1/y$ is clearly not the smallest element of $\mathbb{Q} - \alpha$, so y is in α^{-1} .
2. Clearly $\alpha^{-1} \neq \emptyset$.
3. Since $\alpha > \mathbf{0}$, there is some positive rational number x in α . Then $1/x$ is not in α^{-1} , so $\alpha^{-1} \neq \mathbb{Q}$.
4. Suppose x is in α^{-1} . If $x \leq 0$, there is clearly some y in α^{-1} with $y > x$ because α^{-1} contains some positive rationals. If $x > 0$, then $1/x$ is not in α . Since $1/x$ is not the smallest member of $\mathbb{Q} - \alpha$, there is a rational number y not in α , with $y < 1/x$. Choose a rational z with $y < z < 1/x$. Then $1/z$ is in α^{-1} , and $1/z > x$. Thus α^{-1} does not contain a largest member.

☺

In order to prove that α^{-1} is really the multiplicative inverse of α , it helps to have another lemma, which is the multiplicative analogue of our first lemma.

Lemma 22.1.2 Key Lemma for Multiplicative Inverse

Let α be a real number with $\alpha > \mathbf{0}$, and z a rational number with $z > 1$. Then there are rational numbers x in α , and y not in α , such that $y/x = z$. Moreover, we can assume that y is not the least element of $\mathbb{Q} - \alpha$.

Proof: Suppose first that z is in α . Since $z - 1 > 0$ and

$$z^n = (1 + (z - 1))^n \geq 1 + n(z - 1),$$

it follows that the numbers

$$z, z^2, z^3, \dots$$

cannot all be in α . So there is some k such that $x = z^k$ is in α , and $y = z^{k+1}$ is not in α . Clearly $y/x = z$. Moreover, if y happens to be the least element of $\mathbb{Q} - \alpha$, let $x' > x$ be an element of α , and replace x by x' and y by yx'/x .

If z is not in α , there is a similar proof, based on the fact that the numbers $1/z^k$ cannot all fail to be in α . ☺

Theorem 22.1.14 Multiplicative Inverse Property

If α is a real number and $\alpha \neq \mathbf{0}$, then $\alpha \cdot \alpha^{-1} = \mathbf{1}$.

Proof: It obviously suffices to consider only $\alpha > \mathbf{0}$, in which case $\alpha^{-1} > \mathbf{0}$. Suppose that x is a positive rational number in α , and y is a positive rational number in α^{-1} . Then $1/y$ is not in α , so $1/y > x$; consequently $xy < 1$, which means that xy is in $\mathbf{1}$. Since all rational numbers $x \leq 0$ are also in $\mathbf{1}$, this shows that every member of $\alpha \cdot \alpha^{-1}$ is in $\mathbf{1}$.

To prove the converse assertion, let z be in $\mathbf{1}$. If $z \leq 0$, then clearly z is in $\alpha \cdot \alpha^{-1}$. Suppose $0 < z < 1$. According to the lemma, there are positive rational numbers x in α , and y not in α , such that $y/x = 1/z$; and we can assume that y is not the smallest element of $\mathbb{Q} - \alpha$. But this means that $z = x \cdot (1/y)$, where x is in α , and $1/y$ is in α^{-1} . Consequently, z is in $\alpha \cdot \alpha^{-1}$. ☺

We are almost done! Only the proof of the distributive law remains.

Theorem 22.1.15 Distributive Law

If α , β , and γ are real numbers, then $\alpha \cdot (\beta + \gamma) = \alpha \cdot \beta + \alpha \cdot \gamma$.

Proof: Assume first that $\alpha, \beta, \gamma > \mathbf{0}$. Then both numbers in the equation contain all rational numbers ≤ 0 . A positive rational number in $\alpha \cdot (\beta + \gamma)$ is of the form $x \cdot (y + z)$ for positive x in α , y in β , and z in γ . Since $x \cdot (y + z) = x \cdot y + x \cdot z$, where $x \cdot y$ is a positive element of $\alpha \cdot \beta$, and $x \cdot z$ is a positive element of $\alpha \cdot \gamma$, this number is also in $\alpha \cdot \beta + \alpha \cdot \gamma$. Thus, every element of $\alpha \cdot (\beta + \gamma)$ is also in $\alpha \cdot \beta + \alpha \cdot \gamma$.

On the other hand, a positive rational number in $\alpha \cdot \beta + \alpha \cdot \gamma$ is of the form $x_1 \cdot y + x_2 \cdot z$ for positive x_1, x_2 in α , y in β , and z in γ . If $x_1 \leq x_2$, then $(x_1/x_2) \cdot y \leq y$, so $(x_1/x_2) \cdot y$ is in β . Thus

$$x_1 \cdot y + x_2 \cdot z = x_2[(x_1/x_2)y + z]$$

is in $\alpha \cdot (\beta + \gamma)$. Of course, the same trick works if $x_2 \leq x_1$.

To complete the proof it is necessary to consider the cases when α , β , and γ are not all > 0 . If any one of the three equals 0 , the proof is easy and the cases involving $\alpha < 0$ can be derived immediately once all the possibilities for β and γ have been accounted for. Thus we assume $\alpha > 0$ and consider three cases: $\beta, \gamma < 0$, and $\beta < 0, \gamma > 0$, and $\beta > 0, \gamma < 0$. The first follows immediately from the case already proved, and the third follows from the second by interchanging β and γ . Therefore we concentrate on the case $\beta < 0, \gamma > 0$. There are then two possibilities:

(1) $\beta + \gamma \geq 0$. Then

$$\alpha \cdot \gamma = \alpha \cdot (|\beta + \gamma| + |\beta|) = \alpha \cdot (\beta + \gamma) + \alpha \cdot |\beta|,$$

so

$$\begin{aligned} \alpha \cdot (\beta + \gamma) &= -(\alpha \cdot |\beta|) + \alpha \cdot \gamma \\ &= \alpha \cdot \beta + \alpha \cdot \gamma. \end{aligned}$$

(2) $\beta + \gamma \leq 0$. Then

$$\alpha \cdot |\beta| = \alpha \cdot (|\beta + \gamma| + \gamma) = \alpha \cdot |\beta + \gamma| + \alpha \cdot \gamma,$$

so

$$\alpha \cdot (\beta + \gamma) = -(\alpha \cdot |\beta + \gamma|) = -(\alpha \cdot |\beta|) + \alpha \cdot \gamma = \alpha \cdot \beta + \alpha \cdot \gamma. \quad \ominus$$

Theorem 22.1.16 Least Upper Bound Property of $\mathbb{R}$

If A is a set of real numbers and $A \neq \emptyset$ and A is bounded above, then A has a least upper bound.

Proof: Let $\beta = \{x : x \text{ is in some } \alpha \text{ in } A\}$. Then β is certainly a collection of rational numbers; the proof that β is a real number requires checking four facts.

1. Suppose that x is in β and $y < x$. The first condition means that x is in α for some α in A . Since α is a real number, the assumption $y < x$ implies that y is in α . Therefore it is certainly true that y is in β .
2. Since $A \neq \emptyset$, there is some α in A . Since α is a real number, there is some x in α . This means that x is in β , so $\beta \neq \emptyset$.
3. Since A is bounded above, there is some real number γ such that $\alpha < \gamma$ for every α in A . Since γ is a real number, there is some rational number x which is not in γ . Now $\alpha < \gamma$ means that α is contained in γ , so it is also true that x is not in α for any α in A . This means that x is not in β ; so $\beta \neq \mathbb{Q}$.
4. Suppose that x is in β . Then x is in α for some α in A . Since α does not have a greatest member, there is some rational number y with $x < y$ and y in α . But this means that y is in β ; thus β does not have a greatest member.

These four observations prove that β is a real number. The proof that β is the least upper bound of A is easier. If α is in A , then clearly α is contained in β ; this means that $\alpha \leq \beta$, so β is an upper bound for A . On the other hand, if γ is an upper bound for A , then $\alpha \leq \gamma$ for every α in A ; this means that α is contained in γ , for every α in A , and this surely implies that β is contained in γ . *This*, in turn, means that $\beta \leq \gamma$; thus β is the least upper bound of A . $\odot$

Chapter 23

Additional Practice

Note:-

These problems are grouped by topic rather than by week. Each item cites its original source so that the source paper can be found quickly if a diagram or formatting detail is needed.

23.1 Foundations, Order, Cardinality, and Induction

Question 1: 2025 Homework 1, Question 1

Source: 2025 Homework 1, Question 1. Let S, T be finite sets with cardinalities $|S| < |T|$. Prove that a map $g : T \rightarrow S$ cannot be one-to-one. Hint: use the pigeonhole principle.

Question 2: 2025 Homework 1, Question 2

Source: 2025 Homework 1, Question 2. Adapt Cantor's diagonal argument for $[0, 1]$ to work with binary expansions instead of decimal expansions. For binary expansions, keep in mind that $0.01 = 0.1$ in binary notation.

Question 3: 2025 Homework 1, Question 3

Source: 2025 Homework 1, Question 3. Let S be a set, and let $g : \mathbb{N} \rightarrow S$ be onto. Prove that S is either finite or countably infinite. Hint: for $s \in S$, consider the smallest $n \in \mathbb{N}$ such that $g(n) = s$.

Question 4: 2025 Homework 1, Question 4

Source: 2025 Homework 1, Question 4. Let S be a set. The power set $\mathcal{P}(S)$ is the set of all subsets of S . Prove that $\mathcal{P}(S)$ cannot be countably infinite. Do not appeal to Cantor–Bernstein; instead distinguish the cases where S is finite and infinite.

Question 5: 2025 Homework 2, Question 1

Source: 2025 Homework 2, Question 1. Complete the square, justifying the main order steps from Properties (P10)–(P12).

- (a) Find the smallest possible value of $2x^2 - 5x + 1$.
- (b) Find all values of a such that $x^2 + axy + y^2 > 0$ whenever x and y are not both zero.

Question 6: 2025 Homework 2, Question 2

Source: 2025 Homework 2, Question 2. Let $V = \{(x_1, x_2) : x_1, x_2 \in \mathbb{Q}\}$ with addition

$$(x_1, x_2) \oplus (y_1, y_2) = (x_1 + y_1 + 1, x_2 + y_2 + 1).$$

Demonstrate that Properties (P1)–(P4) hold for $(V, \oplus)$. Optional curiosity: find a multiplication $\odot$ so that Properties (P5)–(P9) also hold.

Question 7: 2025 Homework 2, Question 3

Source: 2025 Homework 2, Question 3. Let $V = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ with the expected addition and multiplication:

$$(a + b\sqrt{2}) \oplus (c + d\sqrt{2}) = (a + c) + (b + d)\sqrt{2},$$

$$(a + b\sqrt{2}) \odot (c + d\sqrt{2}) = (ac + 2bd) + (ad + bc)\sqrt{2}.$$

- (a) Let $a, b \in \mathbb{Q}$ and $n \in \mathbb{N}$. Show that there exist rational c, d such that $(a + b\sqrt{2})^n = c + d\sqrt{2}$.
- (b) Define a positivity set $P \subseteq V$ using conditions on a and b that do not explicitly involve $\sqrt{2}$. Demonstrate that (P10) holds.
- (c) If $a + b\sqrt{2} \in P$ and $c + d\sqrt{2} \in P$, where $a > 0$ and $c < 0$, demonstrate that their sum and product are in P .

Question 8: 2025 Homework 2, Question 4

Source: 2025 Homework 2, Question 4. Prove Bernoulli's inequality: $(1 + h)^n \geq 1 + nh$ for all $n \in \mathbb{N}$ and all real $h > -1$.

Question 9: 2025 Homework 2, Question 5

Source: 2025 Homework 2, Question 5. A natural number n is even if $n = 2k$ for some $k \in \mathbb{N}$, and odd if $n = 2k - 1$. Prove directly by induction that every natural number is either even or odd.

Question 10: 2025 Homework 2, Question 6

Source: 2025 Homework 2, Question 6. In the Towers of Hanoi puzzle there are three pegs and n disks of decreasing diameter stacked on Peg 1. A disk may be moved from one peg to another if it is not placed on top of a smaller disk. Prove that the whole stack can be moved from Peg 1 to Peg 3 in $2^n - 1$ moves, and that fewer moves cannot

suffice.

Question 11: 2025 Homework 3, Question 1

Source: 2025 Homework 3, Question 1. Let $A \subset \mathbb{R}$ be a bounded non-empty subset. Prove that

$$\inf(A) = -\sup\{-x : x \in A\}.$$

Question 12: 2025 Homework 3, Question 2

Source: 2025 Homework 3, Question 2. Prove density statements.

- (a) Given real $x < y$, construct a rational $r = p/q$ with $x < r < y$. Hint: use the Archimedean property on $(y - x)^{-1}$.
- (b) Given rational $r < s$, construct an irrational a with $r < a < s$.

Question 13: 2025 Homework 3, Question 5

Source: 2025 Homework 3, Question 5. Fix $m > 1$. Two integers are congruent modulo m if their difference is an integer multiple of m .

- (a) Write out the first 9 rows of Pascal's triangle modulo 2.
- (b) Prove that, for $n = 0, 1, 2, \dots$,

$$\binom{2^n - 1}{k} \equiv 1 \pmod{2} \quad (0 \leq k \leq 2^n - 1),$$

and

$$\binom{2^n}{k} \equiv \begin{cases} 1 \pmod{2}, & k = 0, 2^n, \\ 0 \pmod{2}, & \text{otherwise.} \end{cases}$$

Question 14: 2025 Homework 3, Question 6

Source: 2025 Homework 3, Question 6. Prove uniqueness in the Fundamental Theorem of Arithmetic: if $n \in \mathbb{N}$ has a prime factorization $n = \prod_{j=1}^m p_j^{\alpha_j}$ with $p_1 < \dots < p_m$ and $\alpha_j > 0$, then this factorization is unique. You may use Euclid's lemma for primes.

Question 15: 2025 Homework 5, Question 6

Source: 2025 Homework 5, Question 6. Let M_n be the number of permutations of $\{1, \dots, n\}$ with no fixed points.

- (a) Prove that $M_{n+1} = n(M_n + M_{n-1})$ for $n \geq 2$, with $M_1 = 0$ and $M_2 = 1$.
- (b) Given the inclusion-exclusion formula $M_n = n! \sum_{k=0}^n (-1)^k / k!$, verify that it satisfies the same recursion by first proving $N_n = nN_{n-1} + (-1)^n$.

Question 16: 2023 Midterm 1, Question 1

Source: 2023 Midterm 1, Question 1. For the binomial coefficients $\binom{n}{k} = n!/(k!(n-k)!)$:

- (a) State the recursion formula.
- (b) Write down the first 6 rows of Pascal's triangle.
- (c) Describe two different interpretations of $\binom{n}{k}$, including the meanings of n and k .

Question 17: 2023 Midterm 1, Question 2

Source: 2023 Midterm 1, Question 2. State the supremum axiom and the Archimedean property. Then derive the Archimedean property from the supremum axiom, using that $\mathbb{R}$ is an ordered field satisfying (P1)–(P12), but without using Dedekind cuts.

Question 18: 2023 Midterm 1, Question 3

Source: 2023 Midterm 1, Question 3. Treat $\mathbb{R}$ as the set of Dedekind cuts with addition, multiplication, and order as defined in class.

- (a) Define what it means for a Dedekind cut α to be positive.
- (b) Prove directly from the definitions that, for every $\alpha > 0$, $\alpha + \alpha = 2 \cdot \alpha$, where $2 = \{q \in \mathbb{Q} : q < 2\}$.

Question 19: 2023 Midterm 1, Question 4

Source: 2023 Midterm 1, Question 4. Prove the arithmetic-geometric mean inequality.

- (a) If $0 < a \leq 1 \leq b$, show that $a + b \geq 1 + ab$.
- (b) If $a_1, \dots, a_n > 0$ and $a_1 \cdots a_n = 1$, prove by induction that $a_1 + \cdots + a_n \geq n$.
- (c) Conclude that, for positive $b_1, \dots, b_n$,

$$\frac{b_1 + \cdots + b_n}{n} \geq (b_1 \cdots b_n)^{1/n}.$$

Question 20: 2023 Midterm 1, Question 6

Source: 2023 Midterm 1, Question 6. Let $I_n = \{x \in \mathbb{R} : a_n \leq x \leq b_n\}$ be closed intervals with $I_{n+1} \subseteq I_n$, i.e. $a_n \leq a_{n+1} \leq b_{n+1} \leq b_n$. Prove that $\bigcap_{n \in \mathbb{N}} I_n \neq \emptyset$.

Question 21: 2024 Quiz 1, Question 1

Source: 2024 Quiz 1, Question 1. Let A be a non-empty set of real numbers bounded above.

- (a) Define $\sup A$.
- (b) If A has no maximum, prove that for every $n \in \mathbb{N}$ there is $a_n \in A$ such that $0 < \sup A - a_n < 1/n$.

Question 22: 2024 Quiz 1, Question 2

Source: 2024 Quiz 1, Question 2. Use completing the square.

- (a) Find all a such that $x^2 + 2xy + ay^2 > 0$ whenever x and y are not both zero.
- (b) Find the smallest possible value of $x + 1/x$ for $x > 0$, and the x where it is achieved.

Question 23: 2024 Quiz 1, Question 3

Source: 2024 Quiz 1, Question 3. Analyze Towers of Hanoi. Prove by induction that the stack of n disks can be moved from Peg 1 to Peg 3 in $2^n - 1$ steps, and prove that this is best possible.

Question 24: 2025 Quiz 1, Question 1

Source: 2025 Quiz 1, Question 1. Using Properties (P1)–(P9), algebraic manipulations, and the fact that squares are nonnegative, prove:

- (a) $x^2 + xy + y^2 > 0$ unless $x = y = 0$.
- (b) If $x < y$, then $x^3 < y^3$.

Whenever you use (P10)–(P12), state which ordering property is being used.

Question 25: 2025 Quiz 1, Question 2

Source: 2025 Quiz 1, Question 2. Write down a one-to-one and onto function $g : \mathbb{Z} \rightarrow \mathbb{N}$, and prove that it has both properties.

Question 26: 2025 Quiz 1, Question 3

Source: 2025 Quiz 1, Question 3. Consider

$$\sum_{j=1}^{2n} \frac{1}{j} > \frac{n}{2}.$$

Write the first three inequalities ($n = 1, 2, 3$) without summation notation, and then prove the inequality for all $n \in \mathbb{N}$ by induction.

Question 27: 2024 Final, Question 1

Source: 2024 Final, Question 1. State the supremum axiom for $\mathbb{R}$, the Mean Value Theorem, the Cauchy criterion for convergence of a sequence, and the integral comparison test for convergence of a series.

Question 28: 2025 Final, Question 1

Source: 2025 Final, Question 1. State the principle of induction, the ϵ - δ definition of continuity of a function at a point, the Intermediate Value Theorem, and the integral comparison test for convergence of a series.

Question 29: 2026 Makeup Final, Question 1

Source: 2026 Makeup Final, Question 1. State the ϵ - δ definition of $\lim_{x \rightarrow a} f(x) = L$, the Bolzano–Weierstrass Theorem, Taylor’s Theorem with Lagrange remainder, and the Weierstrass M -test.

23.2 Functions, Sets, and Inverses

Question 30: 2025 Homework 1, Question 5

Source: 2025 Homework 1, Question 5. Define

$$x(t) = \frac{2t}{t^2 + 1}, \quad y(t) = \frac{t^2 - 1}{t^2 + 1}.$$

Let $S^1 = \{(x, y) : x^2 + y^2 = 1\}$ and remove the north pole $(0, 1)$.

- (a) Verify that $g(t) = (x(t), y(t))$ maps $\mathbb{R}$ to $S^1 \setminus \{(0, 1)\}$.
- (b) Verify that $h(x, y) = x/(1 - y)$ maps $S^1 \setminus \{(0, 1)\}$ to $\mathbb{R}$.
- (c) Verify that g and h are inverses, hence one-to-one and onto.

Question 31: 2025 Homework 1, Question 6

Source: 2025 Homework 1, Question 6. A Pythagorean triple is a triple (a, b, c) of natural numbers satisfying $a^2 + b^2 = c^2$. Restrict to irreducible triples.

- (a) Define a one-to-one and onto map from irreducible Pythagorean triples to positive rational pairs (p, q) with $p^2 + q^2 = 1$.
- (b) Using the map $h : S^1 \setminus \{(0, 1)\} \rightarrow \mathbb{R}$ above, show that p, q are rational if and only if $h(p, q)$ is rational.
- (c) Conclude that there are infinitely many irreducible Pythagorean triples. Identify the part of $\mathbb{R}$ corresponding to the upper right quarter of S^1 .

Question 32: 2025 Homework 3, Question 3

Source: 2025 Homework 3, Question 3. Let X be a set, and define the indicator function I_A of $A \subseteq X$ by $I_A(x) = 1$ for $x \in A$ and $I_A(x) = 0$ otherwise.

- (a) Write $I_{A \cap B}$, $I_{A \cup B}$, and $I_{X \setminus A}$ in terms of I_X, I_A, I_B .
- (b) If f is real-valued on X , prove that $f^2 = f$ if and only if $f = I_A$ for some $A \subseteq X$.

Question 33: 2025 Homework 3, Question 4

Source: 2025 Homework 3, Question 4. Let $f : X \rightarrow Y$ and $g, h : Y \rightarrow X$.

- (a) If $f \circ g = \text{Id}_Y$, prove that f is surjective and g is injective.
- (b) If also $h \circ f = \text{Id}_X$, prove that f is invertible and $g = h = f^{-1}$.

Question 34: 2025 Homework 4, Question 1

Source: 2025 Homework 4, Question 1.

- (a) Prove that no real-valued functions f, g can satisfy either $f(x) + g(y) = xy$ for all $x, y \in \mathbb{R}$, or $f(x)g(y) = x + y$ for all $x, y \in \mathbb{R}$.
- (b) Find real-valued f, g such that $f(x + y) = g(xy)$ for all $x, y \in \mathbb{R}$.

Question 35: 2025 Homework 4, Question 2

Source: 2025 Homework 4, Question 2.

- (a) Find a nonconstant real-valued f with $|f(y) - f(x)| \leq |y - x|$ for all $x, y \in \mathbb{R}$.
- (b) If $f(y) - f(x) \leq (y - x)^2$ for all $x, y \in \mathbb{R}$, prove that $|f(y) - f(x)| \leq (y - x)^2$ and that f is constant.
- (c) Determine other powers α such that $f(y) - f(x) \leq (y - x)^\alpha$ for all x, y forces f to be constant.

Question 36: 2025 Homework 4, Question 3

Source: 2025 Homework 4, Question 3.

- (a) If $f : \mathbb{R} \rightarrow \mathbb{R}$ is injective, find $h : \mathbb{R} \rightarrow \mathbb{R}$ such that $h \circ f = \text{Id}$.
- (b) If $f : \mathbb{R} \rightarrow \mathbb{R}$ is surjective, find $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $f \circ g = \text{Id}$.
- (c) Discuss whether the same arguments work for arbitrary sets X, Y and maps $f : X \rightarrow Y, g, h : Y \rightarrow X$.

Question 37: 2023 Midterm 1, Question 5

Source: 2023 Midterm 1, Question 5. Let $f, g, h : \mathbb{R} \rightarrow \mathbb{R}$. Decide whether the following identities always hold, giving proof or counterexample:

$$(f + g) \circ h = f \circ h + g \circ h, \quad f \circ (g + h) = f \circ g + f \circ h.$$

Question 38: 2024 Quiz 2, Question 1

Source: 2024 Quiz 2, Question 1. Let g, h be functions.

- (a) Define $h \circ g$ and state the conditions needed for this definition to make sense.
- (b) For real-valued $f, g, h : \mathbb{R} \rightarrow \mathbb{R}$, define $f < g$ to mean $f(x) < g(x)$ for all x . If $f < g$, decide whether $f \circ h < g \circ h$ and whether $h \circ f < h \circ g$.
- (c) For each false statement, give a useful condition on h that makes it true.

Question 39: 2025 Quiz 2, Question 1

Source: 2025 Quiz 2, Question 1. Let $f, g : [0, 1] \rightarrow [0, 1]$ be

$$g(x) = \left(x - \frac{1}{2}\right)^2 + \frac{1}{2}, \quad f(x) = \begin{cases} \frac{1}{2}x, & x < \frac{2}{3}, \\ \frac{7}{3} - 2x, & x \geq \frac{2}{3}. \end{cases}$$

Find the range of g , prove that f is a bijection, give an explicit formula for f^{-1} , and give an explicit formula for $f \circ g$.

Question 40: 2025 Quiz 2, Question 2

Source: 2025 Quiz 2, Question 2. Let $A, B \subset \mathbb{R}$ be non-empty bounded sets and define $A + B = \{a + b : a \in A, b \in B\}$. State the definition of $\sup A$ and prove that

$$\sup(A + B) = \sup A + \sup B.$$

23.3 Sequences and Series

Question 41: 2025 Homework 4, Question 4

Source: 2025 Homework 4, Question 4. Let (x_n) be a sequence of positive real numbers. Prove directly from the definitions that

$$\lim x_n = +\infty \iff \lim \frac{1}{x_n} = 0.$$

Question 42: 2025 Homework 4, Question 5

Source: 2025 Homework 4, Question 5. Let $0 < x \leq y$. Define

$$a_1 = y, \quad g_1 = x, \quad a_{n+1} = \frac{a_n + g_n}{2}, \quad g_{n+1} = \sqrt{a_n g_n}.$$

Prove that $a^* = \lim a_n$ and $g^* = \lim g_n$ exist, and show that $a^* = g^*$.

Question 43: 2025 Homework 4, Question 6

Source: 2025 Homework 4, Question 6. A subsequence (y_k) of (x_n) is obtained by $y_k = x_{n_k}$ for a strictly increasing sequence (n_k) in $\mathbb{N}$.

- (a) If (x_n) converges to L , prove that every subsequence also converges to L .
- (b) Prove the converse criterion: if every subsequence of (x_n) has a sub-subsequence converging to the same L , then (x_n) converges to L .

Question 44: 2025 Homework 5, Question 1

Source: 2025 Homework 5, Question 1. Let (a_n) and (x_n) be real-valued sequences with $a_n \leq x_n$ for all n . Prove directly from the definition that if $\lim a_n = +\infty$, then $\lim x_n = +\infty$.

Question 45: 2025 Homework 5, Question 2

Source: 2025 Homework 5, Question 2. Use Bolzano–Weierstrass to prove that every bounded sequence has a nondecreasing subsequence or a nonincreasing subsequence. Hint: start with a convergent subsequence and distinguish the cases relative to its limit.

Question 46: 2025 Homework 5, Question 3

Source: 2025 Homework 5, Question 3. For a real sequence (x_n) , define averages $a_n = \frac{1}{n} \sum_{k=1}^n x_k$.

- (a) Give an example where (a_n) converges but (x_n) does not.
- (b) Prove that if (x_n) converges, then (a_n) converges and $\lim a_n = \lim x_n$.

Question 47: 2025 Homework 10, Question 5

Source: 2025 Homework 10, Question 5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $|f(x) - f(y)| \leq q|x - y|$ for all x, y , where $0 \leq q < 1$.

- (a) For any $p \in \mathbb{R}$, define $x_0 = p$, $x_{n+1} = f(x_n)$. Prove that (x_n) is Cauchy.
- (b) If $x^* = \lim x_n$, show that $f(x^*) = x^*$ and that this fixed point is unique.
- (c) Prove the exponential convergence estimate

$$|x^* - x_n| \leq |x^* - x_0|q^n \leq \frac{|x_1 - x_0|}{1 - q}q^n.$$

For $q = 9/10$ and $|x_1 - x_0| = 12$, find N so that $|x_n - x^*| < 10^{-16}$ for $n > N$.

Question 48: 2025 Homework 10, Question 6

Source: 2025 Homework 10, Question 6. Without using limit comparison or the ratio test, prove that $\sum_{n=1}^{\infty} n2^{-n}$ converges. Conclude that $\sum_{n=1}^{\infty} n \cos(n)2^{-n}$ converges absolutely.

Question 49: 2026 Homework 15, Question 2

Source: 2026 Homework 15, Question 2. Prove that if $\sum a_n$ converges absolutely, then $\sum a_n^3$ converges absolutely.

Question 50: 2026 Homework 15, Question 5

Source: 2026 Homework 15, Question 5. Let (x_n) be bounded and define

$$\limsup_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} \sup_{m \geq n} x_m.$$

- (a) Write an ϵ - N definition of $\limsup x_n = L$, and relate it to limits of subsequences.
- (b) Prove Hadamard's root test:

$$\limsup |x_n|^{1/n} < 1 \Rightarrow \sum x_n \text{ converges absolutely,}$$

and

$$\limsup |x_n|^{1/n} > 1 \Rightarrow \sum x_n \text{ diverges.}$$

Question 51: Undated Quiz 2 practice sheet, Question 1

Source: Undated Quiz 2 practice sheet, Question 1. Decide which of the following limits exist and determine their values:

$$\lim_{n \rightarrow \infty} \frac{n}{2n^2 - 3}, \quad \lim_{n \rightarrow \infty} \frac{n^2}{2n^2 - 3}, \quad \lim_{n \rightarrow \infty} \frac{n^3}{2n^2 - 3}.$$

Explain which limit properties you are using.

Question 52: Undated Quiz 2 practice sheet, Question 2

Source: Undated Quiz 2 practice sheet, Question 2.

- (a) Let (x_n) be a sequence of positive numbers converging to L . Prove directly that $L \geq 0$.
- (b) Prove the squeeze theorem for sequences: if $a_n \leq y_n \leq b_n$ and $\lim a_n = \lim b_n = L$, then $y_n \rightarrow L$.

Question 53: Undated Quiz 2 practice sheet, Question 3

Source: Undated Quiz 2 practice sheet, Question 3. Define $a_1 = 1$ and $a_{n+1} = \sqrt{1 + a_n}$.

- (a) Prove by induction that (a_n) is increasing.
- (b) Prove that (a_n) converges.
- (c) Find $\lim a_n$.

Question 54: Undated Quiz 2 practice sheet, Question 4

Source: Undated Quiz 2 practice sheet, Question 4. Define subsequence, state Bolzano–Weierstrass, and prove that if $x_n \rightarrow L$, then every subsequence also converges to L .

Question 55: 2024 Quiz 2, Question 2

Source: 2024 Quiz 2, Question 2. For $q \in \mathbb{R}$, determine whether $\lim_{n \rightarrow \infty} q^n$ exists and, if it exists, find it. Explain using the sequence limit properties developed in class. Hint: start with $q > 1$ and use Bernoulli's inequality.

Question 56: 2024 Quiz 2, Question 3

Source: 2024 Quiz 2, Question 3. Let (x_n) be bounded and set $y_n = \sup_{k \geq n} x_k$.

- (a) Give an example where (y_n) converges though (x_n) does not.
- (b) Prove that (y_n) always converges.
- (c) If $x_n > 0$ for all n , prove that $\lim x_n = 0$ if and only if $\lim y_n = 0$.

Question 57: 2025 Quiz 2, Question 3

Source: 2025 Quiz 2, Question 3. Let

$$x_1 = \frac{1}{1000}, \quad x_{n+1} = \frac{4}{3}x_n(1 - x_n).$$

Given that $f(x) = \frac{4}{3}x(1 - x) \leq 1/3$ for all x , prove that $x_n > 0$, that $x_n \leq x_{n+1}$ for all n , and that the sequence converges. Determine its limit.

Question 58: 2026 Quiz 4, Question 2

Source: 2026 Quiz 4, Question 2. Define convergence of a series $\sum_{n=1}^{\infty} a_n$, and define subsequence. If $a_n \geq 0$ and (b_n) is a subsequence of (a_n) , prove that $\sum a_n$ convergent implies $\sum b_n$ convergent.

Question 59: 2025 Quiz 6, Question 3

Source: 2025 Quiz 6, Question 3.

(a) Prove summation by parts:

$$\sum_{k=1}^n a_k(b_k - b_{k-1}) = a_n b_n - a_0 b_0 - \sum_{k=0}^{n-1} (a_{k+1} - a_k) b_k.$$

(b) If (a_n) is nonincreasing with $a_n \rightarrow 0$, and the partial sums of (c_n) are bounded, prove that $\sum_{k=1}^{\infty} a_k c_k$ converges.

Question 60: 2024 Final, Question 6

Source: 2024 Final, Question 6. Let (x_n) be nonnegative and subadditive, meaning $x_{n+m} \leq x_n + x_m$. Set $y_n = x_n/n$.

(a) Show directly that (y_{2^k}) converges.

(b) Prove Fekete's lemma:

$$\lim_{n \rightarrow \infty} y_n = \inf_{n \geq 1} y_n.$$

Hint: choose N with y_N close to $\inf y_n$ and write $n = kN + j$.

23.4 Continuity and Compactness

Question 61: 2025 Homework 5, Question 4

Source: 2025 Homework 5, Question 4. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x + y) = f(x) + f(y)$ for all x, y . Prove that if f is continuous at 0, then f is continuous everywhere.

Question 62: 2025 Homework 5, Question 5

Source: 2025 Homework 5, Question 5. Let f be real-valued on a non-empty open interval (a, b) and let $c \in (a, b)$.

- (a) If f is continuous at c and $x_n \rightarrow c$, prove directly that $f(x_n) \rightarrow f(c)$.
- (b) Conversely, if $f(x_n) \rightarrow f(c)$ for every sequence $x_n \rightarrow c$, prove that f is continuous at c .

Question 63: 2025 Homework 6, Question 1

Source: 2025 Homework 6, Question 1. For $f(x) = \sqrt{|x|}$:

- (a) Prove directly that f is continuous at 0.
- (b) Show that $\sqrt{y} - \sqrt{x} \leq \sqrt{y - x}$ for $0 \leq x \leq y$.
- (c) Conclude that f is continuous on $\mathbb{R}$.

Question 64: 2025 Homework 6, Question 2

Source: 2025 Homework 6, Question 2. Let $f : [0, 1] \rightarrow [0, 1]$ be continuous.

- (a) Prove that $f(c) = c$ for some $c \in [0, 1]$.
- (b) Decide whether there must be $d \in [0, 1]$ such that $f(d) = 1 - d^2$.

Question 65: 2025 Homework 6, Question 3

Source: 2025 Homework 6, Question 3. Let $P(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$.

- (a) If n is odd, prove that P has a zero in $\mathbb{R}$.
- (b) If n is even, prove that P attains an extremum on $\mathbb{R}$.

Question 66: 2025 Homework 6, Question 4

Source: 2025 Homework 6, Question 4. Let $[a, b]$ be a closed interval of positive length and let $f : [a, b] \rightarrow [c, d]$ be continuous and bijective. Prove that f is either strictly increasing or strictly decreasing.

Question 67: 2025 Homework 6, Question 5

Source: 2025 Homework 6, Question 5. Let $f : \mathbb{R} \rightarrow \mathbb{R}$.

- (a) If $|f(x) - f(y)| \leq C|x - y|$ for all x, y , prove directly that f is continuous.
- (b) If f is invertible and both f and f^{-1} are Lipschitz with constants C, D , determine what relation must hold between C and D .
- (c) If $CD = 1$, prove that for some x_0 either $f(x) = C(x - x_0)$ for all x , or $f(x) = C(x_0 - x)$ for all x .

Question 68: 2025 Homework 6, Question 6

Source: 2025 Homework 6, Question 6. For $a, b, c, d > 0$, prove the baseball inequality:

ity:

$$\min \left\{ \frac{a}{c}, \frac{b}{d} \right\} \leq \frac{a+b}{c+d} \leq \max \left\{ \frac{a}{c}, \frac{b}{d} \right\}.$$

Hint: write the ratio using $t = c/(c+d)$.

Question 69: 2026 Homework 11, Question 5

Source: 2026 Homework 11, Question 5. Let f be real-valued on $\mathbb{R}$.

- (a) If f is uniformly continuous on an open interval (a, b) , prove that f is bounded there.
- (b) If f is uniformly continuous on $\mathbb{R}$, prove that it grows at most linearly: there exist A, B such that $|f(x)| \leq A|x| + B$ for all x .

Question 70: 2024 Quiz 3, Question 1

Source: 2024 Quiz 3, Question 1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous and satisfy $\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow +\infty} f(x) = 0$.

- (a) Define global maximum.
- (b) Prove that f has a global maximum or a global minimum.
- (c) Prove that if f has both a global maximum and a global minimum, then it has a zero.

State which theorems apply and verify their assumptions.

Question 71: 2025 Quiz 3, Question 2

Source: 2025 Quiz 3, Question 2. State the ϵ - δ definition of continuity of f at x_0 . Suppose that for every open interval (c, d) , the inverse image $f^{-1}((c, d)) = \{x : c < f(x) < d\}$ is again an open interval. Prove that f is continuous on $\mathbb{R}$.

Question 72: 2023 Midterm Test 2, Question 2

Source: 2023 Midterm Test 2, Question 2. State the Extreme Value Theorem and sketch its proof in two or three sentences.

Question 73: 2023 Midterm Test 2, Question 5

Source: 2023 Midterm Test 2, Question 5. Define continuity of $f : \mathbb{R} \rightarrow \mathbb{R}$ at a . If f is not continuous at a , construct a sequence $x_n \rightarrow a$ such that $f(x_n)$ does not converge to $f(a)$.

Question 74: 2023 Midterm Test 2, Question 6

Source: 2023 Midterm Test 2, Question 6. If $f : [a, b] \rightarrow \mathbb{R}$ is continuous and injective, prove that f is either strictly increasing or strictly decreasing on $[a, b]$.

Question 75: 2026 Quiz 5, Question 2

Source: 2026 Quiz 5, Question 2. Let f be continuous on $(0, 1]$. Find when one can choose c so that

$$g(x) = \begin{cases} f(x), & x \in (0, 1], \\ c, & x = 0 \end{cases}$$

is continuous on $[0, 1]$.

- (a) Give an example where no such c exists.
- (b) Define uniformly continuous.
- (c) If f is uniformly continuous on $(0, 1]$, choose c and prove that g is continuous at 0.

23.5 Differentiation and Mean Value Theorems

Question 76: 2025 Homework 7, Question 1

Source: 2025 Homework 7, Question 1. Let $f(x) = \sqrt{x}$ for $x > 0$.

- (a) Assuming differentiability, use $(f(x))^2 = x$ to derive a formula for $f'(a)$.
- (b) Prove from the difference quotient that f is differentiable on $(0, \infty)$.

Question 77: 2025 Homework 7, Question 2

Source: 2025 Homework 7, Question 2. Without using the Mean Value Theorem, let $f : (a, b) \rightarrow \mathbb{R}$ be differentiable and $c \in (a, b)$.

- (a) If $f'(c) > 0$, prove that $f(x) > f(c)$ for x just to the right of c and $f(x) < f(c)$ for x just to the left.
- (b) Decide whether f must be strictly increasing on that whole small interval.
- (c) If $f'(x) > 0$ for all $x \in (a, b)$, prove that f is strictly increasing on (a, b) . What can be said about $\lim_{x \rightarrow b^-} f(x)$?

Question 78: 2025 Homework 7, Question 3

Source: 2025 Homework 7, Question 3. Prove the racetrack principle: if f, g are continuous on $[a, b]$, differentiable on (a, b) , $f(a) \leq g(a)$, and $f'(x) \leq g'(x)$ on (a, b) , then $f(x) \leq g(x)$ on $[a, b]$.

Question 79: 2025 Homework 7, Question 5

Source: 2025 Homework 7, Question 5. Design a method to approximate $\cos(x)$ for $x \in [-10^{16}, 10^{16}]$ using trigonometric facts and the Mean Value Theorem.

- (a) Reduce the problem to approximating $\cos(x)$ on $(0, \pi/2)$.
- (b) If stored values $(a_n, \cos a_n)$ are uniformly spaced and linear interpolation is used, how large should N be to guarantee error below 10^{-6} ?
- (c) Repeat for linear approximation using $\cos(a_k) - \sin(a_k)(x - a_k)$.
- (d) Suggest a smarter way to use these linear approximations and describe the effect on N .

(e) Explain the challenge in approximating $\tan(x)$ on $(0, \pi/2)$ by linear approximations.

Question 80: 2025 Homework 7, Question 6

Source: 2025 Homework 7, Question 6. If $f : (a, b) \rightarrow \mathbb{R}$ satisfies $|f(x) - f(y)| \leq |x - y|^2$ for all x, y , give a short proof that f is constant.

Question 81: 2025 Homework 8, Question 1

Source: 2025 Homework 8, Question 1. Find

$$\lim_{x \rightarrow \infty} \frac{x^2}{x^2 + \sin x},$$

justifying every step.

Question 82: 2025 Homework 8, Question 2

Source: 2025 Homework 8, Question 2. Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and suppose the one-sided derivatives $f'(a+)$ and $f'(b-)$ exist.

- (a) If f has a local minimum at a , prove $f'(a+) \geq 0$; if at b , prove $f'(b-) \leq 0$.
- (b) Conversely, if $f'(a+) > 0$, prove a strict local minimum at a ; if $f'(b-) < 0$, prove one at b .
- (c) Describe the results using the words necessary and sufficient.

Question 83: 2025 Homework 8, Question 3

Source: 2025 Homework 8, Question 3. Let I contain 0, and suppose f is twice differentiable with $mf''(t) + kf(t) = 0$, where $k, m > 0$.

- (a) If $f(0) = f'(0) = 0$, prove $f \equiv 0$ using the energy $E(t) = \frac{1}{2}(kf(t)^2 + mf'(t)^2)$.
- (b) Prove uniqueness for initial data $f(0) = a$, $f'(0) = b$.
- (c) Discuss when kinetic and potential energies are maximal or minimal.

Question 84: 2025 Homework 8, Question 4

Source: 2025 Homework 8, Question 4. Suppose f is continuous on $[a, b]$, differentiable on (a, b) , and $f'(a+), f'(b-)$ exist.

- (a) If $f'(a+) < 0 < f'(b-)$, prove $f'(x) = 0$ for some $x \in (a, b)$.
- (b) If $f'(a+) < t < f'(b-)$, prove $f'(x) = t$ for some $x \in (a, b)$.

This shows that derivatives have the intermediate value property.

Question 85: 2025 Homework 8, Question 5

Source: 2025 Homework 8, Question 5. If f and g are n -times differentiable, prove

Leibniz' formula:

$$(fg)^{(n)}(x) = \sum_{k=0}^n \binom{n}{k} f^{(k)}(x) g^{(n-k)}(x).$$

Question 86: 2025 Homework 8, Question 6

Source: 2025 Homework 8, Question 6. Let f be continuous on $[0, \infty)$ and differentiable on $(0, \infty)$, with $f(0) = 0$ and f' strictly increasing. Prove that $g(x) = f(x)/x$ is strictly increasing on $(0, \infty)$.

Question 87: 2026 Homework 9, Question 2

Source: 2026 Homework 9, Question 2. The population $x(t) > 0$ satisfies

$$\ln |x(t)| - \ln |x(t) - C| = Ct + B.$$

Find $x'(t)$. Determine when $x'(t) < 0$, $x'(t) > 0$, and $x'(t) = 0$, and interpret the result biologically.

Question 88: 2024 Quiz 3, Question 2

Source: 2024 Quiz 3, Question 2. Compute derivatives using differentiation rules:

- (a) $f(x) = \sin(x + \cos(x^2 + e^{x^3}))$.
- (b) $g(x) = e^{-1/x}$ for $x > 0$.
- (c) Show that $g^{(n)}(x) = P_n(1/x)e^{-1/x}$ for a polynomial P_n ; determine $\deg P_n$.

Question 89: 2024 Quiz 3, Question 3

Source: 2024 Quiz 3, Question 3. Let $f(x) = \sqrt{1 + 9x^2}$. Prove that $|f(x) - f(y)| < 3|x - y|$ for all x, y , state the theorem used, and prove that the constant 3 is best possible.

Question 90: 2025 Quiz 3, Question 1

Source: 2025 Quiz 3, Question 1. Using the graph information printed in the source paper: compute $f'(x)$ for $f(x) = \sqrt{3 - x}$, find $(g \circ f)'(1)$ for the displayed piecewise-linear graph of g , give an example of a differentiable function with $f' = 0$ that is not constant on its domain, and identify local and global extrema of a function s from the displayed graph of s' .

Question 91: 2025 Quiz 3, Question 3

Source: 2025 Quiz 3, Question 3. Define the symmetric derivative

$$f'_s(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x-h)}{2h}.$$

- (a) For $f(x) = |x|$, show that $f'_s(0)$ exists and determine it.

- (b) Prove that differentiability at x implies existence of $f'_s(x)$ and $f'_s(x) = f'(x)$.
- (c) For Dirichlet's function $f(x) = 1$ on $\mathbb{Q}$ and 0 otherwise, find all points where f'_s exists.

Question 92: 2023 Midterm Test 2, Question 1

Source: 2023 Midterm Test 2, Question 1. For $f(x) = \sqrt{1+x^2}$, compute $f'(x)$ and $f''(x)$, then find the limits of $f(x)$, $f'(x)$, and $f''(x)$ as $x \rightarrow \infty$.

Question 93: 2023 Midterm Test 2, Question 3

Source: 2023 Midterm Test 2, Question 3. Prove, for $0 < x < y$,

$$\frac{(1 + \frac{1}{2\sqrt{y}})(y-x)}{(1 + \sqrt{y})^2} < \frac{y}{1 + \sqrt{y}} - \frac{x}{1 + \sqrt{x}} < \frac{(1 + \frac{1}{2\sqrt{x}})(y-x)}{(1 + \sqrt{x})^2}.$$

Conclude that

$$\lim_{x \rightarrow \infty} \left(\frac{x+1}{1 + \sqrt{x+1}} - \frac{x}{1 + \sqrt{x}} \right) = 0.$$

Question 94: 2023 Midterm Test 2, Question 4

Source: 2023 Midterm Test 2, Question 4. Let $f(x) = (\sin x \cos(1/x))^2$ for $x \neq 0$ and $f(0) = 0$. Determine all points where f is differentiable and find f' .

Question 95: 2026 Quiz 4, Question 3

Source: 2026 Quiz 4, Question 3. Let $f(x) = xe^x$.

- (a) Sketch f , indicating extrema, convexity, concavity, and behavior at $\pm\infty$.
- (b) Show f is injective on some open interval containing 0.
- (c) Let I be the largest interval containing 0 on which f is injective, and let $J = f(I)$. Identify I and J .
- (d) If $W : J \rightarrow I$ is the inverse, show that

$$W'(y) = \frac{W(y)}{(1 + W(y))y}$$

whenever the ratio makes sense.

Question 96: 2024 Final, Question 4

Source: 2024 Final, Question 4. For $f(x) = xe^x$, show injectivity near 0, find the maximal domain and range of the inverse, derive the derivative formula for the Lambert

W function, and show

$$\int W(y) dy = y \left(W(y) - 1 + \frac{1}{W(y)} \right) + C.$$

Question 97: 2025 Final, Question 3

Source: 2025 Final, Question 3. Let $f, \beta : [a, \infty) \rightarrow \mathbb{R}$ be continuous, with f differentiable on (a, ∞) and $f'(x) \leq \beta(x)f(x)$.

- (a) Verify that $g(x) = \exp(\int_a^x \beta(t) dt)$ satisfies $g'(x) = \beta(x)g(x)$.
- (b) Prove $f(x) \leq f(a) \exp(\int_a^x \beta(t) dt)$.
- (c) Give a simple useful condition on β ensuring that f is bounded above.

Question 98: 2025 Final, Question 4

Source: 2025 Final, Question 4. For $0 \leq \theta \leq \pi/2$ and $0 \leq p \leq 1$, prove

$$(\cos \theta)^p \leq \cos(p\theta).$$

Hint: differentiate with respect to p , and support the argument with a sketch of the two functions of p .

Question 99: 2026 Makeup Final, Question 4

Source: 2026 Makeup Final, Question 4. Prove that for all $x > 0$,

$$\frac{x}{1+x} \leq \ln(1+x) \leq x.$$

Use this to evaluate $\lim_{n \rightarrow \infty} n \ln(1 + 1/n)$ and deduce that $\lim_{n \rightarrow \infty} (1 + 1/n)^n = e$.

23.6 Optimization, Graphing, and Convexity

Question 100: 2025 Homework 7, Question 4

Source: 2025 Homework 7, Question 4. Two adjacent rectangular chicken pens share a side of length L , and each pen must have area A . Find the minimal total fence length and the L where it is achieved.

Question 101: 2025 Homework 8, warm-up graphing prompt

Source: 2025 Homework 8, warm-up graphing prompt. Sketch and analyze each function, indicating local and global extrema, inflection points, and intervals of increase/decrease:

$$x^3 - x^2 - 8x + 1 \text{ on } [-2, 2], \quad \frac{x}{x^2 - 1} \text{ on } [0, 5],$$

$$3x^4 - 8x^3 + 6x^2 \text{ on } \mathbb{R}, \quad x + \frac{1}{x^2} \text{ on } \mathbb{R}.$$

Question 102: 2026 Homework 9, Question 1

Source: 2026 Homework 9, Question 1. Let $f(x) = (x^2 + 2x)e^{-x}$.

- (a) Find the zeroes, the limits at $\pm\infty$, local and global extrema, inflection points, and intervals of concavity/convexity.
- (b) Sketch the graph.
- (c) For each $t \in \mathbb{R}$, determine the cardinality of $f^{-1}(t)$.

Question 103: 2026 Homework 9, Question 3

Source: 2026 Homework 9, Question 3. If $I \subset \mathbb{R}$ is an interval and $f, g : I \rightarrow \mathbb{R}$ are convex, prove that $f + g$ and $\max\{f, g\}$ are convex.

Question 104: 2026 Homework 9, Question 4

Source: 2026 Homework 9, Question 4. Let $f : I \rightarrow \mathbb{R}$ be weakly convex.

- (a) If $p_i > 0$ and $\sum p_i = 1$, prove by induction that

$$f\left(\sum_{i=1}^n p_i x_i\right) \leq \sum_{i=1}^n p_i f(x_i).$$

- (b) Use this to prove the AM-GM inequality.
- (c) Determine exactly when equality occurs in both parts.

Question 105: 2026 Homework 9, Question 5

Source: 2026 Homework 9, Question 5. Prove that a convex function on an open interval is continuous. Give an example of a convex function on a closed interval that is not continuous.

Question 106: 2026 Homework 9, Question 6

Source: 2026 Homework 9, Question 6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be infinitely differentiable, $f'' > 0$, $\lim_{x \rightarrow -\infty} f(x)/x = -\infty$, and $\lim_{x \rightarrow \infty} f(x)/x = \infty$. Define

$$f^*(y) = \sup_{x \in \mathbb{R}} \{xy - f(x)\}.$$

- (a) Compute the Legendre transform of $q(x) = \frac{a}{2}(x - b)^2 + c$.
- (b) Prove the supremum is attained at a unique point $x^* = h(y)$.
- (c) Relate h to f , state the theorem guaranteeing differentiability of h , and find $h'(y)$.
- (d) Show $(f^*)''(y) > 0$.
- (e) Prove $(f^*)^* = f$, assuming f^* has the desired behavior at $\pm\infty$.

Question 107: 2026 Homework 15, Question 1

Source: 2026 Homework 15, Question 1. Let f be four-times differentiable at a , with $f'(a) = f''(a) = 0$.

- (a) Prove that if $f'''(a) > 0$, then f cannot attain a local extremum at a .
- (b) If $f'''(a) = 0$ and $f^{(4)}(a) > 0$, prove that for small $|x - a|$,

$$f(a) + 0.99 \frac{f^{(4)}(a)}{4!} (x - a)^4 \leq f(x) \leq f(a) + 1.01 \frac{f^{(4)}(a)}{4!} (x - a)^4,$$

and hence f has a strict local minimum at a .

Question 108: 2026 Homework 15, Question 6

Source: 2026 Homework 15, Question 6. Two hallways of widths a and b meet at right angles. Find the greatest possible length of a ladder that can be carried horizontally around the corner. Define variables and justify the setup.

Question 109: 2026 Quiz 4, Question 1

Source: 2026 Quiz 4, Question 1. Let $C > 0$ and

$$P = \{(x, y) \in \mathbb{R}^2 : y \leq C - x^2, y \geq 0\}.$$

Find the rectangle enclosed in P with largest possible area, assuming its base lies on the x -axis and all four corners lie on the boundary. Sketch, state the theorem ensuring existence of a positive maximum, formulate the maximization problem, and solve it.

Question 110: 2024 Final, Question 2

Source: 2024 Final, Question 2. Let $f(x) = (x^2 + 2x)e^{-x}$. Find zeroes and limits at $\pm\infty$, determine local and global extrema, and draw a qualitatively correct graph including tangent lines at the zeroes and regions of convexity/concavity.

Question 111: 2026 Makeup Final, Question 3

Source: 2026 Makeup Final, Question 3. Let $f(x) = x/(1 + x^2)$. Find the domain, limits at $\pm\infty$, zeroes, local/global extrema, inflection points, intervals of convexity/concavity, and sketch the graph with tangent lines at zeroes and the horizontal asymptote.

23.7 Riemann Integration and the Fundamental Theorem of Calculus

Question 112: 2026 Homework 10, Question 1

Source: 2026 Homework 10, Question 1. Let f be nonnegative and integrable on $[a, b]$.

- (a) Prove $\int_a^b f \geq 0$.
- (b) Give a nonzero integrable f with $\int_a^b f = 0$.
- (c) If additionally f is continuous and $\int_a^b f = 0$, prove $f \equiv 0$.

Question 113: 2026 Homework 10, Question 2

Source: 2026 Homework 10, Question 2. If $f, g : [a, b] \rightarrow \mathbb{R}$ are integrable, prove that $\max(f, g)$ is integrable.

Question 114: 2026 Homework 10, Question 3

Source: 2026 Homework 10, Question 3. Prove that $f(x) = e^x$ is integrable on $[0, 1]$, and determine the value of the integral.

Question 115: 2026 Homework 10, Question 4

Source: 2026 Homework 10, Question 4. If $f : [a, b] \rightarrow \mathbb{R}$ is bounded and nondecreasing, prove that f is integrable. Hint: use dyadic partitions and estimate $U(f, P_1) - L(f, P_1)$.

Question 116: 2026 Homework 11, Question 1

Source: 2026 Homework 11, Question 1. Use upper and lower sums to prove the transformations

$$\int_{a+c}^{b+c} f(x) dx = \int_a^b f(x+c) dx,$$
$$\int_{\gamma a}^{\gamma b} f(x) dx = \gamma \int_a^b f(\gamma x) dx \quad (\gamma > 0),$$

and

$$\int_{-b}^{-a} f(x) dx = \int_a^b f(-x) dx.$$

Interpret each geometrically.

Question 117: 2026 Homework 11, Question 2

Source: 2026 Homework 11, Question 2. Define $\log a = \int_1^a \frac{1}{x} dx$ for $a > 0$. Prove from basic principles that $\log(ab) = \log a + \log b$.

Question 118: 2026 Homework 11, Question 3

Source: 2026 Homework 11, Question 3. Let $f : [a, b] \rightarrow [c, d]$ be strictly increasing, continuous, and bijective, with inverse $g = f^{-1}$. Use upper and lower sums to prove

$$\int_a^b f(x) dx + \int_c^d g(y) dy = bd - ac.$$

Question 119: 2026 Homework 11, Question 4

Source: 2026 Homework 11, Question 4. If $f : [a, b] \rightarrow \mathbb{R}$ is integrable, prove

$$\int_a^b |f| \geq \left| \int_a^b f \right|$$

in three ways: from upper and lower sums; using positivity; and using additivity with $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$.

Question 120: 2026 Homework 11, Question 6

Source: 2026 Homework 11, Question 6. If f is integrable on $[a, b]$, prove that for every $\epsilon > 0$ there are continuous functions g, h with $g \leq f \leq h$ and

$$\int_a^b h - \int_a^b g < \epsilon.$$

Question 121: 2026 Homework 12, Question 1

Source: 2026 Homework 12, Question 1. Use the Fundamental Theorem of Calculus to find derivatives of functions such as

$$F(x) = \sin \left(\int_0^x \sin \left(\int_0^y \sin^3(t) dt \right) dy \right),$$

nested integrals $H(x)$, parameter-dependent integrals $G(x) = \int_x^b \frac{1}{1+t^2+\sin^2 t} dt$, and a composition $K(x)$ involving $\int_0^{R(x)} \sin^3 t dt$.

Question 122: 2026 Homework 12, Question 2

Source: 2026 Homework 12, Question 2. Compute and verify indefinite integrals:

$$\int \frac{2x^2 + x + 1}{(x+3)(x-1)^2} dx, \quad \int e^{ax} \sin bx dx, \quad \int \frac{\log(\log x)}{x \log x} dx, \quad \int (\log x)^3 dx.$$

Question 123: 2026 Homework 12, Question 3

Source: 2026 Homework 12, Question 3. For continuous $f : [a, b] \rightarrow \mathbb{R}$, define its average $\bar{f} = \frac{1}{b-a} \int_a^b f(x) dx$.

- (a) Prove there is $c \in [a, b]$ such that $f(c) = \bar{f}$.
- (b) Justify why $(x(t_2) - x(t_1))/(t_2 - t_1)$ is called the average rate of change.

Question 124: 2026 Homework 12, Question 4

Source: 2026 Homework 12, Question 4. Let f be continuous on an interval I , $a \in I$, and

$$G(x) = \int_a^x (x - t)f(t) dt.$$

Find $G'(x)$, prove $G(x) = \int_a^x \int_a^u f(t) dt du$ and also by integration by parts, find $G''(x)$, and propose a one-integral formula for a function whose n -th derivative is f .

Question 125: 2026 Homework 12, Question 5

Source: 2026 Homework 12, Question 5.

- (a) If f is integrable on $[0, x]$ for every $x > 0$ and $f(x) \rightarrow a$, prove

$$\lim_{x \rightarrow \infty} \frac{1}{x} \int_0^x f(t) dt = a.$$

- (b) Prove that $\int_0^x \frac{\sin t}{1+t} dt > 0$ for all $x > 0$ and that $\lim_{x \rightarrow \infty} \int_0^x \frac{\sin t}{1+t} dt$ exists.

Question 126: 2026 Homework 12, Question 6

Source: 2026 Homework 12, Question 6. If f is continuous on $[a, b]$ and $\int_c^d f = 0$ for every subinterval $[c, d] \subseteq [a, b]$, prove $f \equiv 0$ in two ways: by upper/lower sums and by the Fundamental Theorem of Calculus.

Question 127: 2026 Quiz 5, Question 1

Source: 2026 Quiz 5, Question 1.

- (a) Give a graphical example of discontinuous integrable f on $[0, 2]$ and the corresponding $F(x) = \int_0^x f(t) dt$.
- (b) Compute $\int_{-1}^1 f(x) dx$ for $f(x) = 2x$ if $x < 0$ and $f(x) = x^2$ if $x \geq 0$.
- (c) Compute $F'(x)$ where $F(x) = \int_0^x \sin^3 t dt$.

Question 128: 2026 Quiz 5, Question 3

Source: 2026 Quiz 5, Question 3. With $\log x = \int_1^x \frac{1}{t} dt$ and $S_n = \sum_{k=1}^n 1/k$, prove

$$\log n \leq S_n \leq \log(n+1).$$

Hint: use upper and lower sums.

Question 129: 2025 Quiz 6, Question 1

Source: 2025 Quiz 6, Question 1. Compute

$$\int_{-1}^1 \frac{1+2x}{(1+x^2)^3} dx.$$

Question 130: 2026 Quiz 6, Question 1

Source: 2026 Quiz 6, Question 1. Compute

$$\int \sin^3 x \cos^2 x \, dx, \quad \int \frac{8-3x}{(1-x)(x^2+4)} dx.$$

For the corresponding definite integral $\int_a^b \frac{8-3x}{(1-x)(x^2+4)} dx$, state the conditions on a, b under which your antiderivative may be used.

Question 131: 2026 Quiz 6, Question 3

Source: 2026 Quiz 6, Question 3. Let $h : (0, 1) \rightarrow \mathbb{R}$ be twice continuously differentiable with $h'(x) > 0$ and $h''(x) > 0$. Prove that for $0 < a < b$,

$$\left| \int_a^b \cos(h(x)) \, dx \right| \leq \frac{2}{h'(a)}.$$

Hint: integrate by parts with $g'(x) = h'(x) \cos(h(x))$.

Question 132: 2024 Final, Question 3

Source: 2024 Final, Question 3. Evaluate

$$\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx, \quad \int \frac{1}{x^2 + 2x + 3} dx, \quad \int x \arctan x \, dx,$$

and

$$\lim_{x \rightarrow \infty} e^{-x^2} \int_1^{x+(\log x)/(2x)} e^{t^2} dt.$$

Question 133: 2025 Final, Question 2

Source: 2025 Final, Question 2. Evaluate $g'(0)$ where g is the inverse of $f(x) = x^2 \log x$ on the stated interval; evaluate

$$\int x^3 e^{-x^2/2} dx, \quad \int \frac{1}{x(x-1)^2} dx, \quad \lim_{x \rightarrow 0} \frac{\sin x - x}{\log(1+x^3)}.$$

Question 134: 2026 Makeup Final, Question 2

Source: 2026 Makeup Final, Question 2. Evaluate

$$\int \frac{x^2}{(1+x^2)^2} dx, \quad \int \frac{\ln x}{x^2} dx,$$

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x - \frac{x^2}{2}}{x^2(\cos x - 1)},$$

and $h'(0)$ where h is the inverse of $f(x) = x + x^3 + x^5$.

23.8 Taylor Theory, Power Series, and Uniform Convergence

Question 135: 2026 Homework 14, Question 1

Source: 2026 Homework 14, Question 1.

(a) Prove

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta},$$

under the usual non-vertical restrictions.

(b) Conclude an arctangent addition formula

$$\arctan x + \arctan y = \arctan \left(\frac{x + y}{1 - xy} \right)$$

and determine for which x, y it is valid.

Question 136: 2026 Homework 14, Question 2

Source: 2026 Homework 14, Question 2. Use Taylor's theorem to evaluate:

$$\lim_{x \rightarrow 0} \frac{e^x - 1 - x - \frac{1}{2}x^2}{x - \sin x},$$

$$\lim_{x \rightarrow 0} \frac{\frac{e^x}{1+x} - 1 - \frac{1}{2}x^2}{x - \sin x},$$

and

$$\lim_{x \rightarrow 0} \left(\frac{1}{\sin^2 x} - \frac{1}{\sin(x^2)} \right).$$

Question 137: 2026 Homework 14, Question 3

Source: 2026 Homework 14, Question 3. Let

$$f(x) = \begin{cases} (e^x - 1)/x, & x \neq 0, \\ 1, & x = 0. \end{cases}$$

Find the Taylor polynomial of degree n at 0, give a bound for the remainder, and approximate $\int_0^1 f(x) dx$ with error less than 10^{-4} .

Question 138: 2026 Homework 14, Question 4

Source: 2026 Homework 14, Question 4. Let f be n -times continuously differentiable on an open interval I , fix $a \in I$, and let $P_{k,a}$ be the k -th Taylor polynomial.

(a) Prove by induction and integration by parts that

$$f(x) = P_{n-1,a}(x) + \int_a^x \frac{(x-t)^{n-1}}{(n-1)!} f^{(n)}(t) dt.$$

(b) Use this to prove $f(x) - P_{n,a}(x) = o((x-a)^n)$ as $x \rightarrow a$.

Question 139: 2026 Homework 14, Question 5

Source: 2026 Homework 14, Question 5. Write the Taylor polynomial P_n of e^x about 0 and prove that $P_n(x) \rightarrow e^x$ for every $x \in \mathbb{R}$.

Question 140: 2026 Homework 14, Question 6

Source: 2026 Homework 14, Question 6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuously differentiable, $f(a) = 0$, and $f'(a) \neq 0$. Define

$$x_{n+1} = x_n - \frac{f(x_n)}{\alpha}.$$

Analyze the iteration graphically for $f(x) = x^2 - 1$ and $\alpha = f'(a)$ or $f'(x_0)$, prove the contraction estimate for $\Phi_\alpha(x) = x - f(x)/\alpha$, prove local exponential convergence when $\alpha = f'(a)$, and discuss whether the recursion is programmable.

Question 141: 2026 Homework 15, Question 4

Source: 2026 Homework 15, Question 4. Let f be $(n+1)$ -times differentiable and define $S(t)$ by

$$f(x) = f(t) + f'(t)(x-t) + \cdots + \frac{f^{(n)}(t)}{n!}(x-t)^n + S(t).$$

Verify $S(a) = R_{n,a}(x)$, $S(x) = 0$, and $S'(t) = -\frac{f^{(n+1)}(t)}{n!}(x-t)^n$. Use MVT and Cauchy MVT to obtain the Cauchy and Lagrange forms of the remainder.

Question 142: 2026 Homework 16, Question 1

Source: 2026 Homework 16, Question 1. For $f(x) = \sin x/x$ with $f(0) = 1$, find $f^{(k)}(0)$. By a similar method, find the derivatives $\arctan^{(k)}(0)$.

Question 143: 2026 Homework 16, Question 2

Source: 2026 Homework 16, Question 2. If

$$f(x) = \sum_{n=0}^{\infty} a_n x^n, \quad g(x) = \sum_{n=0}^{\infty} b_n x^n$$

on $(-a, a)$, use the product rule to prove that

$$f(x)g(x) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k b_{n-k} \right) x^n.$$

Question 144: 2025 Quiz 6, Question 2

Source: 2025 Quiz 6, Question 2. Let f be differentiable on an open interval, P a polynomial of degree $n > 1$, and a in the interval.

(a) If f is n -times differentiable at a , prove that

$$\lim_{x \rightarrow a} \frac{f(x) - P(x)}{(x - a)^n} = 0 \Rightarrow \lim_{x \rightarrow a} \frac{f'(x) - P'(x)}{(x - a)^{n-1}} = 0.$$

(b) Give an example showing the conclusion can fail without n -times differentiability.

Question 145: 2026 Quiz 6, Question 2

Source: 2026 Quiz 6, Question 2. Define pointwise and uniform convergence for functions $f_n : I \rightarrow \mathbb{R}$. Give an example of bounded functions f_n converging pointwise to an unbounded f , and prove that if each f_n is bounded and $f_n \rightarrow f$ uniformly, then f is bounded.

Question 146: 2024 Final, Question 5

Source: 2024 Final, Question 5. Let f be twice differentiable on an interval containing 0 and satisfy $f''(x) = xf(x)$. Suppose $f(0) = 1$ and $f'(0) = 0$, and let $P_n(x) = \sum_{k=0}^n a_k x^k$ be the Taylor polynomial.

(a) Prove by induction that f is infinitely differentiable.

(b) Show $a_2 = 0$ and $a_{n+2} = a_{n-1}/((n+1)(n+2))$, and find $f^{(6)}(0)$.

(c) Prove that $\sum a_n x^n$ converges absolutely for all $x \in \mathbb{R}$.

Question 147: 2025 Final, Question 6

Source: 2025 Final, Question 6. Let (f_n) be nonnegative and nonincreasing on $[0, 1]$.

- (a) Define uniform convergence.
- (b) Prove pointwise convergence to a limiting function g .
- (c) If all f_n and g are continuous, prove that $f_n \rightarrow g$ uniformly.

Question 148: 2026 Makeup Final, Question 5

Source: 2026 Makeup Final, Question 5. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f'' + f = 0$, $f(0) = 0$, and $f'(0) = 1$.

- (a) Prove f is infinitely differentiable.
- (b) Compute Taylor coefficients $a_n = f^{(n)}(0)/n!$ and show $a_{n+2} = -a_n/((n+1)(n+2))$.
- (c) Prove $\sum a_n x^n$ converges absolutely for all x .
- (d) Using the identity theorem for power series, conclude $f(x) = \sin x$.

Question 149: 2026 Makeup Final, Question 6

Source: 2026 Makeup Final, Question 6. For $f_n(x) = nx/(1 + n^2x^2)$ on $[0, 1]$:

- (a) Compute the pointwise limit.
- (b) Determine whether convergence is uniform by computing the sup norm.
- (c) Compare $\lim_n \int_0^1 f_n$ and $\int_0^1 \lim_n f_n$.
- (d) Prove that uniform convergence of continuous g_n on $[a, b]$ implies convergence of integrals.

23.9 Curves, Arc Length, Improper Integrals, and Applications

Question 150: 2026 Homework 15, Question 3

Source: 2026 Homework 15, Question 3.

- (a) Define the improper integral $\int_0^\infty \frac{\sqrt{x}}{1+x^2} dx$ as a limit and prove it exists by estimating the integrand.
- (b) Show that $\int_0^\infty x^{-r} dx$ does not exist for any $r > 0$.
- (c) Determine for which $r > 0$ the series $\sum_{n=1}^\infty n^{-r}$ converges.

Question 151: 2026 Homework 16, Question 3

Source: 2026 Homework 16, Question 3. Consider the cardioid $r(\theta) = 1 - \sin \theta$ for $\theta \in [-\pi, \pi]$.

- (a) Sketch the curve and describe the unit tangent as $\theta \rightarrow \pi/2^\pm$.
- (b) Compute its length.

Question 152: 2026 Homework 16, Question 4

Source: 2026 Homework 16, Question 4. Prove that the improper integral

$$\int_0^{\infty} \frac{|\sin x|}{x} dx$$

diverges, even though $\int_0^{\infty} \frac{\sin x}{x} dx$ exists.

Question 153: 2026 Homework 16, Question 5

Source: 2026 Homework 16, Question 5. For the sphere of radius $r > 0$ in $\mathbb{R}^3$, prove that the surface area between heights $h_1 < h_2$ is $2\pi r(h_2 - h_1)$.

Question 154: 2026 Homework 16, Question 6

Source: 2026 Homework 16, Question 6. Let

$$I = \int_{-\infty}^{\infty} e^{-x^2/2} dx.$$

Prove I is finite, use shells to compute the volume obtained by rotating the area under the graph about the y -axis, then use Cavalieri's principle to show $V = I^2$ and determine I .

Question 155: 2025 Final, Question 5

Source: 2025 Final, Question 5. Define the length of a curve $\gamma : [a, b] \rightarrow \mathbb{R}^2$. Compute and sketch the curve

$$\gamma(t) = (\sin t - t, 1 - \cos t), \quad t \in [-\pi, \pi],$$

and describe what happens to its unit tangent as $t \rightarrow 0^{\pm}$.